Chapter 1

Mathematical Methods for Scientific Computing

The overall aim of scientific computing could be summarised as:

1. Start with some equations (probably an analytically intractable system of equations)

2. Develop a numerical technique for solving these equations

3. Use this numerical technique to solve interesting problems, such that you can then be awarded a degree/publish papers/make your boss and/or customers happy

In going from step 1 to step 2, you may find yourself performing various mathematical techniques to rewrite the underlying system of equations in a suitable formulation. You may then need to ensure that your numerical technique behaves properly (again, through mathematical analysis). And finally, you may need to analyse results to show that they are a true solution to your interesting problem, and not just a pretty picture.

This course then aims to cover many of the key mathematical techniques used in scientific computing. It may be that you have encountered some of these before (possibly even all), but even in this case, with luck this is a good refresher or a slightly different view on these techniques.

1.0.1 Disclaimer

This is an evolving set of notes still, and is perhaps best considered as an advanced first draft of the notes. As a result, there may be typos (I'd be

alarmed if there weren't), and there may be sections which seem missing/low on detail.

If you find any typos, please let me know, I'll correct them and redistribute these notes.

Please also let me know if any area of these notes seems sparse or missing, or even if there are any topics I should be covering in future years.

1.1 Prerequisite knowledge

Whilst most of the maths you have done so far could be considered prerequisite knowledge, there are a couple of expressions worth repeating, so that we can refer to them explicitly.

1.1.1 Fundamental theorems of calculus

The fundamental theorems of calculus relate integrals and derivatives. The first fundamental theorem of calculus is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{x_0}^{x} f(x')\mathrm{d}x'\right) = f(x). \tag{1.1}$$

This states that the derivative of the integral of a function $f(x)$ is simply $f(x)$, though this does require that f is continuous.

The second fundamental theorem of calculus is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{x_0}^{x_1} f(x)\mathrm{d}x\right) = \left(\int_{x_0}^{x_1} \frac{\mathrm{d}f}{\mathrm{d}x}\mathrm{d}x\right) = f(x_1) - f(x_0). \tag{1.2}$$

This states that the derivative of the integral of a function, $f(x)$, between two points, x_0 and x_1, is the same as the integral of the derivative of this function between these points, providing $f(x)$ is differentiable. This also states that in both these cases, the result is simply the difference of $f(x)$ evaluated at these points.

1.1.2 Series

The Taylor series of a function $f(x)$ about the point $x = a$ is

$$f(x) = f(a) + \left.\frac{\partial f}{\partial x}\right|_{x=a}(x-a) + \frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=a}(x-a)^2 + \frac{1}{3!}\left.\frac{\partial^3 f}{\partial x^3}\right|_{x=a}(x-a)^3 + \ldots \tag{1.3}$$

A second useful definition is the Taylor series of $f(x+a)$ about $f(x)$, which is

$$f(x+a) = f(x) + a\,\frac{\partial f}{\partial x} + a^2\,\frac{1}{2}\frac{\partial^2 f}{\partial x^2} + a^3\,\frac{1}{3!}\frac{\partial^3 f}{\partial x^3} + \ldots \tag{1.4}$$

This second form is a direct substitution of $x \to x+a$ and $a \to x$ of the first expression, and is often encountered in numerical methods and approximations.

One particularly useful expression which results from Taylor series is the binomial expansion,

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \ldots \tag{1.5}$$

The Fourier series for a periodic function $g(x)$, with period L, is

$$g(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{2\pi n x}{L}\right) + b_n \sin\left(\frac{2\pi n x}{L}\right)\right) \tag{1.6}$$

where

$$a_n = \frac{2}{L}\int_{x_0}^{x_0+L} g(x)\cos\left(\frac{2\pi n x}{L}\right) \tag{1.7}$$

and

$$b_n = \frac{2}{L}\int_{x_0}^{x_0+L} g(x)\sin\left(\frac{2\pi n x}{L}\right). \tag{1.8}$$

Chapter 2

Vector calculus

It is hopefully not surprising that scientific computing uses both vectors (velocity, relative positions, etc.) and calculus (rates of change etc.), and therefore vector calculus is regularly encountered. An example, based particularly in continuum modelling, is the compressible Euler equations,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{2.1}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I}p) = 0 \tag{2.2}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0 \tag{2.3}$$

We will not go into these equations in detail here, but a high-level summary is that these three equations are conservation of mass, momentum and energy respectively.

However, if we consider the first of these equations, equation (2.1), the conservation of mass, what we are saying with this equation is that the density (mass per unit volume) of a material changes over time according to the gradient of the momentum. Physically, this should seem reasonable, since increasing momentum either involves increasing mass (directly adding more mass) or increasing velocity (adding more mass per unit time), and the gradient of the momentum can then inform you how quickly this happens. In the continuum courses, we will see that this is a limiting case of the mass within a volume being updated by the amount of mass which enters the volume minus the amount of mass which leaves.

Here, the key thing to identify is that we have immediately introduced the operator $\nabla\cdot$, the divergence, or div, operator, and this means that if we need to rearrange our equations at all, we will almost certainly need vector calculus.

2.0.1 Vector equations

Before we go into details about the vector calculus required for scientific computing, we first consider vector equations. These might be familiar in other contexts, for example, simultaneous equations, such as

$$\begin{aligned} ax + by &= c \\ dx + ey &= f \end{aligned} \tag{2.4}$$

can be written in vector form as

$$\begin{pmatrix} a & b \\ d & e \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c \\ f \end{pmatrix}. \tag{2.5}$$

We shall revisit this form of vector equations when we cover specific matrix topics later.

For scientific computing, we also could form vector equations when we have **coupled systems of equations**, i.e. a system of equations where a change in one variable would cause a change in one or more other variables. For example, the compressible Euler equations (2.1)–(2.3) have a velocity term, $\mathbf{v}$ after the divergence operator in each equation. This means any change in velocity with respect to time in equation (2.2) will directly impact the other variables in equations (2.1) and (2.3). Because of this coupled nature, it is common to find these equations written in "vector form",

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho \\ \rho \mathbf{v} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{v} \\ \rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p \\ (E + p)\, \mathbf{v} \end{pmatrix} = 0. \tag{2.6}$$

This notation is not necessarily what you might typically encounter, most notably, the time derivative has introduced a vector within a vector. You might then also notice that the divergence term operates on what now appears to be a matrix, not a vector. This second point is actually a reasonable mathematical operation, and we shall return to that shortly.

The form of the equations in equation (2.6) is primarily one of convenience; it takes up less space than expressing the three PDEs in equations (2.1)–(2.3). In particular, though, it does so whilst being coordinate independent, so would not change if we had Cartesian coordinates,

$$\mathbf{v} = v_x \mathbf{e}_i + v_y \mathbf{e}_j + v_z \mathbf{e}_k \tag{2.7}$$

or spherical coordinates

$$\mathbf{v} = v_r \mathbf{e}_r + v_\theta \mathbf{e}_\theta + v_\phi \mathbf{e}_\phi. \tag{2.8}$$

We could attempt to write this more generally, for example, using

$$\begin{pmatrix} \rho \\ \rho\mathbf{v} \\ E \end{pmatrix} = \begin{pmatrix} \rho \\ \rho v_1 \\ \rho v_2 \\ \rho v_3 \\ E \end{pmatrix} \tag{2.9}$$

but the matrix term would still not look very readable in this form.

2.0.2 Vector operations

We have already talked about the divergence operator being a likely operator you will find in scientific computing, and, perhaps unsurprisingly, so are grad and curl. For the remainder of this section, we shall now assume that we are working in Cartesian coordinates, i.e. everything is given in terms of the x-, y- and z-directions.

Gradient

The gradient, or grad, of a scalar field is the generalisation of the slope of a line. Because the slope of a scalar field depends on the direction you measure it, the gradient itself, of course, a vector. In Cartesian coordinates, it can be defined as

$$\nabla\alpha = \begin{pmatrix} \frac{\partial\alpha}{\partial x} \\ \frac{\partial\alpha}{\partial y} \\ \frac{\partial\alpha}{\partial z} \end{pmatrix} \tag{2.10}$$

The gradient of a scalar field is perpendicular to the field itself, and hence provides the normal vector for the field. In these notes, we shall use nabla, ∇ as the symbol for the gradient, without explicitly showing that it is a vector, but this is not universal. Other options include:

- $\boldsymbol{\nabla}\alpha$, $\vec{\nabla}\alpha$ or $\underline{\nabla}\alpha$ (this last option is more common in hand written notes) - these all emphasise that grad is a vector
- gradα, if you don't want to use a symbol
- $\partial_i\alpha$ - this form is only used when describing equations in index notation (see next section), hence always has a subscript

Divergence

The divergence (sometimes just div) of a vector field can be considered as the rate at which the vector field enters (or leaves) a given volume. Mathematically, this has the definition as the ratio of a closed surface intergral of the field over a volume, V, to V itself, in the limit $V \to 0$,

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \mathrm{d}S \tag{2.11}$$

where S is the closed surface, and $\hat{\mathbf{n}}$ is the normal vector pointing out of the surface.

In Cartesian coordinates, the divergence of a vector field $\mathbf{a}$,

$$\mathbf{a} = a_x \mathbf{e}_x + a_y \mathbf{e}_y + a_z \mathbf{e}_z \tag{2.12}$$

is defined as

$$\nabla \cdot \mathbf{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}. \tag{2.13}$$

An alternative notation for the divergence is

- div $\mathbf{a}$ - arguably, this is more correct, since technically, the divergence is an operator on a vector field, and the dot product requires summation over values. However, for most practical applications, there is no way we get confused about this, and use of the dot product form is probably the most common

Curl

The curl of a vector field is a measure of the circulation of the vector field. This is not directly related to the rotation of the vector field, but rather the rate of rotation (in some older sources, curl is known as rotor). Mathematically, curl has a definition of

$$\nabla \times \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \hat{\mathbf{n}} \times \mathbf{F} \mathrm{d}S. \tag{2.14}$$

In Cartesian coordinates, the curl of a vector field, $\mathbf{a}$ is given by the determinant of a matrix,

$$\nabla \times \mathbf{a} = \begin{vmatrix} \mathbf{e}_i & \mathbf{e}_j & \mathbf{e}_k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix} = \begin{pmatrix} \frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \\ \frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \\ \frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \end{pmatrix}. \tag{2.15}$$

Other notation for curl is

- curl $\mathbf{a}$ (or occasionally rot $\mathbf{a}$) - as with divergence, the use of the cross product is technically incorrect.

Outer product

Equation (2.2) uses the outer product ($\rho\mathbf{v} \otimes \mathbf{v}$). For vectors, this can be considered the multiplication of a column vector by a row vector, i.e.:

$$\mathbf{a} \otimes \mathbf{b} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} (b_x, b_y, b_z) = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix}. \tag{2.16}$$

There is a clear similarity to the dot product (or, more generally, inner product), which is the multiplication of a row vector by a column vector.

From a physical perspective, our usage of the outer product, $\rho\mathbf{v} \otimes \mathbf{v}$, could be considered a measure of how the momentum vector, $\rho\mathbf{v}$, moves. For example, the momentum in the x-direction, ρv_x, within a material with non-zero y-velocity will be translated in the y-direction. This term then allows equation (2.2) to take account of this translation.

The notation for outer product is not universal, and, unfortunately, in this case, sometimes the outer product between $\mathbf{a}$ and $\mathbf{b}$ is written in papers as $\mathbf{ab}$. Whilst it could technically be considered the multiplication of two vectors, and we might assume that the paper has explicitly told us when dot products are used, hence this term would have to be an outer product, this is not ideal notation. This is not the only case where practical applications have the potential for confusing notation.

Divergence of a matrix

We have just identified that the outer product gives a matrix. Therefore, equation (2.2),

$$\frac{\partial \rho\mathbf{v}}{\partial t} + \nabla \cdot (\rho\mathbf{v} \otimes \mathbf{v} + \mathbf{I}p) = 0, \tag{2.17}$$

is taking the divergence of a matrix. Here, the fact that the divergence is an operator, and not a true dot product, becomes important. For example, we could treat the divergence exactly like a dot product, and consider the divergence as a row vector 'multiplying' a matrix. However, this would give a row vector as the result, which would be inconsistent with our definition in equation (2.9) which assumes we are working with column vectors. Therefore there are two other possibilities we could use, and in some applications, these are given different names:

$$\text{div } A = \begin{pmatrix} \frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{xy}}{\partial y} + \frac{\partial A_{xz}}{\partial z} \\ \frac{\partial A_{yx}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{yz}}{\partial z} \\ \frac{\partial A_{zx}}{\partial x} + \frac{\partial A_{zy}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix} \tag{2.18}$$

and

$$\nabla \cdot A = \begin{pmatrix} \frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{yx}}{\partial y} + \frac{\partial A_{zx}}{\partial z} \\ \frac{\partial A_{xy}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{zy}}{\partial z} \\ \frac{\partial A_{xz}}{\partial x} + \frac{\partial A_{yz}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}. \tag{2.19}$$

Whilst this would be a convenient set of notation for these two cases, unfortunately, it is not widely used. For example, in our momentum equation, which we have written in a form standardly used in the literature, the actual divergence of the matrix is given by equation (2.18).

When faced with expressions like this, you are often simply expected to know how terms like this are defined. This is not ideal, and can be an issue with vector and matrix expressions and calculus, one that gets worse as the systems get more complex (for example, elasticity is a '4-dimensional matrix'). When you need to work with this type of equation, new notation is then neeed to make sure you don't end up with mathematical ambiguity, index notation.

2.1 Index notation

Index notation could either be considered a natural extension of arbitrary vector and matrix notation (especially when encountering it for the first time) or a simplification of Ricci calculus (if you consider things very mathematically). We shall use the former description to build it up.

An arbitrary vector, $\mathbf{a}$, can have its components expressed as

$$\mathbf{a} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} \tag{2.20}$$

or

$$\mathbf{a} = a_x\mathbf{e}_x + a_y\mathbf{e}_y + a_z\mathbf{e}_z \tag{2.21}$$

or more generally, if we might not be using Cartesian coordinates,

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3. \tag{2.22}$$

We could then talk about the ith entry of $\mathbf{a}$, which could be written as a_i.

For an arbitrary matrix, A, we could write this matrix as

$$A = \begin{pmatrix} a_{xx} & a_{xy} & a_{xz} \\ a_{yx} & a_{yy} & a_{yz} \\ a_{zx} & a_{zy} & a_{zz} \end{pmatrix} \tag{2.23}$$

or, again, more generally, as

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \tag{2.24}$$

If we wanted to define A as a symmetric matrix, one way to write this would be to state that

$$A = A^T \tag{2.25}$$

(the matrix is equal to its transpose), but we could do this from a component-by-component way,

$$a_{12} = a_{21}, \quad a_{13} = a_{31}, \quad \text{and} \quad a_{23} = a_{32}. \tag{2.26}$$

We could then simplify this by saying A is symmetric if

$$a_{ij} = a_{ji} \; \forall \; i, j \tag{2.27}$$

Note that this also includes expressions like $a_{11} = a_{11}$, which are trivially true.

This final example has expressed arbitrary elements of A as a_{ij}, this is **index notation**; rather than expressing an entire vector, matrix or other objects as a whole, consider individual elements.

Index notation for a single object is straightforward, for example,

- Scalars have no indices, e.g. ρ or 2
- Vectors have one unique index, e.g. velocity, v_i
- Matrices have two unique indices, e.g. stress τ_{ij}

Here, the word 'unique' is important, the letter in the index must not be repeated; τ_{ij} is a matrix, but τ_{ii} is not. We shall see shortly what τ_{ii} actually is.

2.1.1 Vector and matrix operations in index notation

Simply knowing what individual objects are is not very useful, we need to know how to work with quantities in index notation. The first term to introduce here is **rank** - the number of unique indices a quantity has. A scalar is rank 0, a vector is rank 1, and a matrix rank 2 (not to be confused with rank being used in terms of number of rows and columns of a matrix).

Unsurprisingly, addition and subtraction of objects requires them to be the same rank (effectively, saying a vector can be added to a vector, but not to a matrix or scalar, for example). So

$$\mathbf{c} = \mathbf{a} + \mathbf{b} \tag{2.28}$$

becomes

$$c_i = a_i + b_i \tag{2.29}$$

and

$$C = A + B \tag{2.30}$$

becomes

$$C_{ij} = A_{ij} + B_{ij}. \tag{2.31}$$

Clearly, at this point, standard rules of matrix addition must apply; A, B and C must all be the same size. The indices do hint at this, because if these are all $M \times N$ matrices, then we have

$$1 \leq i \leq M \quad \text{and} \quad 1 \leq j \leq N. \tag{2.32}$$

This means the expression

$$D_{ij} = A_{ij} + A_{ji} \tag{2.33}$$

only makes sense if $M = N$, i.e. if i and j exist over the same ranges. If this is not the case, then the order in which the indices appear is important. In practice, a large majority of applications do have square matrices, and this ordering is rarely a concern.

In index notation, the convention when working in three-dimensional, non-relativistic coordinates is to start with index i, and generally, work alphabetically from there. This is not always universal, and is not necessary either. More advanced applications, such as relativity, use other symbols, particularly Greek alphabet, for spacetime indices, and may use latin letters (starting with a) for reduced terms.

Free and repeated indices

We now return to the question of what τ_{ii} is. It is not simply the index form of a diagonal entry of τ, which is not something that regularly comes up in manipulation of equations. The number of times an index appears in an term determines the index type:

- **Free indices** - these appear once. The number of free indices an expression has determines its rank, and what type of object it is

- **Repeated indices** (appearing twice) - these are a shorthand notation for a sum over these indices, through the **Einstein summation convention**
- **Repeated indices** (appearing more than twice) - these are the sign of a mistake. If any index appears three or more times, then you are either not dealing with index notation (and the indices have some other definition), or you have made a mistake in your calculation

When talking about repeated indices, this only applies to those left in symbolic form, so, for example, we have:

- v_i: Vector (one free index)
- τ_{ij}: Matrix (two free indices)
- τ_{ii}: Scalar (no free indices)
- τ_{11}: A specific entry of the matrix τ_{ij}, $i = 1$, $j = 1$
- C_{ijkl}: Rank four object (this is the elasticity tensor)
- C_{iikl}: Matrix (two free indices)
- C_{iiil}: ??? This could be a mistake. It could be a very specific matrix built up from elements of C_{ijkl} where the entries are taken when $i = j = k$, but this should not be used like this without very clear additional explanation.

Einstein summation convention

The Einstein summation convention for repeated indices is a form of shorthand notation for any case where a summation occurs as part of expression manipulation. It could also be expressed as

$$\tau_{ii} = \sum_{i=1}^{3} \tau_{ii} = \tau_{11} + \tau_{22} + \tau_{33} \tag{2.34}$$

where we note here that we actually have

$$\tau_{ii} = \mathrm{Tr}(\tau), \tag{2.35}$$

the trace of the matrix.

The summation of terms comes up frequently, which is why the convention exists in the first place; we can avoid consistently writing summation signs.

Because repeated indices do represent a summation, this means that there is freedom to switch the letter (or symbol) being used in the summation. As a result, repeated indices are sometimes referred to as **dummy indices**. Free indices cannot be manipulated in the same way. For example, the expression

$$C_{ij} = A_{ij} + B_{ij} \tag{2.36}$$

is not the same as

$$\hat{C}_{ij} = A_{ji} + B_{ji}. \tag{2.37}$$

In this case, for example, if we consider the specific entry for $i = 1$, $j = 2$, we have

$$C_{12} = A_{12} + B_{12}, \quad \hat{C}_{12} = A_{21} + B_{21} \tag{2.38}$$

and these two expressions would only be equal if $A_{12} = A_{21}$ and so on, i.e. if A (and B) were symmetric. We could, however, completely relabel **every** index, so

$$C_{ij} = A_{ij} + B_{ij} \quad \text{and} \quad C_{ji} = A_{ji} + B_{ji} \tag{2.39}$$

do describe the same thing, it is simply indices of an individual element which cannot be arbitrarily moved around.

However, the expression

$$\alpha = \tau_{ii} + \tau_{kk} = \tau_{ii} + \tau_{ii} \tag{2.40}$$

is valid. If we were to write this with the summation terms,

$$\alpha = \sum_{i=1}^{3} \tau_{ii} + \sum_{k=1}^{3} \tau_{kk} \tag{2.41}$$

then clearly renaming k to i does not change the summation,

$$\sum_{i=1}^{3} \tau_{ii} = \sum_{k=1}^{3} \tau_{kk} = \tau_{11} + \tau_{22} + \tau_{33} \tag{2.42}$$

so we can directly say

$$\tau_{ii} = \tau_{kk} \tag{2.43}$$

or

$$\alpha = 2\tau_{ii}. \tag{2.44}$$

This is often useful since the freedom in choosing your dummy indices means that they often need to be renamed when using them in practice.

Vector multiplication

There are three types of vector-vector multiplication we could consider; the dot (or inner) product, the outer product and the cross product. We discussed the first two of these briefly in section 2.0.2; here we look at these operations in index form.

The dot product can be written as

$$\mathbf{v} \cdot \mathbf{v} = v_1 v_1 + v_2 v_2 + v_3 v_3 = \sum_{i=1}^{n} v_i v_i = v_i v_i. \tag{2.45}$$

In terms of computing the rank of the dot product, the entire thing should be considered a single object, with a repeated index, and no free index, so, as expected, this is a scalar. Also, we note that because both vector and index notations are simply ways of writing a scalar, the equality between these terms is valid here.

The outer product is also equally straightforward. Here, it is quite easy to see that

$$\mathbf{a} \otimes \mathbf{b} \quad \rightarrow \quad a_i b_j \tag{2.46}$$

holds. By providing specific values for i and j, you will find that the matrix in equation (2.16) is reproduced. Again, considering this as a single object, we also have two free indices, thus we have a matrix.

The cross product can be considered to be a multiplication of two vectors to produce a vector. This is a more difficult operator to define in index notation, since we somehow have to convey that, under the standard definition

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix} \tag{2.47}$$

the 1-component, for example, contains only entries from the 2 and 3 entries of the original vectors.

In index form, the cross product is defined as

$$(\mathbf{a} \times \mathbf{b})_i = \epsilon_{ijk} a_j b_k \tag{2.48}$$

where we have introduced a rank-3 oblect, the **Levi-Civita symbol**, ϵ_{ijk}. This is defined as

$$\epsilon_{ijk} = \begin{cases} 1 & \text{even permutation of } i, j, k \\ -1 & \text{odd permutation of } i, j, k \\ 0 & \text{otherwise} \end{cases} \tag{2.49}$$

where an even permution is $(i, j, k) = (1, 2, 3)$, $(i, j, k) = (2, 3, 1)$ or $(i, j, k) = (3, 1, 2)$, and an odd permutation is $(i, j, k) = (3, 2, 1)$, $(i, j, k) = (2, 1, 3)$ or $(i, j, k) = (1, 3, 2)$.

Although the definition of the Levi-Civita symbol is a bit awkward, it is a powerful object. It can also be visualised reasonably well, and we attempt now to give a brief overview as to how it can be obtained. First we note that within equation (2.48), we have the term $a_j b_k$, which, in isolation, is the outer product, i.e.

$$\begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{pmatrix} \tag{2.50}$$

and that each of these terms appears in the cross product definition in equation (2.47), the aim of the Levi-Civita symbol is then to 'pick' the correct terms. If we now consider the $i = 1$ term, for which we know

$$(\mathbf{a} \times \mathbf{b})_1 = a_2b_3 - a_3b_2 \tag{2.51}$$

we note that we could obtain this through combining the matrix

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \tag{2.52}$$

with the outer product above. This is not a straightforward multiplication, though, but a double-dot product, where we consider the summation of each element of these two matrices multiplied together (also known as the Frobenius inner product). In index notation, in general, this double dot product is simply written as

$$A : B = A_{ij}B_{ij}. \tag{2.53}$$

This introduction of a double dot product demonstrates one of the advantages in working in index notation - we don't need additional symbols (or even names) for more complex combinations of these quantities. This means they are more generally accessible since only the understanding of how index notation can be read is needed.

For the cross product, in index notation, the matrix in equation (2.52) would be ϵ_{1jk}, and now it is clear that $\epsilon_{123} = 1$ and $\epsilon_{132} = -1$, as expected from equation (2.49).

Matrix multiplication

In index notation, the multiplication of two matrices, A and B, can be written as

$$AB \to A_{ik}B_{kj}. \tag{2.54}$$

The presence of the repeated index in this expression is unsurprising; we know that matrix multiplication is effectively the addition of multiplying the entries of one row (row i in this case) with the corresponding entries of one column (column j here). We also see that we have two free indices and one repeated index, thus a matrix is obtained, as expected. We can also see this in practice, where, for simplicity we'll consider 2×2 matrices. Since

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \tag{2.55}$$

so

$$AB = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}. \tag{2.56}$$

We could then write this as

$$AB = \begin{pmatrix} \sum_{k=1}^{2} a_{1k}b_{k1} & \sum_{k=1}^{2} a_{1k}b_{k2} \\ \sum_{k=1}^{2} a_{2k}b_{k1} & \sum_{k=1}^{2} a_{2k}b_{k2} \end{pmatrix} \tag{2.57}$$

or

$$AB = \begin{pmatrix} a_{1k}b_{k1} & a_{1k}b_{k2} \\ a_{2k}b_{k1} & a_{2k}b_{k2} \end{pmatrix} \tag{2.58}$$

if we adopt Einstein summation convention for these sums.

If we now look at the ith row and j^{th} column of AB, we clearly have

$$(AB)_{ij} = a_{ik}b_{kj} \tag{2.59}$$

Matrix transpose

A matrix transpose is simply switching the rows and the columns of a matrix. Since the first index in represents the row and the second represents the column of a matrix entry in index notation, this becomes:

$$A \to A_{ij} \;\; \text{means} \;\; A^T \to (A_{ij})^T = A_{ji} \tag{2.60}$$

Note that by itself, if a matrix is given as A_{ji}, this does not necessarily mean it is a transpose (we are technically free to call our indices anything we like), however, using i for rows and j for columns is a good idea when possible.

However, in context of other matrices this should become apparent. For example, if we see the expression

$$C_{ij} = A_{ij} + B_{ji} \tag{2.61}$$

then we can assume that, in matrix notation, this would be

$$C = A + B^T \tag{2.62}$$

2.1.2 Vector calculus operators

Grad

The grad operator is a vector, therefore in index notation, it has a single index, which can be written

$$\nabla \quad \rightarrow \quad \nabla_i \quad \text{or} \quad \partial_i. \tag{2.63}$$

One of the key advantages of index notation is that the derivative quantities (grad, div and curl) can be treated as standard derivative operators. This means that standard differentiation techniques, like product rule and chain rule, can be used. For example, consider

$$\nabla (\mathbf{a} \cdot \mathbf{b}) \rightarrow \nabla_i (a_j b_j) \tag{2.64}$$

which can be written as

$$\nabla_i (a_j b_j) = (\nabla_i a_j) b_j + a_j (\nabla_i b_j) . \tag{2.65}$$

Note that here, the choice of brackets is for clarity, because it is unclear if the quantity $\nabla_i a_j b_j$ should be read as $\nabla_i (a_j b_j)$ or $(\nabla_i a_j) b_j$. This is not a drawback of index notation, similar uncertainties can appear in normal vector notation too. However, if there are no brackets, then the derivative operator should apply only to the first 'object' it encounters, i.e. the first vector component, or the first thing within brackets. Therefore, the quantity $\nabla_i a_j b_j$ normally should be interpreted as $(\nabla_i a_j) b_j$, but it is not always guaranteed if you see this in the literature. Wherever possible, add some brackets to make things clear!

Of course, because, in index notation, everything is treated as a scalar, so we can also eliminate confusion in this expression by writing the product rule expansion as

$$\nabla_i (a_j b_j) = b_j \nabla_i a_j + a_j \nabla_i b_j. \tag{2.66}$$

Note that it will not always be possible to write an expression in a clear form without brackets.

Div

The divergence acts a lot like a dot product, and, unsurprisingly, the index form of divergence also looks like that for a dot product,

$$\nabla \cdot \mathbf{a} = \nabla_i a_i. \tag{2.67}$$

Curl

The curl acts like a cross product, and, again, this is reflected in the index notation definition:

$$\nabla \times \mathbf{a} \quad \rightarrow \epsilon_{ijk} \nabla_j a_k. \tag{2.68}$$

Curl can require a few extra steps when simplifying or understanding equations because of its anti-symmetric properties, and the fact that indices don't always appear in the order you want them to. For example, when you manipulate equations, you could end up with terms which look like

$$\epsilon_{jki} \nabla_j a_k \quad \text{or} \quad \epsilon_{ijk} \nabla_k a_j \quad \text{or} \quad \nabla_{ijk} \nabla_i a_k \tag{2.69}$$

The first of these is easy to deal with, by using the cyclic relationship, since $\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}$, hence

$$\epsilon_{jki} \nabla_j a_k = \epsilon_{ijk} \nabla_j a_k \rightarrow \nabla \times \mathbf{a}. \tag{2.70}$$

The second of these requires the antisymmetric property of the Levi-Civita symbol as well as the fact that repeated indices can be renamed. This gives

$$\epsilon_{ijk} \nabla_k a_j = \epsilon_{ikj} \nabla_j a_k = -\epsilon_{ijk} \nabla_j a_k \rightarrow -\nabla \times \mathbf{a}. \tag{2.71}$$

Finally, for the third possibility, we note that the free index in this case is j, and not i, so we need to rearrange to ensure that this is the first index in our Levi-Civita symbol,

$$\epsilon_{ijk} \nabla_i a_k = \epsilon_{jki} \nabla_i a_k = -\epsilon_{jik} \nabla_i a_k \rightarrow -\nabla \times \mathbf{a} \tag{2.72}$$

Laplacian

Divergence of a matrix

The divergence of a matrix was initially identified as one of the key reasons index notation could be beneficial. This is because the two forms of the divergence can be written unambiguously;

$$\text{div } A = \begin{pmatrix} \frac{\partial A_{11}}{\partial x} + \frac{\partial A_{12}}{\partial y} + \frac{\partial A_{13}}{\partial z} \\ \frac{\partial A_{21}}{\partial x} + \frac{\partial A_{22}}{\partial y} + \frac{\partial A_{23}}{\partial z} \\ \frac{\partial A_{31}}{\partial x} + \frac{\partial A_{32}}{\partial y} + \frac{\partial A_{33}}{\partial z} \end{pmatrix} \quad \rightarrow \quad \nabla_j A_{ij} \tag{2.73}$$

and

$$\nabla \cdot A = \begin{pmatrix} \frac{\partial A_{11}}{\partial x} + \frac{\partial A_{21}}{\partial y} + \frac{\partial A_{31}}{\partial z} \\ \frac{\partial A_{12}}{\partial x} + \frac{\partial A_{22}}{\partial y} + \frac{\partial A_{32}}{\partial z} \\ \frac{\partial A_{13}}{\partial x} + \frac{\partial A_{23}}{\partial y} + \frac{\partial A_{33}}{\partial z} \end{pmatrix} \quad \rightarrow \quad \nabla_i A_{ij}. \tag{2.74}$$

In the first of these expressions, it is the second index of the matrix that is the dummy index with the grad operator, whilst the second expression has the first index of the matrix as the dummy operator.

2.1.3 The Kronecker delta and changing indices

The Kronecker delta is a matrix defined in index notation as

$$\delta_{ij} = \begin{cases} 1, & i = 1 \\ 0, & i \neq j \end{cases} \tag{2.75}$$

from which, if this were not index notation, we can see that this would be called the identity matrix. The key difference is that in the definition of the Kronecker delta, we didn't specify that the matrix was $N \times N$, the limits of i and j could be different, and this definition would still be meaningful. For use in vector calculus, however, we do typically require the Kronecker delta matrix to be $N \times N$, hence the its properties are those of the identity matrix. For example, where we normally have

$$I\mathbf{a} = \mathbf{a} \qquad \text{or} \qquad IA = A, \tag{2.76}$$

in index notation we have

$$\delta_{ij} a_j = a_i \tag{2.77}$$

and

$$\delta_{ij} A_{jk} = A_{ik}. \tag{2.78}$$

Here, we see an application of the Kronecker delta; it can be used to 'change' indices. We will see many examples of where this is useful. A quick example, though, where we actually work backwards is the expression

$$\nabla_j \rho v_i v_j + \nabla_i p. \tag{2.79}$$

In scientific computing, it can be convenient if the ∇ symbols all had the same index. Here, we could write

$$\nabla_j \rho v_i v_j + \nabla_i p = \nabla_j \left(\rho v_i v_j + \delta_{ij} p \right) \tag{2.80}$$

where we have said that

$$\nabla_i p = \delta_{ij} \nabla_j p = \nabla_j (\delta_{ij} p). \tag{2.81}$$

In this last step, we have used the fact that the entries of the Kronecker delta are always constant, so it can be brought inside the derivative operator.

Other ways the Kronecker delta may be used include

$$\delta_{ij}A_{ij} = A_{ii} \tag{2.82}$$

where both indices are repeated. And more than one index can be changed with multiple Kronecker deltas,

$$\delta_{im}\delta_{jn}B_{mnkl} = B_{ijkl}. \tag{2.83}$$

For this type of simplification, if in doubt, do things one at a time, e.g.

$$\delta_{im}\delta_{jn}B_{mnkl} = \delta_{im}B_{mjkl} = B_{ijkl} \tag{2.84}$$

or

$$\delta_{im}\delta_{jn}B_{mnkl} = \delta_{jn}B_{inkl} = B_{ijkl}. \tag{2.85}$$

2.1.4 Vector calculus identities

There are a lot of standard vector calculus identities and relations, many of which can be proved purely through vector operations (which may require memorising other vector identities). Index notation often offers a simpler approach - memorising relationships is not as necessary, and complex vector constructions can be avoided. We demonstrate this through some examples.

$\nabla \cdot (\mathbf{a} \times \mathbf{b})$

We first write the desired expression in index notation:

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) \to \nabla_i \left(\epsilon_{ijk}a_j b_k\right) \tag{2.86}$$

The advantage of index notation is that at this stage, we can treat everything inside the bracket as a scalar, hence we can use the product rule. Now, because each entry of the Levi-Civita symbol is a constant, it can be brought outside the grad operator, and then we can apply product rule to the remaining components.

$$\begin{aligned} \nabla_i \left(\epsilon_{ijk}a_j b_k\right) &= \epsilon_{ijk}\nabla_i \left(a_j b_k\right) && (2.87)\\ &= \epsilon_{ijk}b_k\nabla_i a_j + \epsilon_{ijk}a_j\nabla_i b_k && (2.88) \end{aligned}$$

We now have to group the terms into those that are part of a curl (Levi-Civita symbols and grad of a vector) and those that don't (the non-derivative vectors).

$$\epsilon_{ijk}b_k\nabla_i a_j + \epsilon_{ijk}a_j\nabla_i b_k = \left(\epsilon_{ijk}\nabla_i a_j\right) b_k + a_j \left(\epsilon_{ijk}\nabla_i b_k\right). \tag{2.89}$$

Now the indices of the Levi-Civita symbols can be rearranged to give quantities which are clearly curls.

$$\left(\epsilon_{ijk}\nabla_i a_j\right) b_k + a_j \left(\epsilon_{ijk}\nabla_i b_k\right) = \left(\epsilon_{kij}\nabla_i a_j\right) b_k + a_j \left(-\epsilon_{jik}\nabla_i b_k\right). \tag{2.90}$$

Here, we used a cyclical permutation of the first Levi-Civita symbol, and anti-symmetry about the first two indices of the second. Although we could now jump straight to the vector form of this equation, for some expressions with a lot of indices, this can be tricky. Instead, it can be done in two parts, by first identifying the standard vector notation for some of the expression (in this case the terms in brackets), which assists with then simplifying the entire system. In this case, we can identify the curls first, which we know are vectors, so we have

$$\left(\epsilon_{kij}\nabla_i a_j\right) b_k + a_j \left(-\epsilon_{jik}\nabla_i b_k\right) = \left(\nabla \times \mathbf{a}\right)_k b_k - a_j \left(\nabla \times \mathbf{b}\right)_j. \tag{2.91}$$

Note that because curl is a vector quantity, we end up with a single index for each of our bracketed quantities. Finally, we can see that we have a single, repeated pair of indices for each term, i.e. we have two dot products

$$\left(\nabla \times \mathbf{a}\right)_k b_k - a_j \left(\nabla \times \mathbf{b}\right)_j \rightarrow \left(\nabla \times \mathbf{a}\right) \cdot \mathbf{b} - \mathbf{a} \cdot \left(\nabla \times \mathbf{b}\right). \tag{2.92}$$

$\nabla \times \mathbf{E}$

As this is a physically motivated result, we also need to use Ohm's law for a perfect conductor;

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B}. \tag{2.93}$$

We then have

$$\begin{aligned}
\nabla \times \mathbf{E} &= \nabla \times \left(-\mathbf{v} \times \mathbf{B}\right) && (2.94)\\
&= \epsilon_{ijk}\nabla_j \left(-\mathbf{v} \times \mathbf{B}\right)_k && (2.95)\\
&= \epsilon_{ijk}\nabla_j \left(-\epsilon_{klm} v_l B_m\right) && (2.96)\\
&= -\epsilon_{ijk}\epsilon_{klm}\nabla_j \left(v_l B_m\right) && (2.97)\\
&= -\epsilon_{kij}\epsilon_{klm}\nabla_j \left(v_l B_m\right) && (2.98)\\
&= -\left(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}\right)\nabla_j \left(v_l B_m\right) && (2.99)\\
&= \nabla_j \left(v_j B_i\right) - \nabla_j \left(v_i B_j\right) && (2.100)\\
&= \nabla_j \left(B_i v_j - v_i B_j\right) && (2.101)\\
&\rightarrow \nabla \cdot \left(\mathbf{B} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{B}\right). && (2.102)
\end{aligned}$$

Here, key steps to note are; initially, in order to get the indices correct, we go from vector form to index notation form in two steps, one for each cross

product. In the fifth step, we are simply reordering the indices on the second Levi-Civita symbol such that it is clear how to construct the double-delta form. The penultimate step is important for getting our indices in the correct positions for converting back to vector form; because we define our divergence operators in this course as

$$\nabla \cdot A \to \nabla_j A_{ij}, \tag{2.103}$$

we need to ensure that the order of entries for each of the terms within the brackets has first the i index, and then the j index. Because we treat each component as a scalar variable, this is trivial, we can just switch the order of v_j and B_i as if they were multiplied.

2.2 Tensors

When using index notation, we have started to use objects which could be considered higher-dimensional generalisations of vectors and matrices, such as the Levi-Civita symbol. When encountering such quantities, the word 'tensors' often appears too. Tensors are indeed a term suitable for the generalisation of vectors, but not necessarily matrices.

2.2.1 Basis vectors and coordinate transformations

When we want to write down a vector, we need to do so with respect to a coordinate system, for example, in 3D,

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3. \tag{2.104}$$

The vector magnitude and direction, though, are independent of this coordinate system (otherwise something very unphysical would be happening), which means we can also write the vector in another system of our choice,

$$\mathbf{a} = a'_1\mathbf{e}'_1 + a'_2\mathbf{e}'_2 + a'_3\mathbf{e}'_3. \tag{2.105}$$

Since these both describe the same vector, we should be able to change between coordinates, which we shall refer to as the original coordinates and the primed coordinates respectively. You might see these two vectors referred to as $\mathbf{a}$ and $\mathbf{a}'$ respectively, such that the coordinate system in which the vector is defined is clear.

In order to define this change in vectors, we need to consider what our coordinates are. To do this, we start from a position vector, e.g.

$$\mathbf{r} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 \tag{2.106}$$

and

$$\mathbf{r} = x'_1\mathbf{e}'_1 + x'_2\mathbf{e}'_2 + x'_3\mathbf{e}'_3 \tag{2.107}$$

for our two coordinate systems. Now, we shall consider a small change, $\mathrm{d}r$ for equation (2.106), this is given by

$$\mathrm{d}r = \mathrm{d}x_1\mathbf{e}_1 + \mathrm{d}x_2\mathbf{e}_2 + \mathrm{d}x_3\mathbf{e}_3. \tag{2.108}$$

We could then consider a specific value for $\mathrm{d}r$, which we shall call $\mathrm{d}r_1$, such that $\mathrm{d}x_2 = \mathrm{d}x_3 = 0$, i.e.

$$\mathrm{d}r_1 = \mathrm{d}x_1\mathbf{e}_1. \tag{2.109}$$

The quantity $\mathrm{d}r_1$ can still be written in terms of the primed coordinates, i.e. there exists

$$\mathrm{d}r_1 = \mathrm{d}x'_1\mathbf{e}'_1 + \mathrm{d}x'_2\mathbf{e}'_2 + \mathrm{d}x'_3\mathbf{e}'_3. \tag{2.110}$$

However, $\mathrm{d}r_1$ is the same in both cases, i.e. we have

$$\mathrm{d}x_1\mathbf{e}_1 = \mathrm{d}x'_1\mathbf{e}'_1 + \mathrm{d}x'_2\mathbf{e}'_2 + \mathrm{d}x'_3\mathbf{e}'_3. \tag{2.111}$$

We can convert this to derivatives by 'dividing' by $\mathrm{d}x_1$, giving

$$\mathbf{e}_1 = \frac{\partial x'_1}{\partial x_1}\mathbf{e}'_1 + \frac{\partial x'_2}{\partial x_1}\mathbf{e}'_2 + \frac{\partial x'_3}{\partial x_1}\mathbf{e}'_3. \tag{2.112}$$

More generally, we can write

$$\mathbf{e}_i = \frac{\partial x'_1}{\partial x_i}\mathbf{e}'_1 + \frac{\partial x'_2}{\partial x_i}\mathbf{e}'_2 + \frac{\partial x'_3}{\partial x_i}\mathbf{e}'_3. \tag{2.113}$$

or, in index notation

$$\mathbf{e}_i = \frac{\partial x'_j}{\partial x_i}\mathbf{e}'_j \tag{2.114}$$

and equivalently

$$\mathbf{e}'_i = \frac{\partial x_j}{\partial x'_i}\mathbf{e}_j. \tag{2.115}$$

It is also useful to be able to convert a vector $\mathbf{a}$ between coordinate systems. Here, a similar approach is taken, this time consider a specific vector, $\mathbf{a}_1$, which can be written as

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3. \tag{2.116}$$

Again, this vector still exists in the original coordinates,

$$\mathbf{a} = a'_1\mathbf{e}'_1 + a'_2\mathbf{e}'_2 + a'_3\mathbf{e}'_3 \tag{2.117}$$

and these two vectors do describe the same thing. Equation (2.114) allowed us to write the basis vectors $\mathbf{e}_i$ in the primed coordinates, therefore equation (2.116) can be written as

$$\begin{aligned}\mathbf{a} = a_1 &\left(\frac{\partial x'_1}{\partial x_1}\mathbf{e}'_1 + \frac{\partial x'_2}{\partial x_1}\mathbf{e}'_2 + \frac{\partial x'_3}{\partial x_1}\mathbf{e}'_3\right) \\ + a_2 &\left(\frac{\partial x'_1}{\partial x_2}\mathbf{e}'_1 + \frac{\partial x'_2}{\partial x_2}\mathbf{e}'_2 + \frac{\partial x'_3}{\partial x_2}\mathbf{e}'_3\right) + a_3\left(\frac{\partial x'_1}{\partial x_3}\mathbf{e}'_1 + \frac{\partial x'_2}{\partial x_3}\mathbf{e}'_2 + \frac{\partial x'_3}{\partial x_3}\mathbf{e}'_3\right)\end{aligned} \tag{2.118}$$

or, if we group this in terms of basis vectors,

$$\begin{aligned}\mathbf{a} = &\left(a_1\frac{\partial x'_1}{\partial x_1} + a_2\frac{\partial x'_1}{\partial x_2} + a_3\frac{\partial x'_1}{\partial x_3}\right)\mathbf{e}'_1 \\ &+ \left(a_1\frac{\partial x'_2}{\partial x_1} + a_2\frac{\partial x'_2}{\partial x_2} + a_3\frac{\partial x'_2}{\partial x_3}\right)\mathbf{e}'_2 + \left(a_1\frac{\partial x'_3}{\partial x_1} + a_2\frac{\partial x'_3}{\partial x_2} + a_3\frac{\partial x'_3}{\partial x_3}\right)\mathbf{e}'_3.\end{aligned} \tag{2.119}$$

Matching these terms with those in equation (2.117) means we can identify

$$a'_1 = a_1\frac{\partial x'_1}{\partial x_1} + a_2\frac{\partial x'_1}{\partial x_2} + a_3\frac{\partial x'_1}{\partial x_3} \tag{2.120}$$

and similarly for a'_2 and a'_3, or, more generally,

$$a'_i = \frac{\partial x'_i}{\partial x_j}a_j \tag{2.121}$$

or, equivalently,

$$a_i = \frac{\partial x_i}{\partial x'_j}a'_j. \tag{2.122}$$

This gives us the ability to transform any vector between two coordinate systems.

2.2.2 Generalisation to tensors

The transformation rules in equations (2.121) and (2.122) apply to all vectors. At no point did we need to specify any properties about the vector $\mathbf{a}$ that

we used in the previous section. Tensors are generalisations of this; they are objects which transform according to a natural extension of these rules, e.g.

$$s_{ij} = R_{ik}R_{jl}s'_{kl}, \tag{2.123}$$

$$t_{ijkl} = R_{im}R_{jn}R_{kp}R_{lq}t'_{mnpq} \tag{2.124}$$

or, in general,

$$p_{i_1,i_2,\cdots,i_n} = R_{i_1j_1}R_{i_2j_2}\ldots R_{i_nj_n}t'_{j_1j_2\cdots j_n}. \tag{2.125}$$

The importance of this definition lies in the fact that these objects can be described in any coordinate system, and we know that they refer to the same thing. This is important if you want to know, for example, stress within a ball. You may start from spherical coordinates, since these are natural for the ball, but then decide you want to model the stress caused by the impact with a second ball. At this point, spherical coodinates could get cumbersome; you need to switch coordinates (perhaps to Cartesians) but not lose any information about the current stress within this ball.

We have stated that all vectors are tensors, but this statement does not generalise; not all matrices are tensors, for example. We shall look at a few of these soon. Rather than creating specific names for any object s_{ij} which obeys equation (2.123), we instead refer to tensors by their rank; s_{ij} is a rank-2 tensor because it has two indices. We can safely say that all rank-2 tensors are matrices, even though the converse does not hold.

A key feature of tensors is that because they describe an object, regardless of coordinate system, they have some fixed ways to describe the object. For example, in any coordinate system, the magnitude of a vector does not change. Such quantities are known as **invariant quantities** or **invariants**. For a rank-2 tensor, invariant quantities are given by finding the eigenvalues of the tensor,

$$\begin{vmatrix} s_{11}-\lambda & s_{12} & s_{13} \\ s_{21} & s_{22}-\lambda & s_{23} \\ s_{31} & s_{32} & s_{33}-\lambda \end{vmatrix} = 0. \tag{2.126}$$

We shall look at determinants and eigenvalues more closely in section 4.1, but for now, the important thing to note is that the determinant can be written as

$$(a_1-\lambda)(a_2-\lambda)(a_3-\lambda) = 0 \tag{2.127}$$

where a_1, a_2 and a_3 may be complex numbers. Alternatively, we can write this as

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0. \tag{2.128}$$

One of the properties of matrices under coordinate transformations, that we shall consider, is that if s_ij obeys equation (2.123), then under a coordinate transformation, the quantity

$$\begin{vmatrix} s'_{11} - \lambda & s'_{12} & s'_{13} \\ s'_{21} & s'_{22} - \lambda & s'_{23} \\ s'_{31} & s'_{32} & s'_{33} - \lambda \end{vmatrix} = 0 \tag{2.129}$$

would still yield

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0. \tag{2.130}$$

In other words, I_1, I_2 and I_3 are invariant of the coordinate system - these are the invariant quantities. We can actually compute these explicitly for this 3×3 case,

$$I_1 = s_{ii} = s_{11} + s_{22} + s_{33} = \mathrm{tr}(S), \tag{2.131}$$

$$I_2 = \frac{1}{2}\left(s_{ii}s_{jj} - s_{ij}s_{ji}\right) = \frac{1}{2}\left[\left(\mathrm{tr}(S)\right)^2 - \mathrm{tr}(S^2)\right], \tag{2.132}$$

and

$$I_3 = \det\left(S\right). \tag{2.133}$$

Again, we shall look at the trace, $\mathrm{tr}(S)$ and determinant, $\det(S)$ in more details in the matrices section.

Some of these invariant quantities are very familiar, for example speed being the magnitude of velocity. Similarly, material stress, which is a tensor quantity often defined as σ_{ij}, and describes how much a material is pushed (or pulled) in a given direction. However, because this could happen through compression and expansion, or shearing, a vector quantity is not sufficient and a rank-2 tensor is needed. Related to stress, however, is pressure (both linguistically and mechanically). This is a scalar quantity which gives information about how much stress a material is under, but without directional information (analagous to speed and velocity). It is probably unsurprising that pressure is related to an invariant quantity, and the definition is

$$p = \frac{1}{3}I_1 = \frac{1}{3}\mathrm{tr}(\sigma) = \sigma_{ii} \tag{2.134}$$

2.2.3 Matrices which aren't tensors

It is not very common to encounter matrices which are not tensors, at least not in a context where you care about coordinate transformations. In most cases, if a physical quantity is defined by a matrix, it is probably a tensor, or it wouldn't have much actual physical meaning. The two key exceptions, though, are:

- The coordinate transformation Jacobian matrix used in

$$\mathbf{e}_i = \frac{\partial x'_j}{\partial x_i}\mathbf{e}'_j \tag{2.135}$$

- The deformation gradient, F_{ij}, which transforms an infinitesimal particle from a reference configuration, d$\mathbf{X}$ to a new position, d$\mathbf{x}$ by deformation alone (no rotational or translational effects are considered)

$$F_{ij} = \frac{\mathrm{d}x_i}{\mathrm{d}X_i} \tag{2.136}$$

These are still important quantities, and it may be that you do need to know them in a new coordinate system. And this can be done, we can show that in order to transform either of these matrices to a new coordinate system, then we actually only need a single transformation; they transform as vectors, e.g.

$$F'_{ij} = R_{ik}F_{kj}. \tag{2.137}$$

This may not be surprising - these two quantities contain components from two coordinate systems, e.g. the original and the primed coordinates. We then only need to transform one of these indices into a new coordinate system. Such quantities do also have a name, **two-point tensors**, or **double vectors**. However, these names are not used universally, many publications will refer to the 'deformation gradient tensor'.

Chapter 3

Multivariable calculus

3.1 Multivariable calculus

Throughout scientific computing, we will often need to compute the relationship between different physical variables, and these will frequently involve derivatives and multivariable calculus.

3.1.1 Independent and dependent variables

When we need to take partial derivatives, one key thing to be aware of is what the **independent variables** are. A typical closed system of equations will have N independent variables for N independent equations. An independent variable is one that cannot be computed from other variables, whilst an independent equation is one that cannot be expressed as a combination of the other equations being considered.

If we did not have N independent variables, we would have an **overdetermined** system - we might not be able to guarantee all equations will be satisfied simultaneously. On the other hand, if we had if we have more independent variables that equations, we have an **underdetermined** system - we don't have enough information to solve for all our variables.

A typical example is the continuum variables of mass (or density), momentum and energy. These three quantities are well known to be 'conserved', i.e. there is an equation for each one describing how it changes. In this case, we have three partial differential equations,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{3.1}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I}p) = 0 \tag{3.2}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0 \tag{3.3}$$

where ρ is density, $\rho\mathbf{v}$ is momentum (a vector quantity) and E is total energy. We shall assume that we are working in three dimensions, i.e. $\rho\mathbf{v} = \rho(v_1, v_2, v_3)$ for some coordinate system. In these equations, density momentum and total energy are independent variables if:

1. We cannot define any of the independent variables purely in terms of the others
2. All other variables, the **dependent variables**, which here are velocity, $\mathbf{v}$, and pressure, p, must be able to be defined as functions of the independent variables

Typically, in continuum modelling, momentum is not given it's own symbol (though sometimes p is used), so care must sometimes be taken to note that $\rho\mathbf{v}$ and ρ are actually independent variables. However, this does mean that it is straightforward to see how velocity is defined from the independent variables,

$$\mathbf{v} = \frac{\rho\mathbf{v}}{\rho}. \tag{3.4}$$

The definition of pressure is not as straightforward - we shall discuss this further in section 3.2. For now, though, we note that a typical definition of pressure is as a function of two variables,

$$p = p(\rho, \varepsilon) \tag{3.5}$$

where another dependent variable, specific internal energy, ε, has been introduced. At this point, though, there is one further relationship,

$$E = \rho\varepsilon + \frac{1}{2}\rho v_i v_i, \tag{3.6}$$

which relates ε to our independent variables. Physically, this states that total energy is the sum of internal energy and kinetic energy.

We can see in this example that ρ, $\rho\mathbf{v}$ and E are suitable independent variables - we can compute $\mathbf{v}$ through equation (3.4), ε through equation (3.6) and then p through equation (3.5).

It would be just as valid to choose other combinations of independent variables, such as ρ, ε and $\mathbf{v}$ or p, ε and $\rho\mathbf{v}$; The choice really depends on application. However, ρ, p and ε would not be a valid choice; firstly equation (3.5) means one of our independent variable is a function of the others, and secondly, we have no information as to how to compute $\rho\mathbf{v}$, or just $\mathbf{v}$.

3.1.2 Partial derivatives

If we now consider how we might take partial derivatives of these quantities; the first thing to note is that in order to take a derivative with respect to an independent variable, all quantities should be expressed entirely as a function of the independent variables. The second thing to note is that the derivative of one independent variable with respect to another one is always zero. Therefore, using the previous example of density, momentum and total energy being independent variables, we have,

$$\frac{\partial E}{\partial(\rho v_1)} = 0 \qquad \text{and} \qquad \frac{\partial(\rho v_1)}{\partial \rho} = 0. \tag{3.7}$$

It can be useful to ensure that independent variables are always placed in brackets to try to avoid accidentally assuming that, for example, ρv_1 is a function of ρ.

We can, of course, get non-zero derivatives of our dependent variables. For example,

$$\left.\frac{\partial v}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} = \left.\frac{\partial}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} \left(\frac{(\rho v_1)}{\rho}\right) = \frac{1}{\rho} \tag{3.8}$$

where the central step in this equations ensures that we write the velocity as a function of **only** the independent variables. For completeness, we ensure that we denote all variables which remain constant, although it is only ρ that actually appears here.

As another example, we have

$$\begin{aligned}\left.\frac{\partial \varepsilon}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} &= \left.\frac{\partial}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} \left(\frac{E}{\rho} - \frac{1}{2} v_i v_i\right) \\ &= \left.\frac{\partial}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} \left(\frac{E}{\rho} - \frac{1}{2}\frac{(\rho v_i)}{\rho}\frac{(\rho v_i)}{\rho}\right)\end{aligned} \tag{3.9}$$

where in the first step, we simply rearranged equation (3.6), before expressing it in terms of the independent variables. Now, we expand out the sum from the repeated indices to get

$$\begin{aligned}&\left.\frac{\partial \varepsilon}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} \\ &\quad = \left.\frac{\partial}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} \left(\frac{E}{\rho} - \frac{1}{2}\frac{(\rho v_1)^2 + (\rho v_2)^2 + (\rho v_3)^2}{\rho^2}\right)\end{aligned} \tag{3.10}$$

and then note that all derivatives of independent variables are zero, so the only non-vanishing term is

$$\left.\frac{\partial\varepsilon}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} = \left.\frac{\partial}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E}\left(-\frac{1}{2}\frac{(\rho v_1)^2}{\rho^2}\right) = -\frac{\rho v_1}{\rho^2}. \quad (3.11)$$

We may well want to stop writing everything in terms of the independent variables at this point, it is now fine to state

$$\left.\frac{\partial\varepsilon}{\partial(\rho v_1)}\right|_{\rho,(\rho v_2),(\rho v_3),E} = -\frac{v_1}{\rho}. \quad (3.12)$$

It is also worth noting that there is no single choice of independent variables - we stated mass (density), momentum and energy because these were the variables we know are conserved, but equally, we could have density, velocity and pressure, since if we know these variables, and the function $p = p(\rho, \varepsilon)$, we can fully define the other quantities. Then, for this choice of independent variables, we now have

$$\left.\frac{\partial(\rho v_1)}{\partial\rho}\right|_{v_1,v_2,v_3,p} = v_1. \quad (3.13)$$

3.1.3 Switching between multiple variables

So far, we have considered partial derivatives where we can directly analyse results. In general, there are a large number of variables which can be used to describe the properties of a physical system, and you might not have a definition for all of them. For example, a continuum system could be quantified through any of the following quantities (and this is not an exhaustive list):

- Pressure (p)
- Temperature (T)
- Volume (V)
- Density (ρ)
- Entropy (S)
- Energy (E)
- Internal energy (e)

- Specific internal energy (ε)
- Momentum ($\rho\mathbf{v}$)
- Velocity ($\mathbf{v}$)
- Enthalpy (h)
- Heat (Q)

Note that a symbol has been given for each quantity, and these will be used consistently in these notes, but not necessarily in all literature.

Often, these variables can be used to express the same physical relationship in ways that may not look identical. For example, the ideal gas law can be expressed as

$$pV = nRT \tag{3.14}$$

where n is the number of moles of the material (assumed constant) and R is the ideal gas constant, or as

$$p = (\gamma - 1)\rho\varepsilon \tag{3.15}$$

where γ is the ratio of specific heats of the material. More generally, if we do not know the relationships between these variables, then we could write

$$p = p(V, T) \tag{3.16}$$

or

$$p = p(\rho, \varepsilon). \tag{3.17}$$

The speed of sound

A more concrete example of why we need to worry about partial derivatives of these variables is the speed of sound. This is an important quantity to be able to calculate; it is fundamental in identifying the motion of acoustic waves. And it is defined by

$$c_s^2 = \left.\frac{\partial p}{\partial \rho}\right|_s \tag{3.18}$$

where s is entropy per unit volume.

It would be straightforward to compute the speed of sound if we know

$$p = p(\rho, s), \tag{3.19}$$

but, unfortunately, this is rarely the case (entropy is not generally a quantity that can be measured, so many expressions are not written in terms of this variable).

We are far more likely to know one of the forms in either equation (3.121) or equation (3.122), i.e. $p = p(V, T)$ or $p = p(\rho, \varepsilon)$. However, for the speed of sound, we have picked independent variables as ρ and s. We note that for the example considered in section 3.1.1, we would also need another variable which relates to velocity or momentum, but for this example, it will not be used, so we drop it from our list of independent variables. This means that all other variables can be written in terms of ρ and s, i.e. we have

$$p = p(V(\rho, s), T(\rho, s)) \tag{3.20}$$

or

$$p = p(\rho, \varepsilon(\rho, s)). \tag{3.21}$$

Technically, here, we actually have

$$p = p(V(\rho), T(\rho, s)) \tag{3.22}$$

since there is a well-defined relationship between V and ρ,

$$V = \frac{M}{\rho} \tag{3.23}$$

where M is the total mass of the material.

We shall revisit this example throughout this section, as we consider how we can compute this partial derivative if we do not know the function $p = p(\rho, s)$, starting with the multivariable chain rule.

3.1.4 Multivariable Taylor series

We will shortly see that the multivariable chain rule requires multivariable Taylor series, so we shall briefly introduce these. In order to derive a Taylor series for multiple variables, we start with the single variable Taylor series

$$f(x + h) = f(x) + h \frac{\partial f}{\partial x} + h^2 \frac{1}{2} \frac{\partial^2 f}{\partial x^2} + h^3 \frac{1}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \tag{3.24}$$

and note that this is also true if we have a function of multiple variables, but only consider a Taylor expansion in the x-direction, treating other variables as constant. For $f(x, y)$, we then have

$$f(x+h, y) = f(x, y) + h \left.\frac{\partial f}{\partial x}\right|_y (x, y) + h^2 \frac{1}{2} \left.\frac{\partial^2 f}{\partial x^2}\right|_y (x, y) + h^3 \frac{1}{3!} \left.\frac{\partial^3 f}{\partial x^3}\right|_y (x, y) + \dots \tag{3.25}$$

and, if we instead considered an equivalent expansion, but in the y-direction, then we have

$$f(x,y+k) = f(x,y)+k\left.\frac{\partial f}{\partial y}\right|_x (x,y)+k^2\,\frac{1}{2}\left.\frac{\partial^2 f}{\partial y^2}\right|_x (x,y)+k^3\,\frac{1}{3!}\left.\frac{\partial^3 f}{\partial y^3}\right|_x (x,y)+\ldots \tag{3.26}$$

Another thing we need to note that we could have a function, $F(x)$, defined as

$$F(x) = \frac{\mathrm{d}f}{\mathrm{d}x} \tag{3.27}$$

and $F(x)$ would have a Taylor series

$$\begin{aligned} F(x+h) &= F(x) + h\,\frac{\partial F}{\partial x} + h^2\,\frac{1}{2}\frac{\partial^2 F}{\partial x^2} + h^3\,\frac{1}{3!}\frac{\partial^3 F}{\partial x^3} + \ldots && (3.28)\\ &= \frac{\partial f}{\partial x} + h\,\frac{\partial^2 f}{\partial x^2} + h^2\,\frac{1}{2}\frac{\partial^3 f}{\partial x^3} + h^3\,\frac{1}{3!}\frac{\partial^4 f}{\partial x^4} + \ldots, && (3.29)\end{aligned}$$

in other words, the Taylor series of a derivative works as expected.

We now consider the Taylor series in equation (3.25), but as the expansion of $f(x+h, y+k)$, with the expansion in the x-direction, i.e. this is a direct substitution of $y \to y+k$

$$f(x+h,y+k) = f(x,y+k)+h\left.\frac{\partial f}{\partial x}\right|_y (x,y+k)+h^2\,\frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_y (x,y+k)+h^3\,\frac{1}{3!}\left.\frac{\partial^3 f}{\partial x^3}\right|_y (x,y+k)+\ldots \tag{3.30}$$

Each of the derivatives in this expression, and $f(x, y+k)$ too, can now undergo a Taylor expansion in the y-direction; we've already considered $f(x, y+k)$ in equation (3.26), and identified how Taylor series of derivatives work. This gives

$$\begin{aligned} f(x+h,y+k) = &\left(f(x,y) + k\left.\frac{\partial f}{\partial y}\right|_x (x,y) + k^2\,\frac{1}{2}\left.\frac{\partial^2 f}{\partial y^2}\right|_x (x,y) + \ldots \right)\\ &+ h\left(\left.\frac{\partial f}{\partial x}\right|_y (x,y) + k\,\frac{\partial^2 f}{\partial x \partial y}\,(x,y) + k^2\,\frac{1}{2}\frac{\partial^3 f}{\partial x \partial y^2}\,(x,y) + \ldots \right)\\ &+ \frac{h^2}{2}\left(\left.\frac{\partial^2 f}{\partial x^2}\right|_y (x,y) + k\,\frac{\partial^3 f}{\partial x^2 \partial y}\,(x,y) + k^2\,\frac{1}{2}\frac{\partial^4 f}{\partial x^2 \partial y^2}\,(x,y) + \ldots \right)\end{aligned} \tag{3.31}$$

Now, if we simplify this expression, and only keep terms quadratic in h and

k (i.e. h^2. hk and k^2), we get the two-variable Taylor series,

$$\begin{aligned} f(x+h,y+k) =& f(x,y) + h\left.\frac{\partial f}{\partial x}\right|_y (x,y) + k\left.\frac{\partial f}{\partial y}\right|_x (x,y) \\ &+ \frac{h^2}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_y (x,y) + hk\,\frac{\partial^2 f}{\partial x \partial y}(x,y) + \frac{k^2}{2}\,\frac{1}{2}\left.\frac{\partial^2 f}{\partial y^2}\right|_x (x,y) + \ldots \end{aligned} \tag{3.32}$$

Note that in the two expressions above, we haven't attempted to denote fixed variables in our mixed derivatives, but these can be easily inferred (e.g. a partial derivative with respect to x has y constant). Also note that this approach could now easily generalise to a function of three or more variables.

3.1.5 Multivariable chain rule

If we have a function of two variables, $f(x,y)$, but x and y are themselves functions of two different variables, i.e. $x = x(u,v)$ and $y = y(u,v)$, then we can write

$$f(x,y) = f(x(u,v), y(u,v)). \tag{3.33}$$

If we now wish to take the derivative

$$\left.\frac{\partial f}{\partial u}\right|_v, \tag{3.34}$$

then we do this through the multivariable chain rule,

$$\left.\frac{\partial f}{\partial u}\right|_v = \left.\frac{\partial f}{\partial x}\right|_y \left.\frac{\partial x}{\partial u}\right|_v + \left.\frac{\partial f}{\partial y}\right|_x \left.\frac{\partial y}{\partial u}\right|_v . \tag{3.35}$$

Whilst this is a fairly natural extension of the single-varible chain rule, it is worth investigating where this comes from.

We first consider how do we obtain an expression for an infinitesimal change in x, $\mathrm{d}x$, in terms of infinitesimal changes in u and v, $\mathrm{d}u$ and $\mathrm{d}v$ respectively. If we start with $x(u,v)$, and make these infinitesimal changes in u and v, we get

$$x(u+\mathrm{d}u, v+\mathrm{d}v) \tag{3.36}$$

and from this we can define

$$\mathrm{d}x = x(u+\mathrm{d}u, v+\mathrm{d}v) - x(u,v) \tag{3.37}$$

Consider now the multivariable Taylor series given in equation (3.32), for $x(u + \mathrm{d}u, v + \mathrm{d}v)$, i.e. with $h = \mathrm{d}u$ and $k = \mathrm{d}v$, then, to first order, we have

$$x(u + \mathrm{d}u, v + \mathrm{d}v) = x(u, v) + \mathrm{d}u \left.\frac{\partial x}{\partial u}\right|_v + \mathrm{d}v \left.\frac{\partial x}{\partial v}\right|_u + \dots \tag{3.38}$$

or, using equation (3.37)

$$\mathrm{d}x = \left.\frac{\partial x}{\partial u}\right|_v \mathrm{d}u + \left.\frac{\partial x}{\partial v}\right|_u \mathrm{d}v \tag{3.39}$$

Note here, that in the infinitesimal limit, we are allowed to make this an equality. We then have similiar results for $\mathrm{d}y$ and $\mathrm{d}f$,

$$\mathrm{d}y = \left.\frac{\partial y}{\partial u}\right|_v \mathrm{d}u + \left.\frac{\partial y}{\partial v}\right|_u \mathrm{d}v \tag{3.40}$$

and

$$\mathrm{d}f = \left.\frac{\partial f}{\partial x}\right|_y \mathrm{d}x + \left.\frac{\partial f}{\partial y}\right|_x \mathrm{d}y \tag{3.41}$$

By inserting the expressions for $\mathrm{d}x$ and $\mathrm{d}y$ into the above expression, we can obtain

$$\mathrm{d}f = \left.\frac{\partial f}{\partial x}\right|_y \left(\left.\frac{\partial x}{\partial u}\right|_v \mathrm{d}u + \left.\frac{\partial x}{\partial v}\right|_u \mathrm{d}v \right) + \left.\frac{\partial f}{\partial y}\right|_x \left(\left.\frac{\partial y}{\partial u}\right|_v \mathrm{d}u + \left.\frac{\partial y}{\partial v}\right|_u \mathrm{d}v \right) \tag{3.42}$$

$$= \left(\left.\frac{\partial f}{\partial x}\right|_y \left.\frac{\partial x}{\partial u}\right|_v + \left.\frac{\partial f}{\partial y}\right|_x \left.\frac{\partial y}{\partial u}\right|_v \right) \mathrm{d}u + \left(\left.\frac{\partial f}{\partial x}\right|_y \left.\frac{\partial x}{\partial v}\right|_u + \left.\frac{\partial f}{\partial y}\right|_x \left.\frac{\partial y}{\partial v}\right|_u \right) \mathrm{d}v \tag{3.43}$$

However, we could also say that because both x and y are functions of u and v, then

$$f = f(u, v) \tag{3.44}$$

which means we can also obtain an expression for $\mathrm{d}f$ using the multivariable Taylor series as

$$\mathrm{d}f = \left.\frac{\partial f}{\partial u}\right|_v \mathrm{d}u + \left.\frac{\partial f}{\partial v}\right|_u \mathrm{d}v. \tag{3.45}$$

By comparing the coefficients of $\mathrm{d}u$ and $\mathrm{d}v$ between equations (3.43) and (3.45), we then obtained the desired chain rule expressions,

$$\left.\frac{\partial f}{\partial u}\right|_v = \left.\frac{\partial f}{\partial x}\right|_y \left.\frac{\partial x}{\partial u}\right|_v + \left.\frac{\partial f}{\partial y}\right|_x \left.\frac{\partial y}{\partial u}\right|_v \tag{3.46}$$

and

$$\left.\frac{\partial f}{\partial v}\right|_u = \left.\frac{\partial f}{\partial x}\right|_y \left.\frac{\partial x}{\partial v}\right|_u + \left.\frac{\partial f}{\partial y}\right|_x \left.\frac{\partial y}{\partial v}\right|_u . \tag{3.47}$$

We will not always be dealing with functions $f(x(u,v), y(u,v))$. Instead, we might have a case where $x = x(u)$ and $y = y(u,v)$ with a function

$$g = g(x(u), y(u,v)). \tag{3.48}$$

Here, it is clear that

$$\left.\frac{\partial x}{\partial v}\right|_u = 0, \tag{3.49}$$

and, perhaps unsurprisingly, our chain rule derivatives simply miss a term for the v derivative,

$$\left.\frac{\partial g}{\partial u}\right|_v = \left.\frac{\partial g}{\partial x}\right|_y \frac{\mathrm{d}x}{\mathrm{d}u} + \left.\frac{\partial g}{\partial y}\right|_x \left.\frac{\partial y}{\partial u}\right|_v \tag{3.50}$$

(note also the change in the derivative of x from partial to full, and hence no need to specify a constant variables) and

$$\left.\frac{\partial g}{\partial v}\right|_u = \left.\frac{\partial g}{\partial y}\right|_x \left.\frac{\partial y}{\partial v}\right|_u . \tag{3.51}$$

Another case that is often considered is if we have the case of $x = u$ and $y = y(u,v) = y(x,v)$. Note that although x and y are independent variables, this does not mean that y can be written as a function of v only. For example, we know that the pressure of a material can be written as a function of specific internal energy and density, i.e.

$$p = p(\rho, \varepsilon) \tag{3.52}$$

but also that internal energy can be written as a function of density and pressure,

$$U = U(\rho, p). \tag{3.53}$$

Using these, we could have internal energy in terms of density and specific internal energy through

$$U = U(\rho, p(\rho, \varepsilon)). \tag{3.54}$$

In the case of $u = x$, in equation (3.46), we have

$$\left.\frac{\partial x}{\partial u}\right|_v \rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\mathrm{d}x}{\mathrm{d}x} = 1 \tag{3.55}$$

and hence, for a function $h = h(x, v(x, y))$, we have

$$\left.\frac{\partial h}{\partial x}\right|_v = \left.\frac{\partial h}{\partial x}\right|_y + \left.\frac{\partial h}{\partial y}\right|_x \left.\frac{\partial y}{\partial x}\right|_v \tag{3.56}$$

and

$$\left.\frac{\partial h}{\partial v}\right|_u = \left.\frac{\partial h}{\partial y}\right|_x \left.\frac{\partial y}{\partial v}\right|_u . \tag{3.57}$$

Finally, we note that for functions of three or more variables, the chain rule expression is straightforward to generalise.

3.1.6 Cyclic and reciprocal rules

If we have $z = z(x, y)$, then we can also write this as $y = y(x, z)$ (we may not be able to do this analytically, but the logic remains; changing x and z will change y). We are able to express changes $\mathrm{d}z$ and $\mathrm{d}y$, as in equation (3.45), as

$$\mathrm{d}z = \left.\frac{\partial z}{\partial x}\right|_y \mathrm{d}x + \left.\frac{\partial z}{\partial y}\right|_x \mathrm{d}y \tag{3.58}$$

and

$$\mathrm{d}y = \left.\frac{\partial y}{\partial x}\right|_z \mathrm{d}x + \left.\frac{\partial y}{\partial z}\right|_x \mathrm{d}z. \tag{3.59}$$

This last equation can be rewritten for $\mathrm{d}z$ as

$$\mathrm{d}z = \frac{1}{\left.\frac{\partial y}{\partial z}\right|_x}\mathrm{d}y - \frac{\left.\frac{\partial y}{\partial x}\right|_z}{\left.\frac{\partial y}{\partial z}\right|_x}\mathrm{d}x \tag{3.60}$$

(assuming the derivative in the denominator is not zero).

Now, comparing equations (3.58) and (3.60), we see, from the $\mathrm{d}y$ terms that

$$\left.\frac{\partial z}{\partial y}\right|_x = \frac{1}{\left.\frac{\partial y}{\partial z}\right|_x} \tag{3.61}$$

i.e. partial derivatives obey a reciprocal rule.

By comparing the $\mathrm{d}x$ terms, we have

$$\left.\frac{\partial z}{\partial x}\right|_y = -\frac{\left.\frac{\partial y}{\partial x}\right|_z}{\left.\frac{\partial y}{\partial z}\right|_x} \tag{3.62}$$

or

$$\left.\frac{\partial z}{\partial x}\right|_y \left.\frac{\partial x}{\partial y}\right|_z \left.\frac{\partial y}{\partial z}\right|_x = -1 \tag{3.63}$$

which is a cyclic rule, where the final step needs to use the reciprocal rule.

3.1.7 Higher derivatives

We can derive higher derivative expressions in partial differentiation, but these do get quite messy. For example, consider the function

$$p = p(V(\rho), T(\rho, s)). \tag{3.64}$$

This is a function with its origins in physics, stating that pressure, p, is a function of volume, V, and temperature, T, but we could also write volume as a function of density, ρ, and temperature as a function of density and entropy, s. Using equation (3.50), we have the partial derivative

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s . \tag{3.65}$$

which we have chosen, because, as detailed in section 3.1.3, this is the speed of sound squared,

$$c_s^2 = \left.\frac{\partial p}{\partial \rho}\right|_s . \tag{3.66}$$

However, we may then wish to consider the second derivative of this equation,

$$\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s . \tag{3.67}$$

At this point, this example stops being directly physically relevant; this is not a quantity you are likely to use. However, it is useful for investigating higher derivatives. We first note that this expression could be written

$$\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s = \left.\frac{\partial}{\partial \rho}\left(\left.\frac{\partial p}{\partial \rho}\right|_s\right)\right|_s = \left.\frac{\partial}{\partial \rho}\left(\left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s\right)\right|_s . \tag{3.68}$$

In this expression, we have deliberately left in all variables being kept constant, because it helps with working out what we do next. The quantities

$$\left.\frac{\partial p}{\partial V}\right|_T \quad \text{and} \quad \left.\frac{\partial p}{\partial T}\right|_V \tag{3.69}$$

are functions of V and T, so taking derivatives of these quantities will need to follow the multivariable chain rule. However, the quantities

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} \quad \text{and} \quad \left.\frac{\partial T}{\partial \rho}\right|_s \tag{3.70}$$

will be functions of ρ and s, hence derivatives of these quantities can be taken directly. Therefore, with this in mind, we can apply the product rule appropriately to equation (3.68), giving

$$\begin{aligned}\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s &= \frac{\partial}{\partial \rho}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_s \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}^2 V}{\mathrm{d}\rho^2} \\ &\quad + \frac{\partial}{\partial \rho}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_s \left.\frac{\partial T}{\partial \rho}\right|_s + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial^2 T}{\partial \rho^2}\right|_s \end{aligned} \tag{3.71}$$

and then applying the multivariable chain rule gives

$$\begin{aligned}\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s &= \left[\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_V \left.\frac{\partial T}{\partial \rho}\right|_s\right] \frac{\mathrm{d}V}{\mathrm{d}\rho} \\ &\quad + \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}^2 V}{\mathrm{d}\rho^2} \\ &\quad + \left[\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_V \left.\frac{\partial T}{\partial \rho}\right|_s\right] \left.\frac{\partial T}{\partial \rho}\right|_s \\ &\quad + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial^2 T}{\partial \rho^2}\right|_s . \end{aligned} \tag{3.72}$$

Whilst this is a general expression for the second derivative, in practice, it is not actually a useful expression. In most cases, it is far simpler to simplify the result of taking the first derivative (using equation (3.65)) and then take additional derivatives.

To illustrate this, we consider an example, which, like this section, is initially based in physics, but we do depart from sensible physics to make things mathematically easier. Firstly, we consider our function, $p(V, T)$, to be the ideal gas law,

$$p = \frac{nRT}{V} \tag{3.73}$$

where n and R are constants (amount of substance in moles, and gas constant, respectively). We also have

$$V = \frac{M}{\rho} \tag{3.74}$$

where M is the total mass, and is also a constant. For an ideal gas, we also can define entropy as

$$s = \frac{Nk_b}{M} \ln \frac{VT^{c_V}}{N\Phi} \tag{3.75}$$

where N is the number of particles, k_B the Boltzmann constant, c_V the specific heat capacity at constant volume and Φ is a constant. However,

inverting this to get $T = T(\rho, s)$ gives an expression which leads to messy partial derivatives. Instead, we are going to take the physics out of this example, and consider a toy problem where we have

$$p = \frac{nRT}{V}, \quad V = \frac{M}{\rho}, \quad T = \rho^2 s^2 \tag{3.76}$$

where n, R and M are still constants, but now they don't have any meaningful physical interpretation.

Now, in this toy problem, we can compute all quantities needed in equation (3.72),

$$\left.\frac{\partial p}{\partial V}\right|_T = -\frac{nRT}{V^2}, \tag{3.77}$$

$$\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_T = \frac{2nRT}{V^3}, \qquad \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_V = -\frac{nR}{V^2}, \tag{3.78}$$

$$\left.\frac{\partial p}{\partial T}\right|_V = \frac{nR}{V}, \tag{3.79}$$

$$\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_T = -\frac{nR}{V^2}, \qquad \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_V = 0, \tag{3.80}$$

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -\frac{M}{\rho^2}, \qquad \frac{\mathrm{d}^2V}{\mathrm{d}\rho^2} = \frac{2M}{\rho^3}, \tag{3.81}$$

and

$$\left.\frac{\partial T}{\partial \rho}\right|_s = 2\rho s^2, \qquad \left.\frac{\partial^2 T}{\partial \rho^2}\right|_s = 2s^2. \tag{3.82}$$

Plugging all these expresions into equation (3.72) gives

$$\begin{aligned}\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s =& \left[-\frac{2nRT}{V^3}\frac{M}{\rho^2} - \frac{nR}{V^2}\left(2\rho s^2\right)\right]\left(-\frac{M}{\rho^2}\right) + \left(-\frac{nRT}{V^2}\right)\frac{2M}{\rho^3} \\ &+ \frac{nR}{V^2}\frac{M}{\rho^2}\left(2\rho s^2\right) + \frac{nR}{V}\left(2s^2\right) \qquad (3.83)\\ =& \frac{2nRM^2T}{V^3\rho^4} + \frac{2nRMs^2}{V^2\rho} - \frac{2nRMT}{V^2\rho^3} + \frac{2nRMs^2}{V^2\rho} + \frac{2nRs^2}{V}. \qquad (3.84)\end{aligned}$$

This is not an ideal form to leave the solution in, since we are taking the derivatives with respect to ρ, we would like the solution in terms of ρ, and the corresponding independent variable s. Substituting for V everywhere gives

$$\begin{aligned}\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s =& \frac{2nRT}{M\rho} + \frac{2nR\rho s^2}{M} - \frac{2nRT}{M\rho} + \frac{2nR\rho s^2}{M} + \frac{2nR\rho s^2}{M} \qquad (3.85)\\ =& \frac{6nR}{M}\rho s^2. \qquad (3.86)\end{aligned}$$

This approach, although thorough, was fairly lengthy, and, as a result, prone to errors when doing the calculation. It is, instead, better to start by first computing the first derivative (the square of the sound speed in equation (3.65)),

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \frac{nRMT}{V^2\rho^2} + \frac{2nR\rho s^2}{V} = \frac{nRT}{M} + \frac{2nR\rho^2 s^2}{M} = \frac{3nR}{M}\rho^2 s^2 \tag{3.87}$$

where we first applied the chain rule, then wrote V in terms of ρ, and finally wrote T in terms of ρ and s. It is now very simple to take the second derivative of this expression;

$$\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s = \frac{6nR}{M}\rho s^2. \tag{3.88}$$

Because we had already expressed our first derivative in terms of ρ and s, we didn't need to consider the multivariable chain rule for the second derivative, and instead could just use standard partial derivatives.

Note here, we could have also expressed the first derivative in terms of T and V, and used a simpler application of the multivariable chain rule. In this case, we would have had

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \frac{nRMT}{V^2\rho^2} + \frac{2nR\rho s^2}{V} = \frac{nRT}{M} + \frac{2nR\rho^2 s^2}{M} = \frac{3nR}{M}T \tag{3.89}$$

Because V dropped out of the expression in this case, we don't even need to use the multivariable chain rule, instead we have

$$\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s = \frac{\partial}{\partial \rho}\left(\frac{3nR}{M}T\right)\bigg|_s = \frac{3nR}{M}\left.\frac{\partial T}{\partial \rho}\right|_s = \frac{6nR}{M}\rho s^2. \tag{3.90}$$

In this case, this was not really necessary, but cases can arise where it is easier to write the expression in terms of a specific set of variables.

3.2 Thermodynamics

One of the key applications of multivariable calculus is computing quantities such as the speed of sound, using expressions like equation (3.65). Recall that this expression considered $p = p(V(\rho), T(\rho, s))$ - it may well be the case that these functions are not well known. In fact, for many materials, at the continuum scale at least, we do not know how to parameterise temperature with respect to density and entropy. Fortunately, however, by combining the

laws of thermodynamics with the properties of multivariable derivatives, we can avoid this.

In thermodynamics, we typically consider four variables, pressure, p, volume, V, temperature, T, and entropy, S. However, we are also able to identify that the overall thermodynamic system has two degrees of freedom, and hence only two of these are needed for a complete thermodynamic description of a system we consider (under some assumptions about the system, which are beyond what we will discuss here). However, if this is the case, then we also need two additional equations to provide closure to this system. One of these is the equation of state; and equation which describes (or describes at a statistically averaged level) the molecular behaviour of the material, such as the ideal gas law we introduced in equation (3.73). The second equation for closure of the system comes from the first law of thermodynamics.

We note here that we have actually introduced some equations of state, for example, we explicitly discussed the ideal gas law in section 3.1.3, and also gave function of the form $p = p(\rho, \varepsilon)$ and $p = p(V, T)$. These are all equations of state, and will be considered in detail in other courses.

Volumetric quantities

As we start to consider thermodynamic relationships, it is worth noting that there are often two ways in which a variable could be defined; a volumetric, or **total** quantitiy, or a quantity **per unit mass**. For a given material, we consider mass a constant (we will not be worrying about chemistry causing phase change or other effects in this section), so these two types of variable are related by a constant, but it can be important to know how your variable is defined (or, equivalently, what units it has).

Perhaps most important to note here is entropy - when we have used this so far, e.g. when defining the expression for the speed of sound, we used the variable s, which is entropy per unit mass. Total entropy has variable S, and this will be used in these notes, but this is not always used consistently in the literature.

Additionally, here, we note that in the speed of sound, because s and S are related by a constant (total mass), and s was kept constant (hence S is also kept constant), it is not always important to make the distinction about how entropy is defined.

3.2.1 First law of thermodynamics

We shall not attempt a full discussion of the laws of thermodynamics here, but simply state the important laws as a reference point. Here, we have the

first law of thermodynamics, which can be written

$$\mathrm{d}Q = \mathrm{d}U + p\mathrm{d}V \tag{3.91}$$

where Q is heat, and U is internal energy, and, for reversible processes, we can also write this as

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V. \tag{3.92}$$

Both of these forms are useful in simplifying expressions with variables we don't want to work with.

Note that we stated we could use the first law of thermodynamics to provide closure to the thermodynamic system, but here we've actually introduced another variable, internal energy. This does need consideration, and we shall look at this in more detail shortly.

3.2.2 Maxwell's relations

One key application of the first law of thermodynamics is that we can use it to find relationships between partial derivatives. If we consider internal energy to be a function of entropy and volume, then by the multivariable chain rule (3.39), we have

$$\mathrm{d}U = \left.\frac{\partial U}{\partial S}\right|_V \mathrm{d}S + \left.\frac{\partial U}{\partial V}\right|_S \mathrm{d}V. \tag{3.93}$$

However, comparing this to the first law of thermodynamics (3.92), we see that we have

$$\left.\frac{\partial U}{\partial S}\right|_V = T, \qquad \left.\frac{\partial U}{\partial V}\right|_S = -p. \tag{3.94}$$

This itself is useful; we might end up with these partial derivatives in an expression, and can replace them with a known quantity. However, we can also consider mixed-second derivatives of the internal energy. As thermodynamic variables can be assumed to be well-behaved functions, we have a symmetry of mixed derivatives, i.e.

$$\frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}\,. \tag{3.95}$$

Taking the second derivatives of both terms in equation (3.94), we have

$$\frac{\partial^2 U}{\partial V \partial S} = \left.\frac{\partial T}{\partial V}\right|_S, \qquad \frac{\partial^2 U}{\partial S \partial V} = -\left.\frac{\partial p}{\partial S}\right|_V \tag{3.96}$$

or

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial p}{\partial S}\right|_V . \tag{3.97}$$

This relationship is known as one of Maxwell's relations, and others can be computed.

We can derive other relationships too; to do this, we first note that we can derive similar relationships to the first law of thermodynamics (3.92), but with different dependent variables (for example, instead of the $T\mathrm{d}S$ term, we might want an $S\mathrm{d}T$ term instead (which won't necessarily have the same sign)). Such relationships are quite straightforward to derive. Consider

$$F = U - TS \tag{3.98}$$

then we can consider the quantity

$$\mathrm{d}F = \mathrm{d}U - \mathrm{d}(TS). \tag{3.99}$$

Leibniz's rule then gives

$$\mathrm{d}(TS) = T\mathrm{d}S + S\mathrm{d}T, \tag{3.100}$$

which follows from equations (3.37) and (3.39). Here, if we defined $x = x(T, S) = TS$, then the derivatives in equation (3.39) can be simply evaluated to give the expression above. From this, we have

$$\mathrm{d}F = \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T = -p\mathrm{d}V - S\mathrm{d}T \tag{3.101}$$

where the final expression is obtained using the first law of thermodynamics to remove $\mathrm{d}U$ from the equation. Now, by considering $F = F(T, V)$, and repeating the process to obtain information about second derivatives, we will find that

$$\left.\frac{\partial S}{\partial V}\right|_T = \left.\frac{\partial p}{\partial T}\right|_V . \tag{3.102}$$

Similarly, the definition $H = U + pV$ will give

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p, \qquad \left.\frac{\partial T}{\partial p}\right|_S = \left.\frac{\partial V}{\partial S}\right|_p \tag{3.103}$$

and $G = H - TS$ will give

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p, \qquad \left.\frac{\partial S}{\partial p}\right|_T = -\left.\frac{\partial V}{\partial T}\right|_p . \tag{3.104}$$

These four relations are then Maxwell's relations, and in deriving them, we introduced new variables, F, G and H. The process in which we moved between independent variables is actually a **Legendre transformation**, and the concept can be generalised mathematically. Additionally, these three variables are not simply mathematical conveniences. They are, along with internal energy, U, **thermodynamic potentials**, and are different ways of measuring thermodynamic potential energy computed under assumptions about how the energy is changed within the system. As such, they have additional applications (such as chemical equilibrium calculations), and we have

F: Helmholtz free energy

G: Gibbs free energy

H: Enthalpy

As with a lot of this section, we shall not go into the full details of what these potentials mean, but use them for their mathematical properties.

3.2.3 Specific heat capacity

As mentioned at the start of section 3.2.1, our attempts to close our system introduced an additional variable, U, the internal energy. To deal with this we need to know one more piece of information about the system; the specific heat capacity, and in particular, the specific heat capacity at constant volume. In practice, if our material has a known equation of state, and we know only a different specific heat capacity, for example, at constant pressure, we will be able to derive the specific heat capacity at constant volume using the laws of thermodynamics.

The heat capacity of a material is the amount of heat that must be added to raise the temperature by one unit, where the unit is dependent on the units of heat capacity itself. For example, if your heat capacity is defined as Joules per Kelvin (J/K), then its value is the heat required to increase the temperature by one Kelvin. **Specific** heat capacity is then the heat capacity per unit mass. Typical notation would be that heat capacity has symbol C, and specific heat capacity $c = C/M$, however, usage of this convention is not universal.

Because the heat required to raise temperature is dependent on how the heat is added, there is not a unique value, hence the typical quantities of (specific) heat capacity at constant volume (the material is not allowed to expand when heat is added) or at constant pressure (the material expands

such that there is no pressure change). There is no reason for these heat capacities to be the same value, hence we always distinguish how heat capacity was calculated, c_V and C_V for constant volume and c_p and C_p for constant pressure. Conveniently, this avoids a variable name that could be confused with the various other uses of c too, such as the speed of light (and sometimes sound).

Specific heat capacity is a useful quantity to close the system because it is usually reasonably straight forward to measure (and can also be computed from first principles approaches). All we then need to do is to relate it to internal energy, for which we start with the definition of specific heat at constant volume,

$$c_V = \frac{1}{M} \left. \frac{\mathrm{d}Q}{\mathrm{d}T} \right|_V . \tag{3.105}$$

If we consider the first law of thermodynamics, as given in equation (3.91), then at constant volume ($\mathrm{d}V = 0$), this simply becomes

$$\mathrm{d}Q = \mathrm{d}U. \tag{3.106}$$

This means c_v is given by

$$c_V = \frac{1}{M} \left. \frac{\mathrm{d}U}{\mathrm{d}T} \right|_V , \tag{3.107}$$

giving a relationship between internal energy and specific heat at constant volume. We can now integrate this expression with respect to temperature,

$$\int c_v \mathrm{d}T = \frac{1}{M} \int \left. \frac{\mathrm{d}U}{\mathrm{d}T} \right|_V \mathrm{d}T \tag{3.108}$$

which, using the fundamental theorem of calculus, gives

$$U = M \int c_v \mathrm{d}T. \tag{3.109}$$

For many materials, c_v is a known function of temperature, and hence this integral can be computed explicitly. This is particularly useful when c_v is constant - in practice this is unlikely to be the case for the full range of thermodynamic states a material can support, but it often can be considered constant over the range of likely values within a simulation, which are unlikely to reach the extremes of the thermodynamic behaviour of a given material. In this case, we have

$$U = M c_v T. \tag{3.110}$$

3.2.4 Dimensional analysis

Although not strictly an application of multivariable calculus, the introduction of internal energy is a good time to look at how dimensional analysis can prove useful. This comes from the fact that there could be multiple quantities loosely referred to as internal energy:

U: Total internal energy or just internal energy, units: Joules (J)

e: Volumetric internal energy (energy per unit volume) or just internal energy (again), units: $\mathrm{J/m^3}$

ε: Specific internal energy (energy per unit mass), units: J/kg

U_m: Molar internal energy (energy per mole of material), units: J/mol

Although we've given a unique symbol to each of these terms, and this is a convention we shall stick to, it is, unfortunately, not a universal convention, and, in particular, all of these variables could be called e. This is made worse by the fact that two of these quantities can be referred to as internal energy. Therefore we may need to work out the units of our internal energy variable, and hence work out which quantity we have.

As an example, we shall consider c_v to be a constant, and use equation (3.110) to investigate the units of internal energy. First, we note that we could do this either in a unit system of choice (e.g. SI units) or more generally. In the general case, any quantity is expressed in terms of:

T: Time

L: Length

M: Mass

Θ: Temperature (absolute temperature)

I: Electric current

N: Amount of substance

J: Luminous intensity

Applying this to our expression for (total) internal energy in equation (3.110), and noting that $1\ \mathrm{J} = 1\ \mathrm{kg\ m^2 s^{-2}}$ we can write

$$[U] = \mathrm{ML^2T^{-2}} \quad \text{or} \quad [U] = \mathrm{kg\ m^2 s^{-2}} \tag{3.111}$$

where the first expression could be considered formal dimensional analysis, and the second a unit-specific version (though works the same way).

Now, when $U = Mc_V T$, we have

$$[M] = \mathrm{M} \quad \text{or} \quad [M] = \mathrm{kg} \tag{3.112}$$

$$[c_V] = \mathrm{L}^2\mathrm{T}^{-2}\Theta^{-1} \quad \text{or} \quad [M] = \mathrm{Jkg}^{-1}\mathrm{K}^{-1} = \mathrm{m}^2\mathrm{s}^{-2}\mathrm{K}^{-1} \tag{3.113}$$

$$[T] = \Theta \quad \text{or} \quad [T] = \mathrm{K}. \tag{3.114}$$

Now we have

$$[Mc_V T] = \mathrm{ML}^2\mathrm{T}^{-2}\Theta^{-1}\Theta = M\mathrm{L}^2\mathrm{T}^{-2} \tag{3.115}$$

or

$$[Mc_V T] = \mathrm{kgm}^2\mathrm{s}^{-2}\mathrm{K}^{-1}\mathrm{K} = \mathrm{kgm}^2\mathrm{s}^{-2} = \mathrm{J}, \tag{3.116}$$

hence we have the same units (Joules) on both sides of this equation, as we would expect.

Whilst this may have been obvious, this allows us to work with other expressions where it is not clear which of the internal energy quantities is being referenced. For example, in the equation

$$\mathcal{E} = c_V T \tag{3.117}$$

when we have

$$[c_V T] = \mathrm{m}^2\mathrm{s}^{-2}\mathrm{K}^{-1}\mathrm{K} = \mathrm{m}^2\mathrm{s}^{-2} = \mathrm{J/kg} \tag{3.118}$$

so the units of $\mathcal{E}$ are energy per unit mass, or $\mathcal{E} = \varepsilon$, the specific internal energy. Note that we worked in units, rather than pure dimensions here, because this makes it easier to combine quantities into energy (Joules), where this quantity can be expressed as a single (non-SI) unit.

Another expression which may be encountered is

$$\hat{\mathcal{E}} = \rho c_V T \tag{3.119}$$

and here we have

$$[\rho c_V T] = \mathrm{kgm}^{-3}\mathrm{m}^2\mathrm{s}^{-2}\mathrm{K}^{-1}\mathrm{K} = \mathrm{kgm}^{-1}\mathrm{s}^{-2} = \mathrm{J/m}^{-3} \tag{3.120}$$

hence here we have $\hat{\mathcal{E}} = e$, the volumetric internal energy.

Note that we can also derive these expressions from equation (3.110),

$$\varepsilon = \frac{U}{M} = \frac{Mc_V T}{M} = c_V T \tag{3.121}$$

and

$$e = \frac{U}{V} = \frac{Mc_V T}{V} = \rho c_V T. \tag{3.122}$$

In this case, we didn't necessarily need to use dimensional analysis to identify which quantity we were working with, but in general, you might be working with other expressions which use an internal energy quantity, and hence this approach could be useful.

3.2.5 Application to the speed of sound

As an example of how the various the techniques for multivariable calculus and thermodynamics can be used, we return to the speed of sound, which, as defined by equation (3.65), is

$$c_s^2 = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_S . \tag{3.123}$$

In order to simplify this expression, we shall initially assume that we have an equation of state of the form $p = p(V, T)$, and an internal energy of the form $U = U(V, T)$ (i.e. we are not assuming constant specific heat at constant volume). By assuming we know these functions, we are saying we would like all partial derivatives to be in terms of these variables, and not others, in particular, not in terms of entropy (ether in the derivative, or as the variable held constant).

First we note that since $V = 1/\rho$, we have

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -\frac{1}{\rho^2}. \tag{3.124}$$

We can also use this relationship to turn derivatives with respect to ρ into derivatives with respect to V. If we have independent variables V and S, we have $T = T(V, S)$, however, we could also use independent variables ρ and S, which means we would have $T = T(V(\rho), S)$. Then, by the chain rule, we have

$$\left.\frac{\partial T}{\partial \rho}\right|_S = \left.\frac{\partial T}{\partial V}\right|_S \frac{\mathrm{d}V}{\mathrm{d}\rho} = -\frac{1}{\rho^2} \left.\frac{\partial T}{\partial V}\right|_S . \tag{3.125}$$

At this stage, the speed of sound expression now becomes

$$c_s^2 = -\frac{1}{\rho^2} \left(\left.\frac{\partial p}{\partial V}\right|_T + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial V}\right|_S \right). \tag{3.126}$$

The two derivative terms in this can be computed from the equation of state, and it is the final term, where entropy is held constant, that requires work. We shall focus on this term by itself; and first, we note that we can use the cyclic property of partial derivatives here (equation (3.63)),

$$\left.\frac{\partial T}{\partial V}\right|_S \left.\frac{\partial V}{\partial S}\right|_T \left.\frac{\partial S}{\partial T}\right|_V = -1 \tag{3.127}$$

which, when combined with the reciprocal rule (3.61) gives

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial S}{\partial V}\right|_T \left.\frac{\partial T}{\partial S}\right|_V . \tag{3.128}$$

The first term on the right hand side of this equation appears in one of Maxwell's relations, equation (3.102), so we have

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial S}\right|_V . \tag{3.129}$$

We then note that since $T = T(V, S)$ and $U = U(V, S)$, we could write $T = T(V, U(S, V))$, and hence, by the chain rule (3.57), we have

$$\left.\frac{\partial T}{\partial S}\right|_V = \left.\frac{\partial T}{\partial U}\right|_V \left.\frac{\partial U}{\partial S}\right|_V \tag{3.130}$$

and then

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial U}\right|_V \left.\frac{\partial U}{\partial S}\right|_V . \tag{3.131}$$

Now, the central derivative term on the right hand side can be determined from our definition of U, but we still have entropy appearing in the final term. However, when we were deriving Maxwells relations, we found, in equation (3.94), that

$$\left.\frac{\partial U}{\partial S}\right|_V = T \tag{3.132}$$

hence

$$\left.\frac{\partial T}{\partial V}\right|_S = -T \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial U}\right|_V . \tag{3.133}$$

This means we can now write the speed of sound entirely in terms of quantities we can compute,

$$c_s^2 = -\frac{1}{\rho^2}\left[\left.\frac{\partial p}{\partial V}\right|_T - T\left(\left.\frac{\partial p}{\partial T}\right|_V\right)^2 \left.\frac{\partial T}{\partial U}\right|_V\right]. \tag{3.134}$$

As a further example, we can apply this to the ideal gas equation of state, for which, we also have constant specific heat at constant pressure, i.e. we have

$$p = \frac{nRT}{V}, \qquad T = \frac{U}{Mc_V}. \tag{3.135}$$

This gives a speed of sound

$$c_s^2 = -\frac{1}{\rho^2}\left[-\frac{nRT}{V^2} - T\left(\frac{nR}{V}\right)^2 \frac{1}{Mc_V}\right] = \frac{1}{\rho^2}\left(\frac{1}{V}\frac{nRT}{V} + \frac{1}{V}\frac{nRT}{V}\frac{nR}{Mc_V}\right) \tag{3.136}$$

which, noting that $\rho V = 1$ and using the ideal gas law gives

$$c_s^2 = \frac{p}{\rho}\left(1 + \frac{nR}{Mc_V}\right). \tag{3.137}$$

Continuum courses will investigate and use the sound speed further.

3.3 Leibniz integral rule

Leibniz's integral rule is used to take the derivative of an integral where the variable of the derivative is present in the limits of the integral. We have an integral

$$G(t) = \int_{a(t)}^{b(t)} f(t,x)\mathrm{d}x \tag{3.138}$$

where we note that for this to be a challenge, the integral must be of a multivariable function. An example application of such an integral could be the the total energy inside a piston as a function of time; the walls of this piston, given by $x = a(t)$ and $x = b(t)$, will move (we'll assume we know how). The total energy will change due to the pressure excerted by the compression (or similar expansion effects), and this would need to be incorporated in our function f. For such a system, we can envision that we may want to know the rate of change of energy within the piston, i.e.

$$\frac{\mathrm{d}G}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int_{a(t)}^{b(t)} f(t,x)\mathrm{d}x. \tag{3.139}$$

This is given by the Leibniz integral rule, which we shall now derive. Firstly, we use a definition which will make things easier,

$$F(t,y) = \int_{u_0}^{y} f(t,u)\mathrm{d}u \tag{3.140}$$

where we can allow y to be a function of t. Now, we note that we can write equation (3.138) in terms of this new function,

$$G(t) = \int_{u_0}^{b(t)} f(t,x)\mathrm{d}x + \int_{a(t)}^{u_0} f(t,x)\mathrm{d}x \tag{3.141}$$

$$= \int_{u_0}^{b(t)} f(t,x)\mathrm{d}x - \int_{u_0}^{a(t)} f(t,x)\mathrm{d}x \tag{3.142}$$

$$= F(t,b(t)) - F(t,a(t)). \tag{3.143}$$

In order to take the derivative of $G(t)$, we now see that we need to take the derivative of a multivariable function, for example, if we consider $F(t, b(t))$, then

$$\frac{\mathrm{d}F\left(t,b(t)\right)}{\mathrm{d}t} = \left.\frac{\partial F\left(t,b(t)\right)}{\partial t}\right|_b + \left.\frac{\partial F\left(t,b(t)\right)}{\partial b}\right|_t \frac{\mathrm{d}b(t)}{\mathrm{d}t}. \tag{3.144}$$

This means we have

$$\begin{aligned}\frac{\mathrm{d}G}{\mathrm{d}t} &= \left.\frac{\partial F\left(t,b(t)\right)}{\partial t}\right|_b + \left.\frac{\partial F\left(t,b(t)\right)}{\partial b}\right|_t \frac{\mathrm{d}b(t)}{\mathrm{d}t} \\ &\quad - \left(\left.\frac{\partial F\left(t,a(t)\right)}{\partial t}\right|_a + \left.\frac{\partial F\left(t,a(t)\right)}{\partial a}\right|_t \frac{\mathrm{d}a(t)}{\mathrm{d}t}\right) \qquad (3.145) \\ &= \left.\frac{\partial F\left(t,b(t)\right)}{\partial b}\right|_t \frac{\mathrm{d}b(t)}{\mathrm{d}t} - \left.\frac{\partial F\left(t,a(t)\right)}{\partial a}\right|_t \frac{\mathrm{d}a(t)}{\mathrm{d}t} \\ &\quad + \left.\frac{\partial}{\partial t}\left(F(t,b(t) - F(t,a(t))\right)\right|_{a,b}. \qquad (3.146)\end{aligned}$$

Here, we have been careful to make sure we clearly show the variables for each function, because $F(t, a(t))$ and $F(t, b(t))$ are the same function, but acting on different variables, hence we must not forget this.

This is like having a function $h(x, y) = x^2y^3$, and $h(x, z) = x^2z^3$ - this is clearly the same function with different variables, and in each case, we have a similar relationship between derivatives,

$$\left.\frac{\partial h(x,y)}{\partial x}\right|_y = 2xy^3, \quad \left.\frac{\partial h(x,z)}{\partial x}\right|_z = 2xz^3. \tag{3.147}$$

If we treated these independently we could call our variables what we like, and we would not worry about whether we used y or z. However, if we were working in 3D space (i.e. with x, y and z coordinates), then we would have to make sure we kept these variables distinct. In this toy example, this might be overly complicated, and we'd be better off calling these two functions different names, e.g. $h_1(x, y)$ and $h_2(x, z)$. However, as we shall see, in our case here, we need to keep the same function name, but have these two different variables.

Returning to equation (3.146), we consider what these various terms are in terms of our the original function we were integrating, $f(t, x)$. First,

consider the final term in equation (3.146),

$$\frac{\partial}{\partial t}\left(F(t,b(t)-F(t,a(t))\right|_{a,b} = \frac{\partial}{\partial t}\left[\int_{u_0}^{b(t)} f(t,x)\mathrm{d}x - \int_{u_0}^{a(t)} f(t,x)\mathrm{d}x\right]\Bigg|_{a,b}$$
$$= \frac{\partial}{\partial t}\left[\int_{a(t)}^{b(t)} f(t,x)\mathrm{d}x\right]\Bigg|_{a,b}. \tag{3.148}$$

We have been very careful to make sure we remember that a and b are being kept constant in this expression; this means that we can bring the partial derivative inside the integral,

$$\frac{\partial}{\partial t}\left(F(t,b(t)-F(t,a(t))\right|_{a,b} = \int_{a(t)}^{b(t)} \left.\frac{\partial f(t,x)}{\partial t}\right|_x \mathrm{d}x. \tag{3.149}$$

Now consider the first term in equation (3.146); here we have

$$\left.\frac{\partial F\left(t,b(t)\right)}{\partial b}\right|_t = \frac{\partial}{\partial b}\left(\int_{u_0}^{b(t)} f(t,x)\mathrm{d}x\right)\Bigg|_t \tag{3.150}$$
$$= f(t,b(t)) \tag{3.151}$$

where this is actually just stating the first fundamental theorem of calculus (because t is constant, we can effectively ignore it).

Piecing everything together, we then get Leibniz's integral theorem,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{a(t)}^{b(t)} f(t,x)\mathrm{d}x\right) = f(t,b(t))\frac{\mathrm{d}b}{\mathrm{d}t} - f(t,a(t))\frac{\mathrm{d}a}{\mathrm{d}t} + \int_{a(t)}^{b(t)} \frac{\partial f(t,x)}{\partial t}\mathrm{d}x. \tag{3.152}$$

Chapter 4

Matrices

4.1 General properties of matrices

It can be useful to think of matrices as a collection of vectors, either row vectors, or column vectors. For example, we might have M column vectors, $\mathbf{v}_i$, $i \in [1, M]$, of length N, which can be grouped together to form an $N \times M$ matrix, which might be written

$$A = (\mathbf{v}_1 | \mathbf{v}_2 | \mathbf{v}_3 | \cdots | \mathbf{v}_M) \tag{4.1}$$

where there is a convention to include the pipe symbol between columns of a matrix defined by a set of vectors. Alternatively, we could have a set of M row vectors, $\mathbf{w}_j$, $j \in [1, M]$, of length N, which can be grouped together to form an $M \times N$ matrix,

$$B = \begin{pmatrix} \mathbf{w_1} \\ \hline \mathbf{w_2} \\ \hline \mathbf{w_3} \\ \hline \vdots \\ \hline \mathbf{w_M} \end{pmatrix} \tag{4.2}$$

4.1.1 Linear dependency

The rows, or columns, of a matrix can be either linearly independent or linearly dependent. If we consider the vectors used in equation (4.1), then we can try to identify a linear combination of these vectors such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v_2} + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v_M} = \mathbf{0} \tag{4.3}$$

where each a_i is a constant scalar value. The vectors $\mathbf{v}_i$ are said to be linearly independent if the only way this linear equation holds is if

$$a_i = 0 \tag{4.4}$$

for all i. If this equation also holds for at least one a_i being non-zero, then we have linearly dependent vectors. If we assume that $a_1 \neq 0$, then we could write

$$\mathbf{v}_1 = \frac{1}{a_1}\left(a_2\mathbf{v_2} + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v_M}\right), \tag{4.5}$$

in other words, $\mathbf{v}_1$ can be written as a linear combination of the other vectors.

4.2 Properties of square matrices

Add examples

Most of the time, when we are using matrices in scientific computing, we will be using square matrices. This is primarily because these matrices are used in solving systems of equations, for which we expect the same number of equations (rows of the matrix) as we do variables (columns of the matrix). There are various properties that a square matrix can (or will) have.

Symmetry

A matrix is symmetric if

$$A = A^T \qquad \text{or} \qquad A_{ij} = A_{ji} \tag{4.6}$$

and it is anti-symmetric if

$$B = -B^T \qquad \text{or} \qquad B_{ij} = -B_{ji}. \tag{4.7}$$

Note that the diagonal entries of an anti-symmetric matrix must be zero, since for the nth row and column, we require $A_{nn} = -A_{nn}$. The matrices have additional useful properties which we will encounter throughout this topic.

Diagonality

A matrix is diagonal if

$$A_{ij} = \begin{cases} a_i & i = j \\ 0 & i \neq j \end{cases}. \tag{4.8}$$

Diagonal matrices also have a lot of useful properties for solving equations. Two related forms of equations are upper triangular matrices,

$$B_{ij} = \begin{cases} b_{ij} & i \leq j \\ 0 & i > j \end{cases} \tag{4.9}$$

and lower triangular matrices,

$$C_{ij} = \begin{cases} c_{ij} & i \geq j \\ 0 & i < j \end{cases} \tag{4.10}$$

which retain some of the useful properties of fully diagonal matrices, whilst being potentially easier to compute.

Trace

The trace of a matrix is simply the sum of the diagonal entries of the matrix,

$$\mathrm{Tr}(A) = A_{ii}. \tag{4.11}$$

Although straightforward to compute, the trace of a matrix does not change under transformations such as rotation, and, as such, is referred to as an invariant quantity. These shall be considered in more detail later.

Inverse

A matrix may have an inverse, where multiplying a matrix by this index gives the identity matrix,

$$AA^{-1} = I \qquad \text{or} \qquad A_{ik}(A^{-1})_{kj} = \delta_{ij}. \tag{4.12}$$

Not all square matrices have an inverse, for example, any matrix multiplied by the zero matrix, e.g.

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \tag{4.13}$$

will always be the zero matrix - no inverse can exist. We shall deal with finding the inverse, and its uses, shortly.

Determinant

The determinant of a matrix, like the trace, is an invariant quantity, and, unlike the trace, does not have a simple expression to describe how it is calculated. One way of writing a general expression for an $N \times N$ matrix, A, is

$$\det(A) = \sum_{k=1}^{N} (-1)^{1+k} M_{1,k} A_{1k} \tag{4.14}$$

where $M_{1,k}$ is a minor of A, and we have included the summation sign because this is not an expresion in index notation form. We might guess this because

k is repeated three times, but it is also only used as an index location once (for A_1k). It is also used as an exponent, and for identifying a scalar, the minor, $M_{i,j}$, of a matrix. For a matrix A, $M_{i,j}$ is the determinant of a matrix formed from A by removing the ith row and the jth column, for example, if

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \tag{4.15}$$

then

$$M_{1,3} = \begin{vmatrix} d & e \\ g & h \end{vmatrix}. \tag{4.16}$$

A related definition of the determinant uses the cofactor, $C_{i,j}$, defined by

$$C_{i,j} = (-1)^{i+j} M_{i,j} \tag{4.17}$$

which gives

$$\det(A) = \sum_{k=1}^{N} C_{1,k} A_{1k}. \tag{4.18}$$

The form of the determinant expression given here is the 'traditional' approach, where the determinant is computed by taking the minors of the first row of the matrix. However, it is equally possible to choose any other row, e.g. for row M

$$\det(A) = \sum_{k=1}^{N} (-1)^{M+k} M_{M,k} A_{Mk} \tag{4.19}$$

or any column P,

$$\det(A) = \sum_{k=1}^{N} (-1)^{k+P} M_{k,P} A_{kP}. \tag{4.20}$$

Formally demonstrating this is not straightforward, but, for an arbitrary 3×3 matrix, it is quick to see that any of these definitions give the same calculation.

It is possible to write the determinant of a matrix in index notation too. When first encountering the cross product, you will have seen how it can be computed as the determinant of a matrix, and then, when introducing index notation, we saw that this could be expressed using the Levi-Civita symbol. It is then perhaps unsurprising that the Levi-Civita symbol is used to express determinants in index notation. For a 3×3 matrix, we have

$$\det(A) = \epsilon_{ijk} A_{1i} A_{2j} A_{3k} \tag{4.21}$$

which can be quickly verified by expanding out the sums in this expression. For an $N \times N$ matrix, this generalises to

$$\det(A) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{2 i_2} A_{3 i_3} \ldots A_{N i_N} \tag{4.22}$$

where we have used a generalisation of the Levi-Civita symbol,

$$\varepsilon_{a_1 a_2 a_3 \ldots a_N} = \begin{cases} +1 & \text{if } (a_1, a_2, a_3, \ldots, a_N) \text{ is an even permutation of} \\ & \qquad (1, 2, 3, \ldots, N) \\ -1 & \text{if } (a_1, a_2, a_3, \ldots, a_N) \text{ is an odd permutation of} \\ & \qquad (1, 2, 3, \ldots, N) \\ 0 & \text{otherwise} \end{cases} . \tag{4.23}$$

Here, a permutation is the switching of two indices, so if, for example, $N = 6$, an even permutation requires an even number of switches to get the entries back to 123456, whilst and odd permutation requires an odd number of switches. Although messy, this index notation formulation can help identify some of the properties of the determinant.

4.3 Elementary row and column operations

When solving systems of simultaneous equations, computing determinants or computing inverse matrices, we may find ourselves performing row and column operations. These are probably most familiar for the first application, simultaneous equations, note that if we have equations

$$\begin{aligned} ax + by + cz &= j \\ dx + ey + fz &= k \\ gx + hy + iz &= l \end{aligned} \tag{4.24}$$

then we could write this as a product of matrices and vectors,

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ k \\ l \end{pmatrix}, \tag{4.25}$$

which is sometimes simply written as

$$A\mathbf{x} = \mathbf{b}. \tag{4.26}$$

When solving our simultaneous equations, there are typically three row (or column) operations we make. Considering only rows here, we have:

1. Multiplying a row by a scalar. For a given row, r_i, we perform the operation $r_i \to \lambda r_i$

2. Adding a multiple of one row to another. For a given row, r_i, we could add a multiple of row r_j, $i \neq j$ through the operation $r_i \to r_i + \mu r_j$ (note that if $i = j$, this is simply multiplying a row by a scalar)

3. Transposing two rows. For two rows, r_i and r_j, $i \neq j$, we switch these through the operation $r_i \leftrightarrow r_j$

For column operations, the same operations hold, and the only change is typically the r notation will become c notation.

add matrix row changes too Considering our system of equations (4.24), we could consider $r_2 \to \frac{g}{d} r_2$:

$$\begin{gathered} ax + by + cz = j \\ gx + \frac{eg}{d} y + \frac{fg}{d} z = \frac{gk}{d} \\ gx + hy + iz = l \end{gathered} \tag{4.27}$$

We could then perform $r_2 \to r_2 - \frac{g}{a} r_1$:

$$\begin{gathered} ax + by + cz = j \\ \left(\frac{eg}{d} - \frac{bg}{a}\right) y + \left(\frac{fg}{d} - \frac{cg}{a}\right) z = \frac{gk}{d} - \frac{gj}{a} \\ gx + hy + iz = l \end{gathered} \tag{4.28}$$

Finally, we could switch the final two rows, $r_2 \leftrightarrow r_3$

$$\begin{gathered} ax + by + cz = j \\ gx + hy + iz = l \\ \left(\frac{eg}{d} - \frac{bg}{a}\right) y + \left(\frac{fg}{d} - \frac{cg}{a}\right) z = \frac{gk}{d} - \frac{gj}{a} \end{gathered} \tag{4.29}$$

This was a rather inefficient approach to remove the x component from the lowest equation, but here, it is used purely as an example.

4.3.1 Elementary row and column matrices

Each of the three elementary row (or column) operations can be described through matrix multiplication. We shall focus on row operations first, and consider each of the operations in turn.

Multiplication of a row by a scalar

Multiplying a given row, a, by a scalar, $r_a \to \lambda r_a$ has an elementary row matrix which in its general form could be written

$$E_{ij} = \begin{cases} 1 & i = j, \quad i \neq a \\ \lambda & i = j = a \\ 0 & i \neq j \end{cases} . \tag{4.30}$$

This description is simply a diagonal matrix where all entries are 1, except the entry on the row (or column) corresponding to the mulitplication, which takes the value of the multiplying scalar. It is probably clearer through example, so we shall consider the three operations on equation (4.24) from the previous section, but this time on the matrix form of the simultaneous equations in equation (4.25). The operation $r_2 \to \frac{g}{d} r_2$ has elementary row matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{g}{d} & 0 \\ 0 & 0 & 1 \end{pmatrix} . \tag{4.31}$$

To perform the operation, we simply multiply both sides of equation (4.25) by this matrix,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{g}{d} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{g}{d} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} j \\ k \\ l \end{pmatrix} \tag{4.32}$$

which simplifies to

$$\begin{pmatrix} a & b & c \\ g & \frac{eg}{d} & \frac{fg}{d} \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ \frac{gk}{d} \\ l \end{pmatrix} \tag{4.33}$$

which is now the matrix form of equation (4.27).

Adding a multiple of another row

Adding a multiple of one row to another, $r_a \to r_a + \mu r_b$, has general form

$$E_{ij} = \begin{cases} 1 & i = j \\ \mu & i = a, j = b \\ 0 & i \neq j \end{cases} . \tag{4.34}$$

Again this is not clear, but the diagonal components of this matrix are 1, and there is one other non-zero value, which takes the multiplying factor according to the row being operated on, and the row where the multiple comes from. Applying this to the example in equation (4.28), with the operation being $r_2 \to r_2 - \frac{g}{a} r_1$, we have elementary row matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ -\frac{g}{a} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.35}$$

which, if we apply this to equation (4.33), gives

$$\begin{pmatrix} 1 & 0 & 0 \\ -\frac{g}{a} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & c \\ g & \frac{eg}{d} & \frac{fg}{d} \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{g}{a} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} j \\ \frac{gk}{d} \\ l \end{pmatrix} \tag{4.36}$$

which then simplifies to

$$\begin{pmatrix} a & b & c \\ 0 & \left(\frac{eg}{d} - \frac{bg}{a}\right) & \left(\frac{fg}{d} - \frac{cg}{a}\right) \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ \frac{gk}{d} - \frac{gj}{a} \\ l \end{pmatrix} \tag{4.37}$$

Switching two rows

The elementary matrix for switching two rows, $r_a \leftrightarrow r_b$ is defined as

$$E_{ij} = \begin{cases} 1 & i = j, i \neq a, i \neq b \\ 1 & (i = a, j = b) \text{ or } (i = b, j = a) \\ 0 & \text{otherwise} \end{cases} . \tag{4.38}$$

This is perhaps the least clear general form to read, for the simplest elementary matrix to describe - this is the identity matrix, but with the two rows that you want to swap also swapped here. Again, for an example, we use the previous case considered in equation (4.29), where, for $r_2 \leftrightarrow r_3$, the elementary matrix is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{4.39}$$

which we now apply to equation (4.37),

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a & b & c \\ 0 & \left(\frac{eg}{d} - \frac{bg}{a}\right) & \left(\frac{fg}{d} - \frac{cg}{a}\right) \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} j \\ \frac{gk}{d} - \frac{gj}{a} \\ l \end{pmatrix} \tag{4.40}$$

and, unsurprisingly, this gives

$$\begin{pmatrix} a & b & c \\ g & h & i \\ 0 & \left(\frac{eg}{d} - \frac{bg}{a}\right) & \left(\frac{fg}{d} - \frac{cg}{a}\right) \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ l \\ \frac{gk}{d} - \frac{gj}{a} \end{pmatrix} \tag{4.41}$$

Add usage section

Column operations

So far we have only considered row operations, partly because our example, simultaneous equations, is generally solved through these, rather than column operations (for example, switching two columns simply switches the order two variables appear in the equation, which doesn't help solve anything!) However, for other cases, column operations can be useful. Column operations still have the same elementary matrices, but they multiply the right hand side of the matrix being operated on, instead of the left. So, if we had elementary matrix E, then we could operate on the rows of a matrix A through

$$A' = EA \tag{4.42}$$

whilst the column operation would be

$$\hat{A} = AE \tag{4.43}$$

For example, consider the elementary matrix for switching rows or columns we defined in equation (4.39). Here, we see that

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} a & c & b \\ d & f & e \\ g & i & h \end{pmatrix}. \tag{4.44}$$

Applying this to the simultaneous equation example is trickier; we can't simply multiply the right hand side of the equation with our elementary matrix, since this would involve multiplying a column vector by a square matrix in a way in which the indices don't match up. Instead, we could consider

$$A\mathbf{x} = AEE^{-1}\mathbf{x} = \mathbf{b} \tag{4.45}$$

since multiplying a matrix by its inverse gives the identity matrix. This means that we perform the column operation on our matrix, A, and perform the operation associated with the inverse elementary matrix on $\mathbf{x}$.

Whilst we haven't talked about inverses yet (though calculating them may well be familiar), for this example, where we are switching rows, we

can easily identify the inverse. If we apply the same row switching operation twice, we get back to where we started, for example,

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}. \tag{4.46}$$

which means that

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.47}$$

and hence the row or column switching elementary matrix is it's own inverse. Applying this to the general simulaneous equation case gives

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} =$$
$$\begin{pmatrix} a & c & b \\ d & f & e \\ g & i & h \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a & c & b \\ d & f & e \\ g & i & h \end{pmatrix} \begin{pmatrix} x \\ z \\ y \end{pmatrix} = \begin{pmatrix} j \\ k \\ l \end{pmatrix} \tag{4.48}$$

where in the first step, we multiply the original matrix by the elementary operation matrix, and in the second step, we multiply the elementary operation matrix by the vector $\mathbf{x}$. In this way, we see that the columns switch, but so do the positions of y and z, which recovers the simultaneous equations with the order of the variables switched around.

4.4 The inverse

We shall now consider the inverse of a matrix in a little more detail. There are several useful properties which are straightforward to derive, using simply the definition of the inverse,

$$AA^{-1} = I \tag{4.49}$$

or, equivalently

$$A^{-1}A = I. \tag{4.50}$$

4.4.1 Properties of the inverse

Inverse of an inverse

The inverse of the inverse of a matrix is the original matrix,

$$(A^{-1})^{-1} = A. \tag{4.51}$$

Note that if we multiply this expression on the left hand side by A^{-1}, we are multiplying a matrix by its inverse, hence

$$A^{-1}(A^{-1})^{-1} = I, \tag{4.52}$$

which, by comparing to equation (4.50), yields the desired result.

Scalar multiplication and the inverse

The inverse of a scalar multiple is given by

$$(kA)^{-1} = \frac{1}{k}A^{-1}. \tag{4.53}$$

By the definition of the inverse, we know that

$$(kA)(kA)^{-1} = I \tag{4.54}$$

however, we can also consider

$$(kA)\left(\frac{1}{k}A^{-1}\right) = \frac{k}{k}AA^{-1} = AA^{-1} = I \tag{4.55}$$

and comparing these two equations gives the result.

Inverse of a matrix product

The inverse of a matrix product is given by

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{4.56}$$

Again, we use the definition of the inverse, which means that

$$(AB)^{-1}(AB) = I \tag{4.57}$$

but also consider

$$B^{-1}A^{-1}(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I \tag{4.58}$$

hence by comparing these two expressions, we have the desired property. Note that this also generalises (which could be demonstrated through repeated application) to

$$(A_1A_2\cdots A_n)^{-1} = (A_n)^{-1}(A_2)^{-1}(A_1)^{-1}. \tag{4.59}$$

4.4.2 Inverse of elementary matrices

In equation (4.45), we decided to use the inverse of our elementary operation matrix. Whilst this does work out to be fine, what we didn't consider was whether the inverse itself actually existed. Fortunately, for all elementary operation matrices, it does. If we consider a single operation through one of the matrices, we have

$$A' = EA. \tag{4.60}$$

However, for our elementary operations, we are able to reverse the operation,

- **Multiplying a row by a scalar** - reversed by dividing a row by a scalar, or $r_a \to \lambda r_a$ reversed by $r_a \to \frac{1}{\lambda} r_a$
- **Adding a multiple of another row** - reversed by subtracting the same multiple of this same row, or $r_a \to r_a + \mu r_b$ reversed through $r_a \to r_a - \mu r_b$
- **Switching two rows** - reversed by switching the same two rows again, or $r_a \leftrightarrow r_b$ reversed through $r_a \leftrightarrow r_b$

This last case was discussed in our example following equation (4.45). Note that each of these elementary operations are reversed by another elementary operation (of the same type). Therefore, if we first apply an operation (through matrix E_1), and then apply the reverse operation (through matrix E_2), we do not change our original matrix, i.e.

$$A = E_2 E_1 A \tag{4.61}$$

which means that

$$E_2 E_1 = I \qquad \text{or} \qquad E_2 = (E_1)^{-1} \tag{4.62}$$

so in all cases, the inverse of our elementary matrices exists.

4.4.3 Non-invertible matrices

Although our argument in using the inverse of elementary matrices was actually fine, we did need to show this, because not all matrices are invertible. Consider the general matrix form for simultaneous equations, seen in equation (4.26),

$$A\mathbf{x} = \mathbf{b}. \tag{4.63}$$

Now, if A has an inverse, then we have

$$A^{-1}A\mathbf{x} = \mathbf{x} = A^{-1}\mathbf{b} \tag{4.64}$$

so if we know the inverse, we can compute unique values for all variables in $\mathbf{x}$. However, we also know that not all simultaneous equations have unique solutions (or even any solutions). For example, consider

$$\begin{array}{c} x+y+z=1 \\ 2x+y=1 \\ y+2z=b \end{array} \tag{4.65}$$

we attempt to solve this through row operations,

$$r_3 \to r_3 + r_2: \qquad \begin{array}{c} x+y+z=1 \\ 2x+y=1 \\ 2x+2y+2z=b+1 \end{array} \tag{4.66}$$

$$\tag{4.67}$$

$$r_3 \to r_3 - 2r_1: \qquad \begin{array}{c} x+y+z=1 \\ 2x+y=1 \\ 0=b-1 \end{array} \tag{4.68}$$

Now, if $b=1$, this becomes two equations for three unknowns - we have an underdetermined system, and if $b \neq 0$, then we have a contradiction in the final equation, we cannot solve the system.

At this point, it might become apparent that there is a connection between linear dependency and non-invertibility. Recall that we could express a matrix as a collection of row vectors,

$$A = \begin{pmatrix} \frac{\mathbf{w_1}}{\mathbf{w_2}} \\ \frac{\mathbf{w_3}}{\vdots} \\ \frac{}{\mathbf{w_M}} \end{pmatrix} \tag{4.69}$$

and that if these vectors are linearly dependent, then we have

$$\mathbf{w}_1 = \frac{1}{a_1}\left(a_2\mathbf{w_2} + a_3\mathbf{w}_3 + \cdots + a_M\mathbf{w_M}\right). \tag{4.70}$$

Now suppose we take this matrix A and perform the row operation $r_1 \to r_1 - \frac{a_2}{a_1}r_2$, this would give us

$$A' = \begin{pmatrix} \frac{\mathbf{w_1} - \frac{a_2}{a_1}\mathbf{w_2}}{\mathbf{w_2}} \\ \frac{\mathbf{w_3}}{\vdots} \\ \frac{}{\mathbf{w_M}} \end{pmatrix}. \tag{4.71}$$

However, because this is simply multiplication by an elementary matrix, which we shall call E_1, we have

$$A' = E_1 A. \tag{4.72}$$

Now we perform the next operation as $r_1 \to r_1 - \frac{a_3}{a_1} r_3$, which gives

$$A'' = \begin{pmatrix} \mathbf{w_1} - \frac{1}{a_1}\left(a_2\mathbf{w_2} + a_3\mathbf{w_3}\right) \\ \hline \mathbf{w_2} \\ \hline \mathbf{w_3} \\ \hline \vdots \\ \hline \mathbf{w_M} \end{pmatrix} \tag{4.73}$$

or, again using an elementary matrix E_2,

$$A'' = E_2 E_1 A. \tag{4.74}$$

If we continue this until we have made $M-1$ row operations, where operation m is $r_1 \to r_1 - \frac{a_m}{a_1} r_m$, we get

$$A^{(M-1)} = \begin{pmatrix} \mathbf{w_1} - \frac{1}{a_1}\left(a_2\mathbf{w_2} + a_3\mathbf{w_3} + a_M\mathbf{w_M}\right) \\ \hline \mathbf{w_2} \\ \hline \mathbf{w_3} \\ \hline \vdots \\ \hline \mathbf{w_M} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \hline \mathbf{w_2} \\ \hline \mathbf{w_3} \\ \hline \vdots \\ \hline \mathbf{w_M} \end{pmatrix} \tag{4.75}$$

and we could also write this as

$$A^{(m-1)} = E_{M-1} \cdots E_2 E_1 A \tag{4.76}$$

or, because $E_{M-1} \cdots E_2 E_1$ can be evaluated to give a single matrix, we could write this as

$$\bar{A} = \bar{E} A \tag{4.77}$$

defining $\bar{E} = E_{M-1} \cdots E_2 E_1$ and $\bar{A} = A^{(M-1)}$. The reason for this last expression is that there is not a unique set of elementary row operations which will get us to this point. For example, we could perform a row swapping operation such that we then always subtract the multiple of the second row from the first row, rather than work through the rows in turn, which would require $2(M-1)$ operations. The key information is that there exists a combination of row operations which allow us to find a matrix $\bar{A}$ where the top row is entirely zero.

Note also that this also implies

$$A = (\bar{E})^{-1}\bar{A} \tag{4.78}$$

and we know the inverse of $\bar{E}$ exists since

$$(\bar{E})^{-1} = (E_{n-1}\cdots E_2E_1)^{-1} = (E_1)^{-1}(E_2)^{-1}\cdots(E_n)^{-1} \tag{4.79}$$

where we use equation (4.59) here. Now, we have already shown that all elementary matrices have inverses, hence $\bar{E}$ must do too.

Now, if A were invertible, we would have

$$AA^{-1} = I, \tag{4.80}$$

and we can then substitute in equation (4.77), giving

$$(\bar{E})^{-1}\bar{A}A^{-1} = I \tag{4.81}$$

or

$$\bar{A}A^{-1} = \bar{E}. \tag{4.82}$$

Now, we know that the top row of $\bar{A}$ is zero everywhere, from equation (4.75). This means that the top row of $\bar{A}A^{-1}$ is also zero. However, we also know that the rows of the identity matrix are linearly independent, hence the series of elementary matrix operations cannot result in cancellation such that the top row of $\bar{E}$ is zero. In other words, if the top row of $\bar{A}$ is zero, this expression is a contradiction, and since we know that $\bar{A}$ and $\bar{E}$ exist, A^{-1} cannot exist.

This is a key result - all matrices with linearly dependent rows (or columns) do not have an inverse.

4.4.4 Invertible matrices

The converse of the result from the previous section is also true - all matrices with linearly independent rows (or columns) have an inverse. This starts from the fact that all matrices can be put in row echelon form; a form where, through row operations, each row's leading entry (the first non-zero entry) is ordered such that it is in at least one column to the right of any leading zeroes in rows above this row, for example,

$$\begin{pmatrix} 2 & a & 4 & \pi \\ 0 & -1 & c & c \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{4.83}$$

It is straightforward to see that any matrix can be put in this form through row operations. If you start with a general matrix,

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \\ a_{m1} & a_{m2} & & a_{mn} \end{pmatrix} \tag{4.84}$$

1. If $a_{11} = 0$, switch this row with any other row $a_{i1} \neq 0$ (if they are all zero, then skip this and the next step - the first column is already reduced).

2. Then subtract off an approriately weighted multiple of the first row from all other rows, i.e. for any row i, $i > 1$, $r_i \to r_i - \frac{a_{i1}}{a_{11}} r_1$. The matrix will then look like

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} - \frac{a_{21}a_{12}}{a_{11}} & \cdots & a_{2n} - \frac{a_{21}a_{1n}}{a_{11}} \\ \vdots & \vdots & \ddots & \\ 0 & a_{m2} - \frac{a_{m1}a_{12}}{a_{11}} & & a_{mn} - \frac{a_{m1}a_{1n}}{a_{11}} \end{pmatrix} \tag{4.85}$$

3. If $a_{22} = 0$, switch this row with another row $a_{j,2} \neq 0$, $j > 2$ (if one exists)

4. Now subtract off an appropriately weighted multiple of the second row from all rows $i > 2$. This can be done through $r_i \to r_i - \frac{a_{i2} - \frac{a_{i1}a_{12}}{a_{11}}}{a_{22} - \frac{a_{21}a_{12}}{a_{11}}} r_2$.

5. Repeat for subsequent columns until row echelon form is obtained.

If we have a square matrix, there are two possibilities,

1. Row echelon form will have (at least) the bottom row as entirely zero, in which case we have at least linearly dependent row, and the matrix has no inverse

2. All rows will have a non-zero entry, and this means we have an upper triangular matrix (or a diagonal matrix if we are lucky), e.g.

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & t & 2 & -2 \\ 0 & 0 & 4 & \omega \\ 0 & 0 & 0 & -2 \end{pmatrix} \tag{4.86}$$

In this second case, it is again straightforward to see that we can convert this to a fully diagonal matrix, through further row operations. This time starting with the right-most row to remove all but the diagonal entry from this row. And once we have a diagonal matrix, we can multiply each row by a suitable scalar to get the identity matrix.

Mathematically, we are saying that if our matrix has only linearly independent rows, we can find elementary matrices such that we have

$$I = E_{M-1} \cdots E_2 E_1 A = \hat{E} A = A^{-1} A \tag{4.87}$$

Note that this also gives us a strategy to find the inverse; where every time an elementary operation is performed on our matrix A, we also apply it to the identity matrix. Then, once we have made the operations such that $A \to I$, we also find we have obained $I \to A^{-1}$. This is again easy to see since, if we begin with

$$A A^{-1} = I \tag{4.88}$$

then applying the steps to both sides means

$$\hat{E} A A^{-1} = \hat{E} I \tag{4.89}$$

but we know $\hat{E} A = I$, so

$$A^{-1} = \hat{E} I = \hat{E}. \tag{4.90}$$

4.5 Determinant properties

Having introduced elementary matrices, and inverses, we now return to determinants, and, in particular, some of the properties of the determinant.

4.5.1 Determinants of diagonal matrices

Add brief statement here

4.5.2 Effects of elementary operations on determinants

Multiplication of a row by a scalar

If we multiply any row of a matrix by a scalar, then the determinant of matrix is also multiplied by a scalar. We can see this from the index notation form of the determinant in equation (4.22),

$$\det(A) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{2 i_2} A_{3 i_3} \ldots A_{N i_N}. \tag{4.91}$$

If we compute the matrix A' by performing the row operation $r_2 \to \lambda r_2$ on the matrix A, then the determinant of A' is given by

$$\begin{aligned}\det(A') =& \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} \left(\lambda A_{2 i_2}\right) A_{3 i_3} \ldots A_{N i_N} && (4.92)\\ =& \lambda \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{2 i_2} A_{3 i_3} \ldots A_{N i_N} = \lambda \det(A) && (4.93)\end{aligned}$$

A directly related result is that if we multiply the entire matrix by a scalar, μ, which is equivalent to multiplying all rows by μ, then

$$\det(\mu A) = \mu^N \det(A) \tag{4.94}$$

where A is an $N \times N$ matrix.

Adding a multiple of another row

Adding a multiple of another row does not change the determinant. Again, we can see this in the index notation form. Here, we shall apply the operation $r_2 \to r_2 + \nu r_1$, for which the determinant of the transformed matrix, A', would be

$$\det(A') = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} \left(A_{2 i_2} + \nu A_{1 i_2}\right) A_{3 i_3} \ldots A_{N i_N}. \tag{4.95}$$

Because all entries are treated as scalars, we can simply expand out this expression,

$$\begin{aligned}\det(A') =& \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{2 i_2} A_{3 i_3} \ldots A_{N i_N} \\ &+ \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} \left(\nu A_{1 i_2}\right) A_{3 i_3} \ldots A_{N i_N} \\ =& \det(A) + \nu \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{1 i_2} A_{3 i_3} \ldots A_{N i_N}. && (4.96)\end{aligned}$$

Since we treat all entries in these index expressions as scalars, we can write

$$\det(A') = \det(A) + \nu \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_2} A_{1 i_1} A_{3 i_3} \ldots A_{N i_N} \tag{4.97}$$

where we have switched the order of $A_{1 i_1}$ and $A_{1 i_2}$. However, because all indices are repeated, we can also rename i_1 as i_2, and at the same time, i_2 as i_1. This gives

$$\det(A') = \det(A) + \nu \epsilon_{i_2 i_1 i_3 \ldots i_N} A_{1 i_1} A_{1 i_2} A_{3 i_3} \ldots A_{N i_N} \tag{4.98}$$

where, because we have switched a pair of indices in our Levi-Civita symbol, we have switched its sign, i.e.

$$\epsilon_{i_2 i_1 i_3 \ldots i_N} = -\epsilon_{i_1 i_2 i_3 \ldots i_N} \tag{4.99}$$

hence we have

$$\det(A') = \det(A) - \nu\epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1} A_{1i_2} A_{3i_3} \ldots A_{Ni_N} \tag{4.100}$$
$$= \det(A) + \nu\epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1} A_{1i_2} A_{3i_3} \ldots A_{Ni_N} \tag{4.101}$$

which can only hold if

$$\epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1} A_{1i_2} A_{3i_3} \ldots A_{Ni_N} = 0 \tag{4.102}$$

hence we have

$$\det(A') = \det(A) \tag{4.103}$$

Switching two rows

Switching two rows will flip the sign of the determinant. Here, the determinant of the matrix resulting from the operation $r_1 \leftrightarrow r_2$ can be written in index form as

$$\det(A') = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{2i_1} A_{1i_2} A_{3i_3} \ldots A_{Ni_N}. \tag{4.104}$$

where we note that the index of the first two matrix entries has changed. The manipulation we do here is very similar to that for the addition of a multiple of a row; first we switch the first two matrix entries,

$$\det(A') = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_2} A_{2i_1} A_{3i_3} \ldots A_{Ni_N} \tag{4.105}$$

and then we can rename i_1 as i_2 and i_2 as i_1,

$$\det(A') = \epsilon_{i_2 i_1 i_3 \ldots i_N} A_{1i_1} A_{2i_2} A_{3i_3} \ldots A_{Ni_N}. \tag{4.106}$$

Again, we see that this is a switch in the first two indices of the Levi-Civita symbol, so, using equation (4.99), we have

$$\det(A') = -\epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1} A_{2i_2} A_{3i_3} \ldots A_{Ni_N} = -\det(A). \tag{4.107}$$

4.5.3 Determinants following row operations

The reason we consider the effects of the elementary operations is that some for some matrices, finding the determinant might not be a simple calculation, for example,

$$\begin{vmatrix} a-E & -b & -b & -b & -b \\ -b & a-E & -b & -b & -b \\ -b & -b & a-E & -b & -b \\ -b & -b & -b & a-E & -b \\ -b & -b & -b & -b & a-E \end{vmatrix} = (a-4b-E)(a+b-E)^4. \tag{4.108}$$

Whilst this is a nice neat result, if we attempted a brute force approach, we would need to simplify five quintic equations to get this, and if we didn't know what we were aiming for, it would be very challenging.

However, if we have a matrix in upper diagonal form (or lower diagonal form, or just pure diagonal form), then computing the determinant is straightforward; it is simply the product of the diagonal entries. It is generally easier to obtain an upper diagonal form through row operations than it is to simplify a large polynomial, either by hand or computationally. That is why we were concerned with the effects that elementary operations have on determinants. If we perform a series of elementary operations, so that we have

$$\hat{A} = \hat{E}A \tag{4.109}$$

then clearly

$$\det(\hat{A}) = \det(\hat{E}A). \tag{4.110}$$

But the effects of the elementary operations on the determinant of A were to multiply it by a scalar (row/column multiplication) or flip its sign (switching rows), since adding a multiple of one row to another left the determinant unchanged. Therefore we can write

$$\det(\hat{A}) = \det(\hat{E}A) = \alpha\det(A) \tag{4.111}$$

where α is a scalar quantity, and is simply the product of each row/column multiplication scalar multiplied by -1^s where s is the number of row/column switches performed.

4.5.4 Determinants and linear dependency

From any definition of the determinant, e.g.

$$\det(A) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1} A_{2i_2} A_{3i_3} \ldots A_{Ni_N}. \tag{4.112}$$

we know that if any row has all entries zero (e.g. $A_{2i_N} = 0$ for all i_N), then this implies the determinant is zero,

$$\det(A) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1i_1}.0.A_{3i_3} \ldots A_{Ni_N} = 0. \tag{4.113}$$

We also know that if a matrix has linearly dependent rows, then attempting to put it in row echelon form, or any diagonal form, through elementary operations, will result in at least one row of zeroes. Since we have, from equation (4.111),

$$\det(\hat{A}) = \alpha\det(A), \tag{4.114}$$

where $\det(\hat{A}) = 0$ is our matrix with a row of zeroes, and $\alpha \neq 0$, this implies $\det(A) = 0$ for any matrix with linearly dependent rows or columns.

The converse is true for a matrix with linearly independent rows; in this case, the determinant is always non-zero. Recall from section 4.4.4 that if B is a matrix with linearly independent rows, elementary row operations exist such that

$$\hat{E}B = I \tag{4.115}$$

and then

$$\det(I) = \det(\hat{E}B) = \beta \det B. \tag{4.116}$$

This gives

$$\det(B) = \frac{1}{\beta} \neq 0 \tag{4.117}$$

since we know $\beta \neq 0$.

4.5.5 Determinant of a product

The determinant of a matrix product is the product of the two individual determinants.

$$\det(AB) = \det(A)\det(B). \tag{4.118}$$

This result itself is not immediately apparent, or simple to derive from index notation, but the properties of matrices covered so far. The key comes from noting that we can transform these matrices, in particular A, using elementary operations; we will once again be using

$$(\hat{E})^{-1}\hat{A} = A. \tag{4.119}$$

We consider two cases, $\det(A) = 0$ and $\det(A) \neq 0$.

Zero determinant of A

If $\det(A) = 0$, then we can show $\det(AB) = 0$. There are two equivalent ways to do this. First, using elementary operations, we know we have

$$\det(AB) = \det((\hat{E})^{-1}\hat{A}B) = \alpha \det(\hat{A}B). \tag{4.120}$$

Where we have used the fact that the inverse elementary operations are still elementary operations, as discussed in section 4.4.2. If $\det(A) = 0$, then we know that we can find $\hat{A}$ such that at least one row contains only zeroes. In this case, we know also that $\hat{A}B$ must also contain a row of zeroes in the same row, i.e.

$$\det(\hat{A}B) = 0. \tag{4.121}$$

and therefore

$$\det(AB) = 0. \tag{4.122}$$

An alternative way to show this is to assume that $\det(AB) \neq 0$. This means that there is an inverse, such that

$$(AB)(AB)^{-1} = I. \tag{4.123}$$

However, from equation (4.56), we would require

$$(AB)(AB)^{-1} = ABB^{-1}A^{-1} = I. \tag{4.124}$$

Since the inverse of A does not exist, we also cannot have $(AB)^{-1}$, and again, $\det(AB) = 0$.

This then means that

$$\det(AB) = \det(A)\det(B). \tag{4.125}$$

now automatically holds, since both sides of this equation are zero, regardless of any properties of B.

Non-zero determinant of A

If $\det(A) \neq 0$, then we know that the rows of this matrix are linearly independent, hence there exists a set of elementary operations such that

$$\hat{E}A = I \tag{4.126}$$

or

$$A = (\hat{E})^{-1}. \tag{4.127}$$

Therefore we have

$$\det(AB) = \det((\hat{E})^{-1}B) = \alpha\det(B). \tag{4.128}$$

As with the zero determinant case, we use the fact that inverse elementary operations are still elementary operations. This is now a relationship that holds for all matrices B and all matrices with non-zero determinant A. All we need to do is identify α, for which we choose a specific example, $B = I$.

$$\det(AI) = \alpha\det(I) \tag{4.129}$$

or

$$\det(A) = \alpha \tag{4.130}$$

which, substituting α into equation (4.128) gives the general result

$$\det(AB) = \det(A)\det(B). \tag{4.131}$$

4.5.6 Calculating the inverse using determinants

We have discussed how a matrix inverse can be computed using row operations, however, an alternative approach uses a lot of determinants instead. Each element of the inverse can be calculated through

$$(A^{-1})_{ij} = \frac{C_{j,i}}{\det(A)} \tag{4.132}$$

where $C_{j,i}$ is the co-factor defined in equation (4.17). This can be derived (we will not do so for this course), see, for example, here.

4.6 Matrices as geometrical operations

When we multiply a vector by a matrix, we are actually performing a geometrical operation (specifically, a linear map) to that vector. We start with the standard Cartesian basis vectors (sticking to three-dimensional space), $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$, where

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.133}$$

from which we could write any other vector, $\mathbf{v}$ as

$$\mathbf{v} = v_1\mathbf{e_1} + v_2\mathbf{e_2} + v_3\mathbf{e_3}. \tag{4.134}$$

Insert picture here.

We now consider the elementary matrix operations as geometrical operations, and consider specific examples to illustrate the effects.

Multiplication by scalar

Consider the elementary matrix

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.135}$$

then we can map our original coordinate system to a new coordinate system through

$$\mathbf{e}'_i = E\mathbf{e_i} \tag{4.136}$$

and in this case, we have

$$\mathbf{e}'_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{e}'_2 = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \qquad \mathbf{e}'_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.137}$$

so our new coordinate system simply stretches vectors along the y-axis. **Insert picture here**

Adding a multiple of another row

For this example, we consider the elementary matrix

$$E = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.138}$$

for which our new coordinate system becomes

$$\mathbf{e}'_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{e}'_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{e}'_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.139}$$

which has effectively 'tilted' one of our coordinate directions. **Insert picture here**

Note that this is not an orthogonal coordinate system, since $\mathbf{e}_1 \cdot \mathbf{e}_2 = 1$, nor is it normalised, since $\mathbf{e}_2 \cdot \mathbf{e}_2 = 2$, so it might not be the most useful coordinate system, but it is still a coordinate system.

Switching two rows

Now we consider the elementary matrix

$$E = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.140}$$

for which our new coordinate system becomes

$$\mathbf{e}'_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e}'_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{e}'_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.141}$$

so we have switched two of our basis vectors. This is a reflection of our coordinate system, moving from a right-handed system to a left-handed system. **Insert picture here**

Rotation of a coordinate system

Rotation of a coordinate system can be considered a consequence of elementary matrix operations. Consider first the operation $r_1 \to r_1 - ar_2$, then $r_2 \to (1+a^2)r_2$ and then $r_2 \to r_2 + ar_1$, we have

$$\hat{E} = E_3E_2E_1 = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1+a^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.142}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & 0 \\ 0 & 1+a^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.143}$$

$$= \begin{pmatrix} 1 & -a & 0 \\ a & -a^2+1+a^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -a & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{4.144}$$

We then apply two row (or column) operators to normalise the values, $r_1 \to \frac{1}{\sqrt{1+a^2}} r_1$ and $r_2 \to \frac{1}{\sqrt{1+a^2}} r_2$, giving

$$E = \begin{pmatrix} \frac{1}{\sqrt{1+a^2}} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{1+a^2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \hat{E} = \begin{pmatrix} \frac{1}{\sqrt{1+a^2}} & -\frac{a}{\sqrt{1+a^2}} & 0 \\ \frac{a}{\sqrt{1+a^2}} & \frac{1}{\sqrt{1+a^2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.145}$$

Applying this to the Cartesian basis vectors gives

$$\mathbf{e}_1' = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} 1 \\ a \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2' = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} -a \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{e}_3' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.146}$$

and it is straightforward to check that these are all unit vectors, and also orthogonal.

If we consider what these basis vectors look like (sticking to a 2D plane, and assuming $\mathbf{e}_3'$ points into the page), we get **include picture**.

It is clear that our new basis vectors are a rotation of our original basis vectors. The matrix E in equation (4.145) is, in fact, just a different way of writing the standard rotation matrix about the z-axis. From figure **ref fig**, it is clear that the angle between the x-axis and $\mathbf{e}_1'$, θ, obeys

$$\sin\theta = \frac{a}{\sqrt{1+a^2}}, \qquad \cos\theta = \frac{1}{\sqrt{1+a^2}} \tag{4.147}$$

hence

$$E = \begin{pmatrix} \frac{1}{\sqrt{1+a^2}} & -\frac{a}{\sqrt{1+a^2}} & 0 \\ \frac{a}{\sqrt{1+a^2}} & \frac{a}{\sqrt{1+a^2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{4.148}$$

It might be worth mentioning how we derived the series of elementary matrix operations which give a rotation matrix. These may not be immediately easy to spot, and, in fact, it is easier to work backwards. To do this, though, we note that the z-components of the calculation above are entirely unchanged, so we shall simplify things to 2D.

By considering a single operation, where we add on a multiple of one row to another, as shown in figure **ref fig**, it is clear that if we normalise the resultant vector, we have a rotation of one of the standard Cartesian basis vectors (for convenience, we assume that it is a rotation of $\mathbf{e_1}$. **Add new picture?** From this, it is straightforward to identify a second, orthogonal vector corresponding to a rotation of $\mathbf{e}_2$ by the same angle. This identifies that our rotation would have the basis vectors

$$\mathbf{e}'_1 = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} 1 \\ a \end{pmatrix}, \qquad \mathbf{e}'_2 = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} -a \\ 1 \end{pmatrix}. \tag{4.149}$$

Because we have identified these as transformations, we know there is some matrix, E, which obeys

$$E\mathbf{e}_1 = \mathbf{e}'_1, \qquad E\mathbf{e}_2 = \mathbf{e}'_2. \tag{4.150}$$

We could write this explicitly as

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} 1 \\ a \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} -a \\ 1 \end{pmatrix} \tag{4.151}$$

which we can trivially solve for

$$\alpha = \frac{1}{\sqrt{1+a^2}}, \quad \beta = \frac{-a}{\sqrt{1+a^2}}, \quad \gamma = \frac{a}{\sqrt{1+a^2}}, \quad \delta = \frac{1}{\sqrt{1+a^2}} \tag{4.152}$$

or

$$E = \frac{1}{\sqrt{1+a^2}} \begin{pmatrix} 1 & -a \\ a & 1 \end{pmatrix}, \tag{4.153}$$

exactly as expected in equation (4.145). Now, we need to identify the elementary operations which give us this matrix. Recall that the matrix E is obtained by applying the elementary operations to the identity matrix; and

whilst this might be something you can identify, it might be tricky (in particular, working out you need the $1 + a^2$ multiplication operation. Instead, however, if we consider finding the inverse of E, we could do this by performing the row operations which go from E to I; reversing these operations would then give us the desired result. Here, we do this thorugh Gauss-Jordan elimination:

$$\left(\begin{array}{cc|cc} 1 & -a & 1 & 0 \\ a & 1 & 0 & 1 \end{array}\right) \tag{4.154}$$

$$r_2 \to r_2 - ar_1: \quad \left(\begin{array}{cc|cc} 1 & -a & 1 & 0 \\ 0 & 1+a^2 & -a & 1 \end{array}\right) \tag{4.155}$$

$$r_2 \to \frac{1}{1+a^2} r_1: \quad \left(\begin{array}{cc|cc} 1 & -a & 1 & 0 \\ 0 & 1 & -\frac{a}{1+a^2} & \frac{1}{1+a^2} \end{array}\right) \tag{4.156}$$

$$r_1 \to r_1 + ar_2: \quad \left(\begin{array}{cc|cc} 1 & 0 & 1+\frac{a^2}{1+a^2} & \frac{a}{1+a^2} \\ 0 & 1 & -\frac{a}{1+a^2} & \frac{1}{1+a^2} \end{array}\right) \tag{4.157}$$

The eventual form of the inverse is unimportant here, what we know is that to obtain E through row operations, we reverse these operations, from bottom to top, i.e. $r_1 \to r_1 - ar_2$, $r_2 \to (1 + a^2)r_2$ and $r_2 \to r_2 + ar_1$, as we saw earlier.

4.7 Eigenvalues and eigenvectors

The standard definition encountered for eigenvalues, λ, and eigenvectors, $\mathbf{v}$, is that they obey the equation

$$A\mathbf{v} = \lambda\mathbf{v}. \tag{4.158}$$

We typically also have the requirement that $\mathbf{v} \neq \mathbf{0}$, since this equation is always satisfied by the zero vector.

Having considered the fact that matrix multiplication can be considered as a geometrical operation, what we are saying is that $\mathbf{v}$ is a vector which, when multiplied by A, does not change direction.

From this definition, we might also ask, what about vectors which obey

$$\mathbf{w}^T A = \lambda\mathbf{w}^T? \tag{4.159}$$

These differ from the vectors $\mathbf{v}$ in that they are row vectors, not column vectors (we have used the transpose notation to make sure this is clear). In general,

$$\mathbf{v} \neq \mathbf{w}, \tag{4.160}$$

but if we consider

$$\left(\mathbf{w}^T A\right)^T = A^T \left(\mathbf{w}^T\right)^T = A^T \mathbf{w} = \left(\lambda \mathbf{w}^T\right)^T = \lambda \mathbf{w}, \tag{4.161}$$

then if $A = A^T$ (i.e. A is symmetric), we do have $\mathbf{v} = \mathbf{w}$.

These two types of eigenvector are known as **left eigenvectors** ($\mathbf{w}^T$) and **right eigenvectors** ($\mathbf{v}$), with the name coming from which side of A they appear. If a reference talks about eigenvectors without specifying left or right, then it can be assumed they mean right eigenvectors; they are the most commonly encountered ones.

4.7.1 Finding eigenvalues and eigenvectors

In order to find these quantities, we first note that equation (4.158) can also be written as

$$(A - \lambda I)\mathbf{v} = \mathbf{0}. \tag{4.162}$$

To identify these solutions, we'll consider the 2×2 matrix, B, which obeys

$$B\mathbf{x} = \mathbf{0} \qquad \text{or} \qquad \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{4.163}$$

Now, expanding out the left hand side gives two equations,

$$b_{11}x_1 + b_{12}x_2 = 0 \tag{4.164}$$

$$b_{21}x_1 + b_{22}x_2 = 0 \tag{4.165}$$

which, if we consider the matrix B to comprise two column vectors, $\mathbf{b}_{(1)} = (b_{11}, b_{21})^T$, and $\mathbf{b}_{(2)} = (b_{12}, b_{22})^T$, then this expression is

$$x_1\mathbf{b}_{(1)} + x_2\mathbf{b}_{(2)} = \mathbf{0}. \tag{4.166}$$

Recall from section 4.1.1 that if $\mathbf{b}_{(1)}$ and $\mathbf{b}_{(2)}$ are linearly independent, then the only solutions to this equation are $x_1 = x_2 = 0$. Therefore, in order for equation (4.163) to have a non-trivial solution, we must have the two vectors being linearly dependent. This also clearly generalises to arbitrary $N \times N$ matrices.

We now note that in section 4.5.4, we identify that matrices with linearly dependent rows or columns always have zero determinant. Therefore equation (4.163) only has non-trivial solutions if

$$\det(B) = 0. \tag{4.167}$$

Returning to our equation for eigenvalues and eigenvectors, we only have solutions to equation (4.158) if

$$\det(A - \lambda I) = 0. \tag{4.168}$$

Note that we also obtain the same expression if we consider the expression for the left eigenvectors, where we have

$$\mathbf{w}^T(A - \lambda I) = \mathbf{0}, \tag{4.169}$$

the only difference is that we show the matrix must be linearly dependent by considering a matrix to comprise row vectors, not column vectors. Therefore, both left and right eigenvectors of a matrix have the same eigenvalues.

For an $N \times N$ matrix, λI will contain λ N times, so the determinant, $\det(A - \lambda I)$, will be a polynomial in λ of order N. Therefore, the matrix will have N eigenvalues which might be real or complex, and there is no reason they are all unique; repeated eigenvalues are often encountered. This polynomial is sometimes called the characteristic polynomial, and we can find eigenvalues by solving

$$\det(A - \lambda I) = P(\lambda) = 0. \tag{4.170}$$

Note that the equation

$$\det(A - \lambda I) = P(\lambda) \tag{4.171}$$

holds for any value of λ, not just eigenvalues (it is only zero for eigenvalues). However, once we know the eigenvalues, we can write the characteristic polynomial as

$$P(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)(\lambda_3 - \lambda)\cdots(\lambda_N - \lambda). \tag{4.172}$$

If we now choose $\lambda = 0$, then we see from these two equations that

$$\det(A) = P(0) = \lambda_1\lambda_2\lambda_3\cdots\lambda_N, \tag{4.173}$$

so the determinant of a matrix is the product of its eigenvalues.

For each eigenvalue, we can compute an eigenvector. Because we have a linearly dependent matrix, however, we know that this eigenvector will not be unique, but we may be able to define, for example, a unique eigenvector with unit magnitude. We shall look at a few examples to clarify this.

Unique eigenvalues

Consider the matrix

$$A = \begin{pmatrix} 0 & -4 & -3 \\ -3 & 1 & 3 \\ -2 & 1 & 4 \end{pmatrix} \tag{4.174}$$

The eigenvalues of this matrix are found by solving

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & -4 & -3 \\ -3 & 1-\lambda & 3 \\ -2 & 1 & 4-\lambda \end{vmatrix} = 0 \tag{4.175}$$

which gives

$$-\lambda^3 + 5\lambda^2 + 17\lambda - 21 = 0 \tag{4.176}$$

or

$$(\lambda - 1)(\lambda + 3)(\lambda - 7) = 0. \tag{4.177}$$

This matrix has eigenvalues

$$\lambda_1 = 1, \qquad \lambda_2 = -3, \qquad \lambda_3 = 7 \tag{4.178}$$

and to find the eigenvectors, we consider each each eigenvalue separately. Starting with $\lambda_1 = 1$, we want to find

$$(A - \lambda_1 I)\,\mathbf{x} = \begin{pmatrix} -1 & -4 & -3 \\ -3 & 0 & 3 \\ -2 & 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0} \tag{4.179}$$

or

$$-x - 4y - 3z = 0 \tag{4.180}$$
$$-3x + 3z = 0 \tag{4.181}$$
$$-2x + y + 3z = 0 \tag{4.182}$$

The central equation gives

$$x = z \tag{4.183}$$

and substituting this into the first, we have

$$-x - 4y - 3z = -4x - 4y = 0 \tag{4.184}$$

so we have

$$x = -y. \tag{4.185}$$

As we would hope, this then satisfies the final equation,

$$-2x + y + 3z = -2x - x + 3x = 0. \tag{4.186}$$

This means any vector of the form

$$\mathbf{a}_1 = \begin{pmatrix} x \\ -x \\ x \end{pmatrix} \tag{4.187}$$

would satisfy the eigenvector equation, and typically we do not include the x term, and state our eigenvector as

$$\mathbf{r_1} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \qquad \text{or} \qquad \hat{\mathbf{r}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \tag{4.188}$$

where the second of these expressions has been normalised. Note also that we could also have

$$\tilde{\mathbf{r}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \tag{4.189}$$

which would still be a correct, normalised eigenvector. The choice of sign only matters if you are trying to use your three eigenvectors to form a coordinate system, in which case you need to know whether you have a right-handed or left-handed system (and it will be the relative signs between the eigenvectors which matters).

For this system, we can also compute two other eigenvectors, for $\lambda = -3$, we have

$$\hat{\mathbf{r}}_2 = \frac{1}{\sqrt{35}} \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \tag{4.190}$$

and for $\lambda = 7$, we have

$$\hat{\mathbf{r}}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}. \tag{4.191}$$

We could also find the left eigenvectors, e.g. for $\lambda = 1$, we would solve

$$\mathbf{x}^T (A - \lambda I) = (x, y, z) \begin{pmatrix} -1 & -4 & -3 \\ -3 & 0 & 3 \\ -2 & 1 & 3 \end{pmatrix} = \mathbf{0}. \tag{4.192}$$

Solving the left eigenvectors for this system would give us

$$\lambda_1 = 1, \qquad \hat{\mathbf{l}}_1 = \frac{1}{\sqrt{26}}(1, -3, 4) \tag{4.193}$$

$$\lambda_2 = -3, \qquad \hat{\mathbf{l}}_2 = \frac{1}{\sqrt{2}}(1, 1, 0) \tag{4.194}$$

$$\lambda_3 = 7, \qquad \hat{\mathbf{l}}_3 = \frac{1}{\sqrt{6}}(1, -1, -2) \tag{4.195}$$

$$\tag{4.196}$$

Unique eigenvalues, diagonal matrices

For a diagonal, upper triangular or lower triangular matrix, the eigenvalues of this matrix are very simple to calculate. Consider the definition of determinant for a 3×3 matrix,

$$\det(A) = \epsilon_{ijk} A_{1i} A_{2j} A_{3k} \tag{4.197}$$

and if A is upper triangular, then

$$A_{ij} = \begin{cases} a_{ij} & i \leq j \\ 0 & i > j \end{cases}. \tag{4.198}$$

Note that this means the determinant is only non-zero if $k = 3$,

$$\det(A) = \epsilon_{ij3} A_{1i} A_{2j} A_{33}. \tag{4.199}$$

We also need $j \geq 2$ for a non-zero determinant, but for ε_{ij3} to be non-zero, we need $j \neq 3$, hence we must have $j = 2$,

$$\det(A) = \epsilon_{i23} A_{1i} A_{22} A_{33}. \tag{4.200}$$

It is now clear that the only $i = 1$ makes the Levi-Civita symbol non-vanishing, hence, for a diagonal matrix,

$$\det(A) = \epsilon_{123} A_{11} A_{22} A_{33} = A_{11} A_{22} A_{33}. \tag{4.201}$$

This logic extends naturally to $N \times N$ matrices; and also to lower triangular matrices (and, of course, fully diagonal matrices). In all these cases, we have

$$\det(A_{\text{tri}}) = A_{11} A_{22} A_{33} \dots A_{NN}. \tag{4.202}$$

When it comes to eigenvectors, things are just as straightforward; $A-\lambda I$ only changes the diagonal entries of the matrix, it does not change whether the matrix is upper or lower triangular. In this case we have

$$\det(A-\lambda I)=(a_{11}-\lambda_1)(a_{22}-\lambda_2)(a_{33}-\lambda_3)\cdots(a_{NN}-\lambda_N) \tag{4.203}$$

and the eigenvalues of the matrix are simply the diagonal entries of the matrix.
This means that if you don't want to compute the determinant directly, you can use rwo and column operations to simplify things. We shall use the matrix in equation (4.175) as an example, although it is not a very practical example (this approach is generally best for large matrices, where the characteristic polynomial would be challenging to factorise). However, for this example, we have:

$$\det(A-\lambda I)=\begin{vmatrix}-\lambda & -4 & -3\\ -3 & 1-\lambda & 3\\ -2 & 1 & 4-\lambda\end{vmatrix} \tag{4.204}$$

$$\begin{array}{l}r_2\to r_2-\frac{3}{\lambda}r_1:\\ r_3\to r_3-\frac{2}{\lambda}r_1:\end{array}\quad=\begin{vmatrix}-\lambda & -4 & -3\\ 0 & 1-\lambda+\frac{12}{\lambda} & 3+\frac{9}{\lambda}\\ 0 & 1+\frac{8}{\lambda} & 4-\lambda+\frac{6}{\lambda}\end{vmatrix} \tag{4.205}$$

$$\begin{array}{l}r_2\to\lambda r_2:\\ r_3\to\lambda r_3:\end{array}\quad=\frac{1}{\lambda^2}\begin{vmatrix}-\lambda & -4 & -3\\ 0 & -\lambda^2+\lambda+12 & 3\lambda+9\\ 0 & \lambda+8 & -\lambda^2+4\lambda+6\end{vmatrix} \tag{4.206}$$

$$\begin{array}{l}r_1\to\frac{1}{\lambda}r_1:\\ r_3\to r_3-\frac{\lambda+8}{-\lambda^2+\lambda+12}r_2:\end{array} \tag{4.207}$$

$$=\frac{1}{\lambda}\begin{vmatrix}-1 & -\frac{4}{\lambda} & -\frac{3}{\lambda}\\ 0 & -\lambda^2+\lambda+12 & 3\lambda+9\\ 0 & 0 & -\lambda^2+4\lambda+6-\frac{(\lambda+8)(3\lambda+9)}{-\lambda^2+\lambda+12}\end{vmatrix} \tag{4.208}$$

$$\begin{array}{l}r_2\to\frac{1}{-\lambda^2+\lambda+12}r_2:\\ r_3\to(-\lambda^2+\lambda+12)r_3:\end{array} \tag{4.209}$$

$$=\frac{1}{\lambda}\begin{vmatrix}-1 & -\frac{4}{\lambda} & -\frac{3}{\lambda}\\ 0 & 1 & \frac{3\lambda+9}{-\lambda^2+\lambda+12}\\ 0 & 0 & (-\lambda^2+4\lambda+6)(-\lambda^2+\lambda+12)-(\lambda+8)(3\lambda+9)\end{vmatrix} \tag{4.210}$$

Note that here, when multiplying rows by a constant, we have been careful not to change the overall value of the determinant, according to the rules of elementary operations. Whilst this is a bit of a mess, we actually find that

the determinant (or the trace) of this right triangular matrix is

$$-\frac{1}{\lambda}\left[(-\lambda^2+4\lambda+6)(-\lambda^2+\lambda+12)-(\lambda+8)(3\lambda+9)\right]=(-\lambda^3+5\lambda^2+17\lambda-21) \tag{4.211}$$

as we would expect.

Linearly dependent matrices

Recall that the determinant of a matrix is the product of its eigenvalues,

$$\det(A)=P(0)=\lambda_1\lambda_2\lambda_3\cdots\lambda_N. \tag{4.212}$$

This then means that if the matrix had linearly dependent rows or columns, which we know means that $\det(A)=0$, then at least one of the eigenvalues must be zero in order for this equation to hold. However, this does not prevent you from finding eigenvectors (left or right) of this matrix.

Repeated eigenvalues

Consider the matrix

$$A=\begin{pmatrix}5&-2&0\\0&4&0\\3&-5&5\end{pmatrix} \tag{4.213}$$

from which we have eigenvalues

$$\begin{vmatrix}5-\lambda&-2&0\\0&4-\lambda&0\\3&-5&5-\lambda\end{vmatrix}=(5-\lambda)(4-\lambda)(5-\lambda). \tag{4.214}$$

This means we have repeated eigenvalues,

$$\lambda_1=4,\qquad \lambda_{2,3}=5. \tag{4.215}$$

We can still find eigenvectors for this matrix, though. For the non-repeated eigenvalue, we have

$$\lambda_1=4:\qquad \begin{pmatrix}1&-2&0\\0&0&0\\3&-5&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\mathbf{0} \tag{4.216}$$

The top row of this system tells us

$$x-2y=0,\qquad \Longrightarrow\ x=2y \tag{4.217}$$

and then the third row tells us

$$3x - 5y + z = 6y - 5y + z = y + z = 0 \qquad \Longrightarrow \qquad z = -y. \tag{4.218}$$

The unit normal eigenvector associated with λ_1 is then

$$\hat{\mathbf{r}}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}. \tag{4.219}$$

Note that the row of zeroes in the matrix doesn't cause any issues at all.

For the repeated eigenvalue we have

$$\lambda_{2,3} = 5 : \qquad \begin{pmatrix} 0 & -2 & 0 \\ 0 & -1 & 0 \\ 3 & -5 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}. \tag{4.220}$$

Here, both the top and middle rows tell us that

$$y = 0 \tag{4.221}$$

and the bottom row then tells us

$$3x - 5y = 3x = 0 \qquad \Longrightarrow \qquad x = 0. \tag{4.222}$$

This then means that the z component of the eigenvector has no constraints, i.e. we have

$$\hat{\mathbf{r}}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{4.223}$$

So for this system, we have only two eigenvectors (we also find there are only two left eigenvectors).

As a second example, consider the matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \tag{4.224}$$

which has eigenvalues

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda \left[\lambda^2 - 1\right] - (-\lambda - 1) + (1 + \lambda) = -\lambda^3 + 3\lambda + 2 = 0. \tag{4.225}$$

It is clear that $\lambda = -1$ is a solution here, and we find that

$$-\lambda^3 + 3\lambda + 2 = -(\lambda - 2)(\lambda + 1)^2 \tag{4.226}$$

so

$$\lambda_1 = 2, \qquad \lambda_{2,3} = -1. \tag{4.227}$$

Now, the eigenvector for λ_1 is

$$\lambda_1 = 1 : \qquad \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}. \tag{4.228}$$

From the top row, we have

$$-2x + y + z = 0 \qquad \Longrightarrow \qquad z = 2x - y \tag{4.229}$$

and from the second row we have

$$x - 2y + z = x - 2y + 2x - y = 3x - 3y = 0 \qquad \Longrightarrow \qquad y = x \tag{4.230}$$

from which we find

$$z = 2x - y = 2x - x = x \tag{4.231}$$

hence we have eigenvector

$$\hat{\mathbf{r}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \tag{4.232}$$

For $\lambda_{2,3}$ we have

$$\lambda_{2,3} = -1 : \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}. \tag{4.233}$$

From any one of these equations, we have

$$x + y + z = 0 \qquad \Longrightarrow \qquad z = -x - y \tag{4.234}$$

but this is all we can do. This says that any vector

$$\mathbf{v} = \begin{pmatrix} x \\ y \\ -x - y \end{pmatrix} \tag{4.235}$$

satisfies equation (4.233). However, we can write this as

$$\begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \tag{4.236}$$

where x and y can take any value. Since 'any value' includes $x = 0$ or $y = 0$, we can say we have two eigenvectors,

$$\hat{\mathbf{r}}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \hat{\mathbf{r}}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}. \tag{4.237}$$

For an eigenvalue repeated n times, we find that we have at least one eigenvector, and may have up to n eigenvectors. This has consequences for other properties of the matrix which we shall consider shortly.

Identity matrix

It is unlikely that you would need to find eigenvectors for the identity matrix, but it is possible. Firstly, we note that the eigenvectors for the identity matrix must obey

$$A\mathbf{v} = \lambda\mathbf{v} \qquad \text{or} \qquad I\mathbf{v} = \mathbf{v} = \lambda\mathbf{v} \tag{4.238}$$

hence it is clear that $\lambda = 1$. This is also obvious because it is a diagonal matrix, and therefore the diagonal entries are eigenvalues. So from this, we have

$$\lambda_{1,2,3} = 1 \tag{4.239}$$

and hence eigenvectors must obey

$$\begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}. \tag{4.240}$$

so all vectors are eigenvectors. Again, this is not surprising, since the relationship

$$I\mathbf{v} = \lambda\mathbf{v} = \mathbf{v} \tag{4.241}$$

is simply the definition of the identity matrix; all vectors will obey this. We can, however, find three linearly independent eigenvectors, the most obvious choice being

$$\hat{\mathbf{r}}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{\mathbf{r}}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{\mathbf{r}}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{4.242}$$

4.7.2 Linear independence of eigenvectors

When we were considering eigenvectors, we started to talk about linear independence of the vectors. You will find that all eigenvectors in the last section are linearly independent. In the case of repeated eigenvectors, the decomposition of the vector which satisfies $A\mathbf{v} = \lambda\mathbf{v}$ was chosen such that the component vectors were linearly independent. Note here that equation (4.235) describes the equation of a plane, so we know we can find two perpendicular vectors on that plane (i.e. they linearly independent), and equation (4.241) was simply all space, so finding three perpendicular vectors is also straightforward (and the Cartesian axes chosen are the most obvious ones).

It can also be shown that any eigenvectors from different eigenvalues will also be linearly independent. Here, we demonstrate this for three dimensions, but the proof generalises. We start by assuming that matrix A has three distinct eigenvalues, $\lambda_1 \neq \lambda_2 \neq \lambda_3$, and three eigenvectors, $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{r}_3$. We then assume that these eigenvectors are linearly dependent (which we will eventually show is a contradiction), which means

$$\mathbf{r}_3 = a_1\mathbf{r}_1 + a_2\mathbf{r_2}. \tag{4.243}$$

We now multiply both sides of this equation with A, and use the definition of an eigenvector

$$A\mathbf{r}_3 = a_1A\mathbf{r}_1 + a_2A\mathbf{r_2} \quad \implies \lambda_3\mathbf{r_3} = a_1\lambda_1\mathbf{r_1} + a_2\lambda_2\mathbf{r_2}. \tag{4.244}$$

However, we could also multiply the entirety of equation (4.243) by λ_3, giving

$$\lambda_3\mathbf{r}_3 = a_1\lambda_3\mathbf{r}_1 + a_2\lambda_3\mathbf{r_2}. \tag{4.245}$$

Subtracting these last two equations gives

$$a_1(\lambda_1 - \lambda_3)\mathbf{r}_1 + a_2(\lambda_2 - \lambda_3)\mathbf{r_2} = 0. \tag{4.246}$$

There are now two possible cases here:

1. $\mathbf{r}_1$ and $\mathbf{r}_2$ are linearly independent - this means (since each eigenvalue is distinct) that this equation can only be satisfied if

$$a_1 = a_2 = 0 \tag{4.247}$$

2. $\mathbf{r}_1$ and $\mathbf{r}_2$ are linearly dependent, which means $\mathbf{r}_2 = \alpha\mathbf{r}_1$. In this case,

we repeat the steps above:

$$\mathbf{r}_3 = a\mathbf{r}_1 \tag{4.248}$$
$$A\mathbf{r}_3 = aA\mathbf{r}_1 \tag{4.249}$$
$$\lambda_3\mathbf{r_3} = a\lambda_1\mathbf{r_1} \tag{4.250}$$
$$\lambda_3\mathbf{r_3} = a\lambda_3\mathbf{r_1} \tag{4.251}$$
$$a(\lambda_1 - \lambda_3)\mathbf{r_1} = 0 \tag{4.252}$$
$$a = 0 \tag{4.253}$$

In either case, however, equation (4.243) becomes

$$\mathbf{r}_3 = \mathbf{0} \tag{4.254}$$

which means either our assumption that $\mathbf{r}_3$ is a non-zero vector is false (and we wouldn't have found it as a solution to our eigenvector equation), or equation (4.243) is not true, and all our eigenvectors are linearly independent.

4.7.3 Diagonalisation

For a given $N \times N$ matrix, A, with eigenvalues λ_i, we know that a diagonal matrix with entries λ_i, often called Λ, would have the same eigenvalues. It may be possible to find a relationship between A and Λ, and we find that this relationship is related to the eigenvectors of A.

Consider a matrix with columns comprising the right eigenvectors of A,

$$S = (\mathbf{r}_1|\mathbf{r}_2|\mathbf{r}_3|\dots|\mathbf{r}_N)\,. \tag{4.255}$$

If we now consider the product AS, note that the entries in each column of this vector are simply the matrix A multiplied by the the column vector, i.e.

$$AS = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & \vdots & \ddots & \\ a_{N1} & a_{N2} & & a_{NN} \end{pmatrix} \begin{pmatrix} (\mathbf{r}_1)_1 & (\mathbf{r}_2)_1 & \cdots & (\mathbf{r}_N)_1 \\ (\mathbf{r}_1)_2 & (\mathbf{r}_2)_2 & \cdots & (\mathbf{r}_N)_2 \\ \vdots & \vdots & \ddots & \\ (\mathbf{r}_1)_N & (\mathbf{r}_N)_N & & (\mathbf{r}_N)_N \end{pmatrix} \tag{4.256}$$

$$= \begin{pmatrix} a_{11}(\mathbf{r}_1)_1 + a_{12}(\mathbf{r}_1)_2 + \cdots + a_{1N}(\mathbf{r}_1)_N & \cdots \\ a_{21}(\mathbf{r}_1)_1 + a_{22}(\mathbf{r}_1)_2 + \cdots + a_{2N}(\mathbf{r}_1)_N & \cdots \\ \vdots & \ddots \\ a_{N1}(\mathbf{r}_1)_1 + a_{N2}(\mathbf{r}_1)_2 + \cdots + a_{NN}(\mathbf{r}_1)_N & \cdots \end{pmatrix} \tag{4.257}$$

$$= \begin{pmatrix} a_{1i}(\mathbf{r}_1)_i & a_{1i}(\mathbf{r}_2)_i & \cdots & a_{1i}(\mathbf{r}_N)_i \\ a_{2i}(\mathbf{r}_1)_i & a_{2i}(\mathbf{r}_2)_i & \cdots & a_{2i}(\mathbf{r}_N)_i \\ \vdots & \vdots & \ddots & \\ a_{Ni}(\mathbf{r}_1)_i & a_{Ni}(\mathbf{r}_2)_i & \cdots & a_{Ni}(\mathbf{r}_N)_i \end{pmatrix}. \tag{4.258}$$

The use of index notation in the entries of this matrix is a little bit informal, but it allows everything to be written neatly within the matrix. However, note that the multiplication of a vector by a matrix, in index form is

$$B\mathbf{v} = B_{ij}v_j \tag{4.259}$$

hence the columns of this matrix are the result of a vector-matrix multiplication, i.e.

$$AS = (A\mathbf{r}_1|A\mathbf{r}_2|A\mathbf{r}_3|\cdots|A\mathbf{r}_N)\,. \tag{4.260}$$

and through equation (4.158), we have

$$AS = (\lambda_1\mathbf{r}_1|\lambda_2\mathbf{r}_2|\lambda_3\mathbf{r}_3|\cdots|\lambda_N\mathbf{r}_N)\,. \tag{4.261}$$

We also consider the multiplication $S\Lambda$,

$$S\Lambda = \begin{pmatrix} (\mathbf{r}_1)_1 & (\mathbf{r}_2)_1 & \cdots & (\mathbf{r}_N)_1 \\ (\mathbf{r}_1)_2 & (\mathbf{r}_2)_2 & \cdots & (\mathbf{r}_N)_2 \\ \vdots & \vdots & \ddots & \\ (\mathbf{r}_1)_N & (\mathbf{r}_N)_N & & (\mathbf{r}_N)_N \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & & \lambda_N \end{pmatrix} \tag{4.262}$$

$$= \begin{pmatrix} \lambda_1(\mathbf{r}_1)_1 & \lambda_2(\mathbf{r}_2)_1 & \cdots & \lambda_N(\mathbf{r}_N)_1 \\ \lambda_1(\mathbf{r}_1)_2 & \lambda_2(\mathbf{r}_2)_2 & \cdots & \lambda_N(\mathbf{r}_N)_2 \\ \vdots & \vdots & \ddots & \\ \lambda_1(\mathbf{r}_1)_N & \lambda_2(\mathbf{r}_2)_N & \cdots & \lambda_N(\mathbf{r}_N)_N \end{pmatrix}. \tag{4.263}$$

We see that this can be written as

$$S\Lambda = (\lambda_1\mathbf{r}_1|\lambda_2\mathbf{r}_2|\lambda_3\mathbf{r}_3|\cdots|\lambda_N\mathbf{r}_N) \tag{4.264}$$

hence we have

$$AS = S\Lambda \tag{4.265}$$

or

$$A = S\Lambda S^{-1} \tag{4.266}$$

or

$$\Lambda = S^{-1}AS. \tag{4.267}$$

We now note that this definition started with an assumption in equation (4.255) that there were N eigenvectors for our $N \times N$ matrix. As we saw previously, this is not always the case. As such, not all matrices are diagonalisable.

We demonstrated diagonalisability with the right eigenvectors, but we could just have easily used the left eigenvectors. If we have N right eigenvectors, then we also have N left eigenvectors, and we could consider a matrix

$$T = \begin{pmatrix} \mathbf{l_1} \\ \hline \mathbf{l_2} \\ \hline \mathbf{l_3} \\ \hline \vdots \\ \hline \mathbf{l_N} \end{pmatrix}. \tag{4.268}$$

By the same set of operations as used for the right eigenvector case, we can show that

$$TA = \begin{pmatrix} \lambda_1\mathbf{l_1} \\ \hline \lambda_2\mathbf{l_2} \\ \hline \lambda_3\mathbf{l_3} \\ \hline \vdots \\ \hline \lambda_N\mathbf{l_N} \end{pmatrix} \tag{4.269}$$

and also

$$\Lambda T = \begin{pmatrix} \lambda_1\mathbf{l_1} \\ \hline \lambda_2\mathbf{l_2} \\ \hline \lambda_3\mathbf{l_3} \\ \hline \vdots \\ \hline \lambda_N\mathbf{l_N} \end{pmatrix}. \tag{4.270}$$

Therefore, in this case, we have

$$A = T^{-1}\Lambda T \tag{4.271}$$

or

$$\Lambda = TAT^{-1}. \tag{4.272}$$

From the two expressions using left and right eigenvectors, $A = S\Lambda S^{-1}$ and $A = T^{-1}\Lambda T$, we would expect there to be a relationship between T and S. Clearly, if $T = S^{-1}$, then

$$A = T^{-1}\Lambda T = (S^{-1})^{-1}\Lambda S^{-1} = S\Lambda S^{-1}, \tag{4.273}$$

however, we can't state this in general. Note that the matrices S and T are themselves not unique; firstly we can switch columns of S, or rows of T, and still have a matrix comprising the eigenvectors of A and secondly, each eigenvector is only defined up to a constant factor. In order for $T = S^{-1}$ to hold, we need:

1. That column i of S and the row i of T corresponds to the same eigenvalue, λ_i

2. The eigenvectors are appropriately normalised

Here, appropriately normalised does not necessarily mean unit magnitude. For example, consider the eigenvectors found in section 4.7.1,

$$\hat{\mathbf{r}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \qquad \hat{\mathbf{r}}_2 = \frac{1}{\sqrt{35}} \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \qquad \hat{\mathbf{r}}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \tag{4.274}$$

and

$$\hat{\mathbf{l}}_1 = \frac{1}{\sqrt{26}} (1, -3, 4), \qquad \hat{\mathbf{l}}_2 = \frac{1}{\sqrt{2}} (1, 1, 0), \qquad \hat{\mathbf{l}}_3 = \frac{1}{\sqrt{6}} (1, -1, -2). \tag{4.275}$$

If, instead, we scale the right eigenvalues as

$$\bar{\mathbf{r}}_1 = \frac{\sqrt{26}}{8} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \qquad \bar{\mathbf{r}}_2 = \frac{\sqrt{2}}{8} \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \qquad \bar{\mathbf{r}}_3 = \frac{\sqrt{6}}{8} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \tag{4.276}$$

we will find that

$$\begin{pmatrix} \hat{\mathbf{l}}_1 \\ \hline \hat{\mathbf{l}}_2 \\ \hline \hat{\mathbf{l}}_3 \end{pmatrix} (\bar{\mathbf{r}}_1 | \bar{\mathbf{r}}_2 | \bar{\mathbf{r}}_3) = I. \tag{4.277}$$

Add use of diagonalisation for matrix powers here

4.7.4 Similar matrices

In section 4.7.3, we obtained our diagonal matrix through e.g.

$$\Lambda = SAS^{-1}. \tag{4.278}$$

However, more generally, if we have an invertible matrix, B, then we could define

$$C = BAB^{-1} \tag{4.279}$$

and any two matrices A and C related this way are known as **similar matrices**.

Similar matrices have several convenient properties, which we may need to use:

1. Similar matrices have the same determinant:

$$\begin{aligned}\det(C) =& \det(BAB^{-1}) \\ =& \det(B)\det(A)\det(B^{-1}) \\ =& \det(B)\det(B^{-1})\det(A) \\ =& \det(BB^{-1})\det(A) \\ =& \det(I)\det(A) \\ =& \det(A)\end{aligned} \tag{4.280}$$

2. Similar matrices have the same characteristic polynomial, and hence the same eigenvalues

$$\begin{aligned}\det(C-\lambda I) =& \det(BAB^{-1}-\lambda I) \\ =& \det(BAB^{-1}-\lambda BIB^{-1}) \\ =& \det(B(A-\lambda I)B^{-1}) \\ =& \det(B)\det(A-\lambda I)\det(B^{-1}) \\ =& \det(B)\det(B^{-1})\det(A-\lambda I) \\ =& \det(BB^{-1})\det(A-\lambda I) \\ =& \det(A-\lambda I)\end{aligned} \tag{4.281}$$

3. Similar matrices do not have the same eigenvectors, but they are related by a single matrix transform

$$C\mathbf{v}=\lambda\mathbf{v} \implies BAB^{-1}\mathbf{v}=\lambda\mathbf{v} \implies AB^{-1}\mathbf{v}=\lambda B^{-1}\mathbf{v}. \tag{4.282}$$

Here, $B^{-1}\mathbf{v}$ is a vector, hence is an eigenvector of A, and is a transform of $\mathbf{v}$, an eigenvector of C.

4.7.5 Symmetric matrices

So far, we've worked with general square matrices, but often in physical applications, we not only have square matrices, but also symmetric ones. Symmetric matrices have some additional properties which can be useful.

Real eigenvalues

So far, we have not worried about whether the eigenvalues of our matrices are real, but if you were to try and compute these quantities cas part of a numerical algorithm, it is almost certainly going to be a concern. Symmetric matrices, however, can be shown to always have real eigenvalues. If all

entries A are real, and symmetric, we can take the complex conjugate of equation (4.158),

$$A\mathbf{v} = \lambda\mathbf{v} \qquad \Longrightarrow \qquad A\mathbf{v}^* = \lambda^*\mathbf{v}^* \tag{4.283}$$

where, because A has entirely real entries, $A^* = A$, and the complex conjugate vector $\mathbf{v}^*$ simply has elements which are the complex conjugate of the elements of $\mathbf{v}$. We now multiply both sides of the complex conjugate equation with $(\mathbf{v}^*)^T$,

$$(\mathbf{v}^*)^T A\mathbf{v} = \lambda(\mathbf{v}^*)^T\mathbf{v} \tag{4.284}$$

and note that because $A = A^T$, we have

$$(\mathbf{v}^*)^T A\mathbf{v} = (\mathbf{v}^*)^T A^T\mathbf{v} = (A\mathbf{v}^*)^T\mathbf{v} = (\lambda^*\mathbf{v}^*)^T\mathbf{v} \tag{4.285}$$

where in the final statement in this line, we use equation (4.283). These two equations show that

$$\lambda(\mathbf{v}^*)^T\mathbf{v} = (\lambda^*\mathbf{v}^*)^T\mathbf{v} \tag{4.286}$$

and, since λ^* is a scalar, we can write this

$$(\lambda - \lambda^*)(\mathbf{v}^*)^T\mathbf{v} = 0 \tag{4.287}$$

The quantity $(\mathbf{v}^*)^T\mathbf{v}$ is a scalar,

$$(\mathbf{v}^*)^T\mathbf{v} = \sum_{i=1}^{N} v_i^* v_i = \sum_{i=1}^{N} |v_i|^2 . \tag{4.288}$$

Note then that since this quantity is always greater than zero (assuming, as ever, we are interested in non-trivial eigenvectors), then this implies

$$\lambda = \lambda^* \tag{4.289}$$

hence the eigenvalues are real.

Eigenvectors are orthogonal

Any two eigenvectors from two different eigenvalues for a symmetric real matrix, $\mathbf{v}_1$ and $\mathbf{v}_2$, will obey

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0. \tag{4.290}$$

Here, we start with the definition of two eigenvectors,

$$A\mathbf{v}_1 = \lambda_1\mathbf{v}_1, \qquad A\mathbf{v}_2 = \lambda_2\mathbf{v}_2. \tag{4.291}$$

If we now consider multiplying our expression for $\mathbf{v}_1$ by $(\mathbf{v}_2)^T$, we get, for the left hand side,

$$(\mathbf{v}_2)^T A\mathbf{v}_1 = (\mathbf{v}_2)^T A^T \mathbf{v}_1 = (A\mathbf{v_2})^T \mathbf{v}_1 = (\lambda_2 \mathbf{v}_2)^T \mathbf{v}_1 \tag{4.292}$$

hence we have

$$(\lambda_2 \mathbf{v}_2)^T \mathbf{v}_1 = \lambda_1 (\mathbf{v}_2)^T \mathbf{v}_1 \tag{4.293}$$

or

$$(\lambda_2 - \lambda_1)(\mathbf{v}_2)^T \mathbf{v}_1 = 0. \tag{4.294}$$

Since we started with two different eigenvalues ($\lambda_1 \neq \lambda_2$), this means that we must have

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0. \tag{4.295}$$

4.8 LU decomposition

At various times, we have discussed the use of matrices to solve simultaneous equations,

$$A\mathbf{x} = \mathbf{b}. \tag{4.296}$$

In scientific computing, the systems of equations which we need to solve can become large, e.g. considering hundreds of atoms, or discretised points on a continuum domain. This means the matrix A is very large, and, in general, attempting to perform either row reduction or to invert the matrix becomes computationally intensive. However, if A displays diagonal properties, either being upper or lower triangular, or fully diagonal, the situation is easier. For example, consider the equation

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}. \tag{4.297}$$

Firstly, because the matrix is upper triangular, we know that $\det(A) = 2$, and hence the matrix is invertible, and a unique solution exists. However, we can also immediately read that

$$z = 2. \tag{4.298}$$

This means we can quickly identify y, since we get

$$2y + z = 2y + 2 = 1 \quad \Longrightarrow \quad y = -\frac{1}{2} \tag{4.299}$$

and then, equally quickly we have

$$z + y + z = x - \frac{1}{2} + 2 = 1 \quad \Longrightarrow \quad x = -\frac{1}{2}. \tag{4.300}$$

In other words, with an upper triangular matrix, it is straightforward to solve the system sequentially. The lower triangular matrix case is similar, you just start with the first equation.

Now, for any matrix, A, we can perform elementary operations until we obtain an upper triangular matrix, U,

$$E_n E_{n-1} \cdots E_2 E_1 A = U. \tag{4.301}$$

Note here that the order we apply these operators means that we have the right-to-left ordering of the elementary matrix multiplication, and also we do not require A to be invertible for this to be true; some of the rows of U could be entirely zeroes and the matrix would still be upper triangular. We can also write this such that

$$A = (E_n E_{n-1} \cdots E_2 E_1)^{-1} U = (E_1)^{-1}(E_2)^{-1} \cdots (E_{n-1})^{-1}(E_n)^{-1} U \tag{4.302}$$

since we know all elementary operation matrices have inverses.

We shall now consider a special case - that none of the elementary row operations are switching row. This is not true for all matrices, for example

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{4.303}$$

cannot be written in upper diagonal form without switching rows. However, if it is, then

$$A = (E_1)^{-1}(E_2)^{-1} \cdots (E_{n-1})^{-1}(E_n)^{-1} U = LU \tag{4.304}$$

where L is a lower diagonal matrix. To see this, we need to consider two things. Firstly, that if no row switching operations are needed, the upper triangular form can be obtained purely by adding the multiple of one row to another. This uses the process in section 4.4.4, and not using row switching means that we never have the case where

$$(E_m E_{m-1} \cdots E_2 E_1 A)_{m+1,m+1} = 0 \tag{4.305}$$

meaning that after m operations, the next leading diagonal entry is always non-zero. If this is the case, then we are always adding a row to a lower

row, i.e. our matrix E_k will always be lower triangular, as in our example in equation 4.35.

Secondly, we note that the inverse of adding a multiple of one row to another ($r_a \to r_a + \mu r_b$), is subtracting one row from another, ($r_a \to r_a - \mu r_b$), hence if the first operation is lower triangular, then the second operation must be too. Therefore $(E_k)^{-1}$ will always be lower triangular.

Finally, we note that if two matrices, B and C are lower triangular, then BC is too. Note that we have

$$B_{ij} = \begin{cases} b_{ij} & i \geq j \\ 0 & i < j \end{cases} \qquad C_{ij} = \begin{cases} c_{ij} & i \geq j \\ 0 & i < j \end{cases} \tag{4.306}$$

and

$$(BC)_{ij} = B_{ik}C_{kj}. \tag{4.307}$$

For B and C upper triangular, this is only non-zero for the entries where $i \geq k$ and $k \geq j$, i.e. if $i \geq j$. Therefore BC must be upper triangular, or, applied to our example, $L = (E_1)^{-1}(E_2)^{-1} \cdots (E_{n-1})^{-1}(E_n)^{-1}$ is lower triangular.

Now we return to our simultaneous equations,

$$A\mathbf{x} = \mathbf{b} \tag{4.308}$$

which we can write as

$$LU\mathbf{x} = \mathbf{b}. \tag{4.309}$$

Note that the quantity $U\mathbf{x}$ is a vector, so we set

$$U\mathbf{x} = \mathbf{y} \tag{4.310}$$

and

$$L\mathbf{y} = \mathbf{b}. \tag{4.311}$$

Now because L is lower triangular, we can sequentially solve for all elements of $\mathbf{y}$, and once we have this, we can solve equation (4.310) for $\mathbf{x}$.

This is one of the key aims of LU decomposition - we can make the actual solution of simultaneous equations algebraically straightforward. Of course it did take quite a lot of effort to get here, but many physical examples exist where this might make sense. It is quite common that the matrix A might be constant, but the matrix $\mathbf{b}$ might vary. Such a system may require repeated solution of $A\mathbf{x} = \mathbf{b}$ with different $\mathbf{b}$ vectors. This would be expensive to do through row operations, or matrix inversion, every time, but converting A to its LU decomposition can be considered a one-off expense, and then the substitution to find values is relatively cheap.

4.8.1 Dealing with row switching

In the section so far, we have made one very large assumption - no row switching was necessary. This was important for maintaining lower-diagonal form of our elemetary operation matrices; row switching can easily change this, e.g. if we have

$$(E_2)^{-1}(E_1)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{\mu} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ \frac{1}{\mu} & 1 & 0 \end{pmatrix} \tag{4.312}$$

the resultant matrix is no longer lower diagonal. However, it can be shown that if we know all the row switching operations that are required to obtain row echelon form, and then apply these first as an elementary matrix operation, E_s, then we can write

$$E_s A = LU, \tag{4.313}$$

i.e. the resultant matrix has an LU decomposition. Therefore it is straightforward to solve

$$(E_s A)\mathbf{x} = E_s \mathbf{b}. \tag{4.314}$$

Because LU decomposition is an important part of scientific computing, algorithms exist for computing L and U without explicitly calculating each elementary matrix, which will be covered in other courses.

Chapter 5

Partial differential equations

In this course, we will not go into great detail about how partial differential equations (PDEs) can be solved analytically. The main aim of scientific computing is to solve equations computationally, and thus complex analytical methods are not something we need to focus on. This is not to say that they don't have their place; one of the key things we need to do for any computational method is validate it, we must demonstrate that it can get a known correct solution. And one key approach to validation is reproduction of exact solutions, for which we would need to be able to solve our physical model analytically, and thus, may need some of these techniques. The reason we don't cover these, however, is because any technique is often system-specific, and we want to cover the general concepts needed for scientific computing.

5.1 Classification of partial differential equations

When it comes to solving a system of partial differential equations, there is no single numerical technique which can be applied. This is both due to the very different physics that a system may involve, but also very different mathematics through which the governing equations are formulated. When it comes to PDEs, we are able to **classify** them, such that we can identify key mathematical behaviour. This not only governs the techniques we may then employ or develop to solve the equations, but also can check whether we might be unable to capture some of the important physics that we intended our equations to model.

One way in which we might classify a PDE is whether they apply to an equilibrium problem or a steady state problem:

- **Equilibrium problems:** These are problems to which there is a fixed solution in time, a 'balance' of all states at different locations in space. Examples include:
  - Air flow around an object in constant inflow conditions - this is used, for example, to compute lift and drag of an object
  - Minimum energy states of the atoms in a molecular configuration
  - Plasma equilibrium configurations in a tokamak reactor - stable configurations exist where pressure gradients balance with magnetic field
- **Time varying problems:** These are problems where there is no fixed solution in time, something will always be changing. Examples here include:
  - Air flow around an object as it moves through the real world - atmospheric conditions are always changing, so the air flow would change too
  - Evolution of atoms in a system according to molecular dynamics
  - Evolution of plasma instabilities in a tokamak reactor - a wide range of plasma instabilities occur in real world situations because perfect force balance is not practically possible

Whilst these terms do classify PDEs to some extent, they don't help us solve them. They do give some hints though; in a time varying problem, to capture the evolution in time, there must be some information which travels at a finite speed, which would not be necessary in a steady state problem. A better classification might be:

1. Equations where **all** information propagates infinitely fast
2. Equations where **some** information propagates infinitely fast, but some travels with a finite speed
3. Equations where **all** information travels at a finite speed

There is clearly a bit of a physical issue here with infinite propagation speeds; relativity states that nothing travels faster than the speed of light, x. However, in this course, we shall work solely in the Newtonian limit, i.e. where relativistic effects are unimportant, and in this limit we have $c \to \infty$, hence our classifications do make some sense.

Mathematically, identifying information propagation speed is done by considering the **characteristics** of the underlying equation. This is quite a broad field which we shall not cover in this course, though some other courses may consider specific applications. The term will also be used again in section 5.2.5, where we consider a way of writing equations based on characteristics (but finding this form indirectly).

Characteristics allow us to classify equations as:

1. **Elliptic** - entirely infinite information propagation speed
2. **Parabolic** - some information propagates infinitely fast
3. **Hyperbolic** - entirely finite information propagation speed

Elliptic problems (physically realistic ones, anyway) are steady state problems, whilst parabolic and hyperbolic problems are typically time varying. Between the two time-varying types, the infinite propagation of some information in parabolic equations acts as a **diffusion** effect - information is rapidly 'spread out' through the domain. Hyperbolic equations do not have this diffusion, and this leads to different mathematical behaviour we shall discuss shortly.

5.1.1 Techniques to classify PDEs

Conic sections

The names of the classification types, elliptic, parabolic and hyperbolic, might already be familiar from conic sections, the slices through a cone. Here, a slice can be fitted to the equation

$$Ax^2 + Bxy + Cx^2 + Dx + Ey + F = 0. \tag{5.1}$$

The sign of the discriminant, $D = B^2 - 4AC$, determines what curve is described by this equation,

- $D < 0$: Ellipse
- $D = 0$: Parabola
- $D > 0$: Hyperbola

A similar formulation of PDE exists, which, for an evolved variable, u, is expressed as

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + F = 0. \tag{5.2}$$

For this type of PDE, we have

- $D < 0$: Elliptic equation
- $D = 0$: Parabolic equation
- $D > 0$: Hyperbolic equation

This would be possible to demonstrate through characteristics.

Systems of first order PDEs

A lot of PDEs cannot be expressed as in equation (5.2). Many interesting systems are functions of multiple variables, more than two-dimensional, non-linear in the coefficients and/or don't contain second derivatives. Here, we consider how equations of the form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0. \tag{5.3}$$

We are able to use the chain rule on this to get

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \nabla \cdot \mathbf{u} = 0 \tag{5.4}$$

where $\frac{\partial \mathbf{f}}{\partial \mathbf{u}}$ is a Jacobian matrix. The classification of the equation might then be governed by the eigenvalues and eigenvectors of this Jacobian:

- All purely complex: Elliptic
- All real and distinct: Hyperbolic

Note here that we have not made any statements about parabolic equations. The analysis of eigenvalues is only sufficient to classify equations in certain cases, otherwise a full characteristic analysis is needed, which is beyond what we shall cover in this course. In some of the continuum courses, demonstration that a system of equations is hyperbolic can be extremely important for identifying stable numerical methods, and understanding the system behaviour.

At this point, it might be clear that classifying equations is not always trivial. This is made worse when equations are nonlinear - the Jacobian will be a function of the variables in $\mathbf{u}$ and there is no reason why the eigenvalues will always be, for example, real and distinct. In many cases, this leads to equations, or classes of equation, whose behaviour is still an active research topic.

5.2 Ways of expressing PDEs

Whilst the mathematical properties of an equation are important for identifying computational techniques, they may also be dependent on how the equation is written. It is therefore important to be able to manipulate equations into a desirable form. In this section, most of our work will be with equations of the form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{s}(\mathbf{u}). \tag{5.5}$$

This is a very general form, and we also allow the function $\mathbf{s}$ to contain additional derivatives. However, many physical systems can be written in this form including the Navier-Stokes equations, Maxwell's equations and Einstein's field equations (it's not always easy to do so though). In this section, we also consider an example to demonstrate our manipulation, a system of equations which is reasonably simple, but at the same time, not a trivial system. This is the shallow water equations.

5.2.1 Shallow water equations

The shallow water equations are named because they can describe the height of water over a non-trivial geometry in cases where the scale of the problem is such that the length and the width of the water is much greater than the height (i.e. the water is shallow). This allows us to consider height as a variable in the equations, and not as a coordinate. In one spatial dimension, the shallow water equations can be written as

$$\frac{\partial}{\partial t}\begin{pmatrix} \phi \\ \phi u \end{pmatrix} + \frac{\partial}{\partial x}\begin{pmatrix} \phi u \\ \phi u^2 + \frac{1}{2}\phi^2 \end{pmatrix} = \begin{pmatrix} 0 \\ g\phi\frac{\partial h}{\partial x} \end{pmatrix} \tag{5.6}$$

where ϕ is the geopotential, u is the velocity, h is the height (or depth) of the surface beneath the water and g is gravitational acceleration. The geopotential is the only one of these variables which is not regularly encountered, and it is defined by

$$\phi = gH = g(\eta + h) \tag{5.7}$$

where η is the height of the water above an initial surface, and therefore H is the total height of the water. For all examples in this section, we shall consider that the surface beneath the water is uniform, such that

$$\frac{\partial h}{\partial x} = 0. \tag{5.8}$$

5.2.2 Conservation form

At various times in this course already, we have shown equations in the form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}(\mathbf{u}) = \mathbf{0}, \tag{5.9}$$

or, in one dimension,

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial}{\partial x}\mathbf{f}(\mathbf{u}) = 0. \tag{5.10}$$

This form is known as conservation form, the reason for which goes back to the definition of the divergence as

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \mathrm{d}S. \tag{5.11}$$

This definition stated that divergence can be considered as the rate at which a vector field enters (or leaves) a given volume. This is the principle behind conservation, the amount of $\mathbf{u}$ in the volume doesn't change, **unless** some of it leaves, or some more enters, the volume. In other courses, the importance of this form will become apparent.

This form also leads to some standard naming conventions:

- $\mathbf{u}$: This is the **vector of conserved variables** (because they can be described through a conservation law)
- $\mathbf{f}$: This is the **flux vector** (because it describes the 'flux' of $\mathbf{u}$ through the volume)

Balance law form

At the start of this section, we wrote a system of equations as

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{f}\left(\mathbf{u}\right) = \mathbf{s}\left(\mathbf{u}\right). \tag{5.12}$$

The left hand side of the equation looks like conservation form, but now there are additional terms on the right hand side. This form of equations is often known as balance law form - the changes in $\mathbf{u}$ which are not due to material entering or leaving from the volume are **balanced** by the terms in $\mathbf{s}(\mathbf{u})$.

Shallow water equation example

The equations we initially gave for the shallow water equation (5.6) are in balance law form,

$$\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \phi u\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\phi u\\ \phi u^2+\frac{1}{2}\phi^2\end{pmatrix}=\begin{pmatrix}0\\ g\phi\frac{\partial h}{\partial x}\end{pmatrix} \tag{5.13}$$

and, under the assumption that h is uniform, they are in conservation law form.

Here we have the vector of conserved variables,

$$\mathbf{u}=\begin{pmatrix}\phi\\ \phi u\end{pmatrix} \tag{5.14}$$

and the flux vector

$$\mathbf{f}(\mathbf{u})=\begin{pmatrix}\phi u\\ \frac{(\phi u)^2}{\phi}+\frac{1}{2}\phi^2\end{pmatrix}. \tag{5.15}$$

In this definition, we have written our flux vector specifically such that it is a function of our two independent variables, ϕ, and ϕu.

5.2.3 Non-conservation form

Conservation form can be very useful for some computational techniques, but other forms of the equations may be more intuitive, or have other uses. Non-conservation form is something of a general term, since it covers any other way we could write our system of equations. However, we shall use it here to discuss an alternative form of the equations we have already considered in equation (5.4),

$$\frac{\partial \mathbf{u}}{\partial t}+\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\nabla\cdot\mathbf{u}=0. \tag{5.16}$$

In one dimension (and in index notation) we could write this as

$$\frac{\partial u_i}{\partial t}+\frac{\partial f_i}{\partial u_j}\frac{\partial u_j}{\partial x}=0 \tag{5.17}$$

which might be written more abstractly as

$$\frac{\partial u_i}{\partial t}+A_{ij}\frac{\partial u_j}{\partial x}=0. \tag{5.18}$$

This precise non-conservation form, as we saw in section 5.1.1, can be used to try to classify the PDE system. However, it is primarily used as an example of how equation manipulation may begin.

Shallow water equation example

For the shallow water equations, we have

$$\mathbf{f}(\mathbf{u}) = \begin{pmatrix} \phi u \\ \frac{(\phi u)^2}{\phi} + \frac{1}{2}\phi^2 \end{pmatrix}. \tag{5.19}$$

Computing the Jacobian of this expression is straightforward, but we must remember that since ϕ and ϕu are independent variables,

$$\frac{\partial(\phi u)}{\partial \phi} = 0. \tag{5.20}$$

We the have

$$\begin{aligned} A &= \begin{pmatrix} \frac{\partial f_\phi}{\partial \phi} & \frac{\partial f_\phi}{\partial(\phi u)} \\ \frac{\partial f_{\phi u}}{\partial \phi} & \frac{\partial f_{\phi u}}{\partial(\phi u)} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial \phi}(\phi u) & \frac{\partial}{\partial(\phi u)}(\phi u) \\ \frac{\partial}{\partial \phi}\left(\frac{(\phi u)^2}{\phi} + \frac{1}{2}\phi^2\right) & \frac{\partial}{\partial(\phi u)}\left(\frac{(\phi u)^2}{\phi} + \frac{1}{2}\phi^2\right) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -\frac{(\phi u)^2}{\phi^2} + \phi & \frac{2\phi u}{\phi} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -u^2 + \phi & 2u \end{pmatrix} \end{aligned} \tag{5.21}$$

where in the final expression, we have simplified the expressions given in conserved variables to a more natural form.

5.2.4 Primitive variable form

Another very broad description of a system of equations is primitive variable form. Here, if we have a vector of n conserved variables, $\mathbf{u}$, then a vector of primitive variables, $\mathbf{w}$ would be n variables which close the system governed by the underlying PDE, but at least one variable in $\mathbf{w}$ is not also in $\mathbf{u}$. This is equivalent to choosing our independent variables discussed during multivariable calculus lectures, and once we have our primtitive variables, we know we could write

$$\mathbf{u} = \mathbf{u}(\mathbf{w}) \tag{5.22}$$

and also

$$\mathbf{f} = \mathbf{f}(\mathbf{w}). \tag{5.23}$$

The reason for choosing a different set of variables is often for intuitive understanding, for example, we know momentum is a conserved quantity, and therefore could be a conserved variable, but perhaps velocity is more intuitive to work with. Alternatively, it might make some equation analysis easier, which we shall return to shortly.

Firstly, we note that when we have chosen our primitive varibales, we will try to write our PDE as

$$\frac{\partial w_i}{\partial t} + B_{ij}\frac{\partial w_j}{\partial x} = 0. \tag{5.24}$$

To do this, we start with our equation in conservation form,

$$\frac{\partial u_i(\mathbf{w})}{\partial t} + \frac{\partial f_i(\mathbf{w})}{\partial x} = 0. \tag{5.25}$$

By the chain rule, this means we have

$$\frac{\partial u_i}{\partial w_j}\frac{\partial w_j}{\partial t} + \frac{\partial f_i(\mathbf{w})}{\partial w_j}\frac{\partial w_j}{\partial x} = 0. \tag{5.26}$$

If we multiply this equation by the inverse (which we know exists, because we must be able to also get $\mathbf{w} = \mathbf{w}(\mathbf{u})$), we get

$$\left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial u_k}{\partial w_j}\frac{\partial w_j}{\partial t} + \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_j(\mathbf{w})}{\partial w_j}\frac{\partial w_j}{\partial x} = 0. \tag{5.27}$$

where, noting that

$$\left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial u_k}{\partial w_j} = \delta_{ij}, \tag{5.28}$$

this simplifies to

$$\frac{\partial w_i}{\partial t} + \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k(\mathbf{w})}{\partial w_j}\frac{\partial w_j}{\partial x} = 0 \tag{5.29}$$

hence we have

$$B_{ij} = \left(\frac{\partial u_i}{\partial w_k}\right)^{-1}\frac{\partial f_k(\mathbf{w})}{\partial w_j}. \tag{5.30}$$

We did not have to start from conservative form to derive our primitive variable form. Instead, we could have started from non-conservative form, equation (5.17), which can be written

$$\frac{\partial u_i(\mathbf{w})}{\partial t} + \frac{\partial f_i(\mathbf{u})}{\partial u_j}\frac{\partial u_j(\mathbf{w})}{\partial x} = 0. \tag{5.31}$$

By the chain rule, we have

$$\frac{\partial u_i}{\partial w_j}\frac{\partial w_j}{\partial t} + \frac{\partial f_i(\mathbf{w})}{\partial u_k}\frac{\partial u_k}{\partial w_j}\frac{\partial w_j}{\partial x} = 0. \tag{5.32}$$

and multiplying through by the inverse gives

$$\frac{\partial w_i}{\partial t} + \left(\frac{\partial u_i}{\partial w_l}\right)^{-1} \frac{\partial f_l(\mathbf{w})}{\partial u_k} \frac{\partial u_k}{\partial w_j} \frac{\partial w_j}{\partial x} = 0. \tag{5.33}$$

If we now define

$$\frac{\partial u_i}{\partial w_j} = C_{ij} \tag{5.34}$$

then we have

$$B_{ij} = \left(\frac{\partial u_i}{\partial w_l}\right)^{-1} \frac{\partial f_l(\mathbf{w})}{\partial u_k} \frac{\partial u_k}{\partial w_j} = (C_{il})^{-1} A_{lk} C_{kj} \tag{5.35}$$

which means that A and B are similar matricies. This then means that they have the same eigenvalues, and this can be another use of primitive variable forms. It might be easier to find the eigenvalues and eigenvectors of B than it is of A.

In general, computing the matrix C and inverting it, then performing the matrix multiplication will always allow you to find your matrix B. In practice, this is often cumbersome, and actually direct manipulation of the equation such that you can write it in primitive variable form is often easier.

Shallow water equation example

For the shallow water equations, neither the geopotential, ϕ, nor the conserved quantity ϕu are natural quantities to work with. It is likely much more intuitive to work with height, H and velocity, u. Our two conserved variables can be written entirely in terms of these primitive variables,

$$\phi = gH, \qquad \phi u = gHu \tag{5.36}$$

where g is the gravitational constant. We can directly manipulate our equations, from any previous form, to write them in primitive variable form. Here, we start from conservative form, and the first step is to express all quantities in terms of the primitive variables, hence we have a closed system, as required,

$$\frac{\partial \phi}{\partial t} + \frac{\partial}{\partial x}(\phi u) = \frac{\partial}{\partial t}(gH) + \frac{\partial}{\partial x}(gHu) = 0 \tag{5.37}$$

and

$$\frac{\partial}{\partial t}(\phi u) + \frac{\partial}{\partial x}\left(\phi u^2 + \frac{1}{2}\phi^2\right) = \frac{\partial}{\partial t}(gHu) + \frac{\partial}{\partial x}\left(gHu^2 + \frac{1}{2}g^2H^2\right) = 0. \tag{5.38}$$

Because g is a constant, we can immediately simplify these expressions,

$$\frac{\partial H}{\partial t} + \frac{\partial}{\partial x}(Hu) = 0 \tag{5.39}$$

and

$$\frac{\partial}{\partial t}(Hu) + \frac{\partial}{\partial x}\left(Hu^2 + \frac{1}{2}gH^2\right) = 0. \tag{5.40}$$

Since in equation (5.39) the time derivative is already one of our primitive variables, this is now very simple to write in primitive variable from; we simply use the product rule on the space derivative,

$$\frac{\partial H}{\partial t} + u\frac{\partial H}{\partial x} + H\frac{\partial u}{\partial x} = 0. \tag{5.41}$$

Equation (5.40) is a little more effort; the time derivative needs simplifying too. The first step, though, is to apply the product rule everywhere, so that all derivatives are derivatives of a single primitive quantity,

$$u\frac{\partial H}{\partial t} + H\frac{\partial u}{\partial t} + u^2\frac{\partial H}{\partial x} + 2Hu\frac{\partial u}{\partial x} + gH\frac{\partial H}{\partial x} = 0. \tag{5.42}$$

We now need to get rid of the time derivative of H. This can clearly be done by using equation (5.41) to make the substitution

$$\frac{\partial H}{\partial t} = -u\frac{\partial H}{\partial x} - H\frac{\partial u}{\partial x} \tag{5.43}$$

giving

$$-u^2\frac{\partial H}{\partial x} - Hu\frac{\partial u}{\partial x} + H\frac{\partial u}{\partial t} + u^2\frac{\partial H}{\partial x} + 2Hu\frac{\partial u}{\partial x} + gH\frac{\partial H}{\partial x} = 0 \tag{5.44}$$

which then simplifies to

$$H\frac{\partial u}{\partial t} + Hu\frac{\partial u}{\partial x} + gH\frac{\partial H}{\partial x} = 0 \tag{5.45}$$

and dividing by H gives

$$\frac{\partial u}{\partial t} + g\frac{\partial H}{\partial x} + u\frac{\partial u}{\partial x} = 0. \tag{5.46}$$

It is also possible to simplify equation (5.47) by factoring out the primitive variable equation for H, writing it as

$$u\left(\frac{\partial H}{\partial t} + u\frac{\partial H}{\partial x} + H\frac{\partial u}{\partial x}\right) + H\frac{\partial u}{\partial t} + Hu\frac{\partial u}{\partial x} + gH\frac{\partial H}{\partial x} = 0. \tag{5.47}$$

because, again from equation (5.41), we know that the entire term in brackets is zero.

By either method, we have the shallow water equations in primitive form, which we could write as

$$\frac{\partial}{\partial t}\begin{pmatrix} H \\ u \end{pmatrix} + \begin{pmatrix} u & H \\ g & u \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} H \\ u \end{pmatrix} = 0 \tag{5.48}$$

5.2.5 Characteristic form

A very specific primitive variable form of a PDE is when it can be written as

$$\frac{\partial \mathcal{V}_i}{\partial t} + \Lambda_{ij}\frac{\partial \mathcal{V}_j}{\partial x} = 0 \tag{5.49}$$

where Λ is a diagonal matrix. This form of the equations is known as **characteristic form**, because the variables $\boldsymbol{\mathcal{V}}$, known as characteristic variables, can also be obtained through characteristic analysis. As we described earlier, this is complex, and we will not consider general approaches.

However, a few things are worth noting, firstly, we know from equation (5.35) that we have

$$\Lambda = C^{-1}AC \tag{5.50}$$

which means to obtain this, we had to diagonalise A. We also know that this is not possible for all matrices, and not all equations will have a characteristic form. This is directly related to the classification of equations - if information travels infinitely fast, it does not have an associated characteristic. However, if A can be diagonalised, then in doing so, we would compute C, which we know is a matrix of the eigenvectors of A. From this, we then have

$$C_{ij} = \frac{\partial u_i}{\partial \mathcal{V}_j} \tag{5.51}$$

which is a system of differential equations which we may be able to solve. If we can do this, we obtain our characteristic variables, $\boldsymbol{\mathcal{V}}$.

Secondly, if we have an equation in primitive variable form,

$$\frac{\partial w_i}{\partial t} + B_{ij}\frac{\partial w_j}{\partial x} = 0. \tag{5.52}$$

and we know that $\mathbf{u} = \mathbf{u}(\mathbf{w})$ and $\mathbf{u} = \mathbf{u}(\boldsymbol{\mathcal{V}})$, then we also have

$$\mathbf{w} = \mathbf{w}(\boldsymbol{\mathcal{V}}). \tag{5.53}$$

This means that, by performing the same manipulation as in section 5.2.4, we can obtain

$$\frac{\partial \mathcal{V}_i(\mathbf{w})}{\partial t} + \left(\frac{\partial w_i}{\partial \mathcal{V}_l}\right)^{-1} B_{lk} \frac{\partial w_k}{\partial \mathcal{V}_j} \frac{\partial \mathcal{V}_j(\mathbf{u})}{\partial x} = 0 \tag{5.54}$$

in which case, we are diagonalising B instead of A. Both these methods would give the same characteristic form.

Shallow water equation example - eigenvalues from conserved variables

We shall first demonstrate that the eigenvalues from equations using both conserved and primitive variable vectors are the same. To do so, we first consider conserved variables, and, starting from the Jacobian matrix derived in equation (5.21),

$$A = \begin{pmatrix} 0 & 1 \\ -u^2 + \phi & 2u \end{pmatrix} \tag{5.55}$$

we first find the eigenvalues,

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 \\ -u^2 + \phi & 2u - \lambda \end{vmatrix}. \tag{5.56}$$

Taking the determinant gives

$$\begin{aligned} &-\lambda(2u - \lambda) + u^2 - \phi = 0 \\ &\lambda^2 - 2u\lambda + u^2 - \phi = 0 \\ &\lambda = \frac{2u \pm \sqrt{4u^2 - 4(u^2 - \phi)}}{2} \\ &\lambda = u \pm \frac{1}{2}\sqrt{4\phi} = u \pm \sqrt{\phi} \end{aligned} \tag{5.57}$$

Shallow water equation example - primitive variables

We now consider primtive variables, and in this case, will use the primitive variables to identify the characteristic form. Starting from the matrix in equation (5.48),

$$B = \begin{pmatrix} u & H \\ g & u \end{pmatrix} \tag{5.58}$$

we find the eigenvalues

$$|B - \lambda I| = \begin{vmatrix} u - \lambda & H \\ g & u - \lambda \end{vmatrix} \tag{5.59}$$

Taking the determinant gives

$$\begin{aligned}(u-\lambda)^2 - gH &= 0\\ (u-\lambda)^2 &= gH\\ u-\lambda &= \pm\sqrt{gH}\\ \lambda &= u \pm \sqrt{gH} = u \pm \sqrt{\phi}.\end{aligned} \tag{5.60}$$

In the final line, we use $\phi = gH$ to check that our result is the same as in equation (5.57).

Note that for the shallow water equations, finding the eigenvalues is straightforward for both the primitive and the conservative variables; this is not always the case.

The next step is to compute the eigenvectors. Starting from $\lambda_1 = u + \sqrt{gH}$, we find the first eigenvector by solving

$$\begin{pmatrix} -\sqrt{gH} & H \\ g & -\sqrt{gH} \end{pmatrix} \begin{pmatrix} r_{11} \\ r_{12} \end{pmatrix} = \mathbf{0} \tag{5.61}$$

The first of these equations gives

$$-\sqrt{gH}r_{11} + Hr_{12} = 0 \tag{5.62}$$

or

$$r_{12} = \frac{1}{H}\sqrt{gH}r_{11}. \tag{5.63}$$

For consistency, we can also check that the second equation gives the same result;

$$\begin{aligned} gr_{11} - \sqrt{gH}r_{12} &= 0\\ r_{12} &= \frac{1}{\sqrt{gH}}gr_{11}\\ r_{12} &= \frac{H}{H}\frac{1}{\sqrt{gH}}gr_{11} = \frac{1}{H}\frac{gH}{\sqrt{gH}}r_{11}\\ r_{12} &= \frac{1}{H}\sqrt{gH}r_{11}\end{aligned} \tag{5.64}$$

This gives our first eigenvector as

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ \frac{1}{H}\sqrt{gH} \end{pmatrix} \tag{5.65}$$

or, alternatively,

$$\mathbf{r}_1 = \begin{pmatrix} \frac{H}{\sqrt{gH}} \\ 1 \end{pmatrix}. \tag{5.66}$$

At this point, there is no preference for using one of these eigenvectors over the other. However, as we work through the derivation of characteristic variables, one of these will have some obvious advantages.

Now, for the second eigenvector, associated with $\lambda_2 = u - \sqrt{gH}$, we need to solve

$$\begin{pmatrix} \sqrt{gH} & H \\ g & \sqrt{gH} \end{pmatrix} \begin{pmatrix} r_{21} \\ r_{22} \end{pmatrix} = \mathbf{0}. \tag{5.67}$$

The first of these equations gives

$$\sqrt{gH} r_{21} + H r_{22} = 0 \tag{5.68}$$

or

$$r_{22} = -\frac{1}{H}\sqrt{gH} r_{21}, \tag{5.69}$$

hence the second eigenvector is

$$\mathbf{r}_2 = \begin{pmatrix} 1 \\ -\frac{1}{H}\sqrt{gH} \end{pmatrix} \tag{5.70}$$

or, under a different choice of scaling,

$$\mathbf{r}_2 = \begin{pmatrix} -\frac{H}{\sqrt{gH}} \\ 1 \end{pmatrix}. \tag{5.71}$$

Piecing the two eigenvectors together gives us our matrix of eigenvectors, which, based on the expressions above, could be wrtten as

$$C_1 = \begin{pmatrix} 1 & 1 \\ \frac{1}{H}\sqrt{gH} & -\frac{1}{H}\sqrt{gH} \end{pmatrix} \tag{5.72}$$

or

$$C_2 = \begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix}. \tag{5.73}$$

This reveals an interesting feature about using eigenvectors to identify characteristic variables; we have previously stated that

$$C = \frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}}. \tag{5.74}$$

However, if both equations (5.72) and (5.73) could both define characteristic variables, $\boldsymbol{\mathcal{V}}$, this would mean that these variables are not uniquely defined. At the moment, we are only considering different ways a PDE could be expressed. Looking at how characteristic forms of the PDE provide information will help us deal with this issue **reference appropriate place/course**. For

now, however, we note that the columns of the matrix in equation (5.74) will be eigenvectors of equation (5.58), but we don't necessarily know what scaling has been applied to the eigenvectors. Which means an arbitrary matrix of right eigenvectors may not allow us to compute the characteristic variables directly.

For the remainder of this example, we shall see that equation (5.73) does allow us to compute characteristic variables directly, i.e.

$$C = \frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}} = \begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix}. \tag{5.75}$$

To compute the characteristic variables, we also need the inverse of this matrix. As a 2×2 matrix, this is straightforward, first noting that

$$\det(C) = \frac{2H}{\sqrt{gH}} \tag{5.76}$$

means we have

$$C^{-1} = \frac{1}{\frac{2H}{\sqrt{gH}}} \begin{pmatrix} 1 & \frac{H}{\sqrt{gH}} \\ -1 & \frac{H}{\sqrt{gH}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix}. \tag{5.77}$$

At this point, it is worth checking our calculation, we should have that

$$C^{-1}BC = \Lambda. \tag{5.78}$$

Checking this (for the first case), we get

$$\begin{aligned} C^{-1}BC &= \frac{1}{2} \begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix} \begin{pmatrix} u & H \\ g & u \end{pmatrix} \begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix} \begin{pmatrix} \frac{Hu}{\sqrt{gH}} + H & -\frac{Hu}{\sqrt{gH}} + H \\ \frac{gH}{\sqrt{gH}} + u & -\frac{gH}{\sqrt{gH}} + u \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} u + \sqrt{gH} + \frac{gH}{\sqrt{gH}} + u & -u + \sqrt{gH} - \frac{gH}{\sqrt{gH}} + u \\ -u - \sqrt{gH} + \frac{gH}{\sqrt{gH}} + u & u - \sqrt{gH} - \frac{gH}{\sqrt{gH}} + u \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 2u + 2\sqrt{gH} & 0 \\ 0 & 2u - 2\sqrt{gH} \end{pmatrix} \\ &= \begin{pmatrix} u + \sqrt{gH} & 0 \\ 0 & u - \sqrt{gH} \end{pmatrix} = \Lambda \end{aligned} \tag{5.79}$$

as expected.

We now have a relationships for $\boldsymbol{\mathcal{V}}$, either through

$$C = \frac{\partial \mathbf{w}}{\partial \boldsymbol{\mathcal{V}}} = \begin{pmatrix} \frac{H}{\sqrt{gH}} & -\frac{H}{\sqrt{gH}} \\ 1 & 1 \end{pmatrix} \tag{5.80}$$

or

$$C^{-1} = \frac{\partial \boldsymbol{\mathcal{V}}}{\partial \mathbf{w}} = \frac{1}{2} \begin{pmatrix} \frac{\sqrt{gH}}{H} & 1 \\ -\frac{\sqrt{gH}}{H} & 1 \end{pmatrix}. \tag{5.81}$$

To identify the characteristic variables, we now need to integrate. This is easiest done in index form, we start with

$$\frac{\partial \mathcal{V}_i}{\partial w_j} = C^{-1}_{ij}. \tag{5.82}$$

We now integrate with respect to w_i,

$$\int \frac{\partial \mathcal{V}_j}{\partial w_i} \mathrm{d}w_i = \int C^{-1}_{ij} \mathrm{d}w_i. \tag{5.83}$$

The left hand side of this equation, by the fundamental theorem of calculus, is simply $\mathcal{V}_i$. The right hand side is best computed now for individual i. Therefore, we have

$$\mathcal{V}_1 = \int C^{-1}_{1j} \mathrm{d}w_j = \int C^{-1}_{11} \mathrm{d}w_1 + \int C^{-1}_{12} \mathrm{d}w_2 \tag{5.84}$$

where we have explicitly expanded the sum over the repeated index. First we note that because of our starting point in equation (5.48), we have

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} H \\ u \end{pmatrix} \tag{5.85}$$

We can now use the entries of equation (5.81) to compute this integral,

$$\begin{aligned} \mathcal{V}_1 &= \int C^{-1}_{11} \mathrm{d}H + \int C^{-1}_{12} \mathrm{d}u \\ &= \int \frac{\sqrt{gH}}{2H} \mathrm{d}H + \int \frac{1}{2} \mathrm{d}u \\ &= \frac{1}{2} \left(\int \frac{\sqrt{g}}{\sqrt{H}} \mathrm{d}H + u \right) \\ &= \frac{1}{2} \left(2\sqrt{gH} + u \right) \\ &= \sqrt{gH} + \frac{1}{2} u. \end{aligned} \tag{5.86}$$

Note that in this equation, we have ignored the constant of integration. Because we will eventually use $\boldsymbol{\mathcal{V}}$ in our PDE, the derivative of this constant is zero, so it can be ignored.

The second characteristic variable is

$$\begin{aligned}\mathcal{V}_2 &= \int C_{21}^{-1}\mathrm{d}H + \int C_{22}^{-1}\mathrm{d}u \\ &= \int -\frac{\sqrt{gH}}{2H}\mathrm{d}H + \int \frac{1}{2}\mathrm{d}u \\ &= -\sqrt{gH} + \frac{1}{2}u. \end{aligned} \tag{5.87}$$

where similarities to the $\mathcal{V}_1$ calculation mean we skip most explicit calculation here.

This now gives us characteristic form of the shallow water equations,

$$\frac{\partial}{\partial t}\begin{pmatrix} \sqrt{gH}+\frac{1}{2}u \\ \sqrt{gH}-\frac{1}{2}u \end{pmatrix} + \begin{pmatrix} u+\sqrt{gH} & 0 \\ 0 & u-\sqrt{gH} \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} \sqrt{gH}+\frac{1}{2}u \\ \sqrt{gH}-\frac{1}{2}u \end{pmatrix} = \mathbf{0}. \tag{5.88}$$

For analysis of PDEs, this form can be extremely valuable, though by itself here, the characteristic variables are not particularly meaningful. Note that we could write this equation entirely in terms of $\mathcal{V}_1$ and $\mathcal{V}_2$. First note that

$$\mathcal{V}_1 - \mathcal{V}_2 = \sqrt{gH} + \frac{1}{2}u - \left(\sqrt{gH} - \frac{1}{2}u\right) = u \tag{5.89}$$

and

$$\frac{1}{2}\left(\mathcal{V}_1 + \mathcal{V}_2\right) = \frac{1}{2}\left(\sqrt{gH} + \frac{1}{2}u + \sqrt{gH} - \frac{1}{2}u\right) = \sqrt{gH}. \tag{5.90}$$

Therefore the eigenvalues can be expressed in terms of characteristic variables,

$$u + \sqrt{gH} = \mathcal{V}_1 - \mathcal{V}_2 + \frac{1}{2}\left(\mathcal{V}_1 + \mathcal{V}_2\right) = \frac{3}{2}\mathcal{V}_1 - \frac{1}{2}\mathcal{V}_2 \tag{5.91}$$

and

$$u - \sqrt{gH} = \mathcal{V}_1 - \mathcal{V}_2 - \frac{1}{2}\left(\mathcal{V}_1 + \mathcal{V}_2\right) = \frac{1}{2}\mathcal{V}_1 - \frac{3}{2}\mathcal{V}_2 \tag{5.92}$$

hence we have

$$\frac{\partial}{\partial t}\begin{pmatrix} \mathcal{V}_1 \\ \mathcal{V}_2 \end{pmatrix} + \begin{pmatrix} \frac{3}{2}\mathcal{V}_1 - \frac{1}{2}\mathcal{V}_2 & 0 \\ 0 & \frac{1}{2}\mathcal{V}_1 - \frac{3}{2}\mathcal{V}_2 \end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix} \mathcal{V}_1 \\ \mathcal{V}_2 \end{pmatrix} = \mathbf{0}. \tag{5.93}$$

5.2.6 Second-order PDEs

One final example of how PDEs can be manipulated uses a different starting point; the scalar wave equation,

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial t^2} \tag{5.94}$$

where a scalar field, u evolves according to a propagation speed, a. Such an equation could apply to sound waves under certain assumptions, and if so, the scalar field would be density, and the propagation speed the speed of sound.

It is possible to write the scalar wave equation as a system of first-order PDEs, instead of a single second-order PDE. To do so, we introduce two new variables,

$$v = \frac{\partial u}{\partial t}, \qquad w = \frac{\partial u}{\partial x}. \tag{5.95}$$

If we now take a further derivative of v with respect to t we get

$$\frac{\partial v}{\partial t} = \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial w}{\partial x} \tag{5.96}$$

and if we take a derivative of w with respect to t we get

$$\frac{\partial w}{\partial t} = \frac{\partial}{\partial t}\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\frac{\partial u}{\partial t} = \frac{\partial v}{\partial x}. \tag{5.97}$$

We have now two equations, which could be written in vector form as

$$\frac{\partial}{\partial t}\begin{pmatrix} v \\ w \end{pmatrix} - \frac{\partial}{\partial x}\begin{pmatrix} a^2 w \\ v \end{pmatrix} = 0. \tag{5.98}$$

This may not be the best way to express this equation (to recover u, we'd need to solve an integral of some sort), but it can be done.

Alternative approaches to obtain more useful first-order systems of PDEs for the wave equation need to start from the physical origins of the equation. Here we shall demonstrate a case which reduces to the wave equation, but for which, working backwards is more difficult. If we consider two variables, $\bar{u}$ and $\bar{v}$ which are governed by

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial \bar{v}}{\partial x} = 0 \tag{5.99}$$

and

$$\frac{\partial \bar{v}}{\partial t} + a^2 \frac{\partial \bar{u}}{\partial x} = 0, \tag{5.100}$$

and take further partial derivatives, we get

$$\frac{\partial^2 \bar{u}}{\partial t^2} + \frac{\partial^2 \bar{v}}{\partial t \partial x} = 0 \tag{5.101}$$

and

$$\frac{\partial^2 \bar{v}}{\partial x \partial t} + a^2 \frac{\partial^2 \bar{u}}{\partial x^2} = 0. \tag{5.102}$$

Now if we subtract these two equations, because the mixed partial derivative terms cancel, we get

$$\frac{\partial^2 \bar{u}}{\partial t^2} - a^2 \frac{\partial^2 \bar{u}}{\partial x^2} = 0 \tag{5.103}$$

which is again the wave equation

$$\frac{\partial^2 \bar{u}}{\partial t^2} = a^2 \frac{\partial^2 \bar{u}}{\partial x^2}. \tag{5.104}$$

The advantage in this case is that both sets of equations contain the variable $\bar{u}$, and we don't need to try to recompute it after solving either approach.

5.3 Discontinuities

Partial differential equations and discontinuities are not always things you want to find together; a discontinuity (an infinite-gradient jump in a quantity) is not differentiable. This means that the partial differential equation is no longer valid, and there are implications for computational methods too (which may stop solving your equations). However, the underlying behaviour modelled by the PDEs may actually lead to the formation of discontinuities (or you could have them in your initial data). Mathematically the formation of these features is admissible for **hyperbolic PDEs**, and will be demonstrated in further courses. In this course, we consider the mathematical implications of the formation of a discontinuity.

This is not just a mathematical feature, though, discontinuities do exist in the real world. The most obvious example is the discontinuity between two materials, but these are not what we are concerned with - they are typically not the outcome of the solution of our underlying equations. Perhaps the most obvious example within a single material, then, is a **shock wave**. These are generated when something moves faster than information propagation speeds (and hence why they occur in hyperbolic systems, where these speeds are finite). As a result, we must consider how we would deal with a case in which a discontinuity forms.

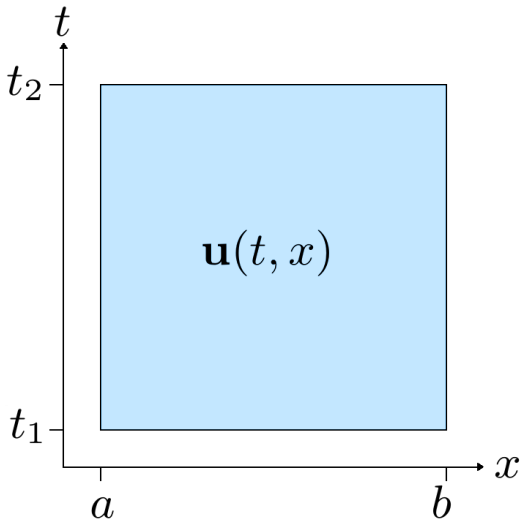


Figure 5.1: Illustration of an integral domain, in which our vector of variables $\mathbf{u}(x,t)$ is being solved for $t \in [t_1, t_2]$ and $x \in [a, b]$.

At this point, it is perhaps worth mentioning that technically, true discontinuities are not actually encountered; across a material change, there is an atomic-scale transition region, and across a shock wave, there will be a small amount of thermal diffusion. The mathematical models which govern these systems will likely not include these effects, but they are still there. This is a valid approach, however, because scientific computing requires that whatever domain you are solving on is **discretised**, such that computational methods can be used. This introduces minimum length scales, and if these are larger than, e.g. the width across a shock wave, then as far as our computational model is concerned, we have a true discontinuity.

5.3.1 Integral equations

The key to dealing with discontinuities and PDEs is that, due to the nature of differentiation and integration, we can rewrite our PDEs as **integral equations**. Integrals can deal with discontinuous data, and as such, discontinuities would not be a problem. To derive this, we first consider an **integral domain**, a region in space and time over which we want to solve our PDE.

A simple integral domain is shown in figure 5.1, where a vector of variables, $\mathbf{u}$ exists in time and a single space dimension, $t \in [t_1, t_2]$ and $x \in [a, b]$. In practice, this might represent a situation where we know the solution $\mathbf{u}(t_1, x)$ for $x \in [a, b]$ (i.e. initial value problem). It might then be that our

computational method is designed to compute the solution at t_2, still for $x \in [a, b]$. However, since we have $\mathbf{u}(t, x)$, we can integrate this over our domain, and similarly, we could integrate our PDE over this domain.

If we consider a scalar PDE

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \tag{5.105}$$

and integrate this over the domain, we get

$$\int_a^b \int_{t_1}^{t_2} \frac{\partial u(t,x)}{\partial t} + \frac{\partial f(u(t,x))}{\partial x} \mathrm{d}t\mathrm{d}x = 0. \tag{5.106}$$

If we assume, for now, that u is continuous within the integral domain, then we can re-order derivatives freely, i.e. we can write

$$\int_a^b \int_{t_1}^{t_2} \frac{\partial u(t,x)}{\partial t} \mathrm{d}t\mathrm{d}x + \int_{t_1}^{t_2} \int_a^b \frac{\partial f(u(t,x))}{\partial x} \mathrm{d}x\mathrm{d}t = 0. \tag{5.107}$$

In this form, we can apply the fundamental theorem of calculus, since the first term is the time integral of a time derivative, and the second term is the space integral of a space derivative, giving

$$\int_a^b u(t_2, x) - u(t_1, x) \mathrm{d}x + \int_{t_1}^{t_2} f(u(t, b)) - f(u(t, a)) \mathrm{d}t = 0. \tag{5.108}$$

This is known as the integral form of our underlying PDE.

If we relate this equation back to some previous terminology, we note that we wrote the PDE in equation (5.105) in a conservation form, as described in section 5.2.2. In this case, the function $f(u)$ could be considered a flux function, since it originated from a divergence of a quantity. Now, in the integral form of the equation (5.108), we have that the first term is the difference in u between times t_1 and t_2 over the domain $x \in [a, b]$. We can interpret this as the amount of u that has left, or entered, the domain in this time interval. The second term is the integral of $f(u)$ over time for $x = a$ and $x = b$, in the normal direction of our volume (hence the negative sign at $x = a$). This can be interpreted as the flux of u over the time interval at the domain boundaries. The overall equation is then making the statement that the change in the amount of u in our domain is goverened purely by the flux through the boundaries (which itself, is a statement of conservation).

5.3.2 Weak solutions

Any solution which satisfies a PDE, such as in equation (5.105), will also satisfy the integral form of this equation (5.108). However, within the integral

form of the equation, u could be discontinuous, hence a solution which satisfies the integral form will not necessarily satisfy the PDE. Such a solution is known as a **weak solution**.

Computational techniques which deal with systems where discontinuities can form need to be able to identify weak solutions, and do so by solving these equations in integral form. Sometimes, this is referred to as solving the weak-form of the equations. We won't consider any examples here, but at this point, it is worth noting that weak solutions are not necessarily unique, and some care must be taken in finding the correct solution.

Chapter 6

Numerical discretisation

6.1 Discretisation and error

Scientific computing requires that whatever system of equations you are solving on is discretised in some manner, such that it can be placed on a CPU or GPU for solving. This could be discretisation in space, in time, in Fourier space, some other transformation of space and time, or a combination of these. All solutions on discrete space will result in some computational error - this could be due to how the discretisation is taken, or simply from numerical round-off error. Fundamentally, this is not an issue, provided that you understand the sources of this error, and how it will affect your simulations. Most importantly, you need to know that this error will not grow unboundedly.

The error within a given computational technique typically needs to be quantified at the same time as introducing the technique, and therefore we shall not go into that sort of detail in this section, but the appropriate lecture courses will cover this material. Here, we shall look at some fundamental techniques employed in discretisation and error analysis.

To understand why numerical error is important, figure 6.1 shows the same computational simulation for two different methods, one which generates significantly more error than the other. This case, which is the Orszag-Tang vortex in magnetohydrodynamics, demonstrates the transition from smooth changes to chaotic features, and it is clear that the accuracy of your method affects the features you can see. In order to understand the error in a computational method, we need to be able to quantify it, and the first step to achieving this is to understand how any derivatives you are using have been approximated.

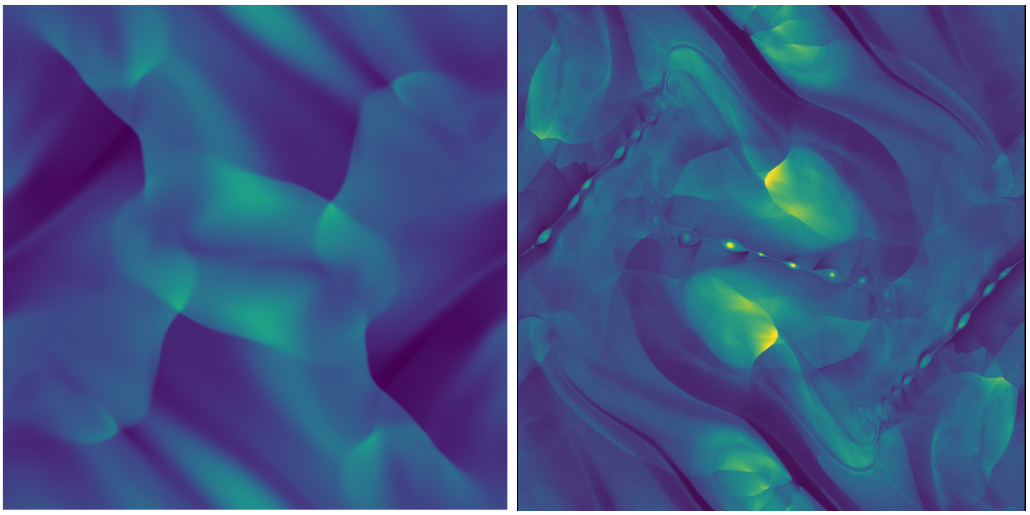


Figure 6.1: Numerical solution of a test problem (Orzsag-Tang vortex in MHD) for two different numerical methods. The method on the left generates larger numerical error than the method on the right, and the difference in the features is clearly evident.

6.1.1 Approximations to derivatives

Whenever we approximate something, we have to make a choice when to decide our approximation is 'good enough' for our application. Hopefully, this decision is mathematically based too. Approximation of derivatives is a good example here, where we quantify 'good enough' as the **order of accuracy** of the method. One key method we have for approximating dervatives is Taylor series. These start from the standard definition,

$$f(x) = f(a) + \left.\frac{\partial f}{\partial x}\right|_{x=a} (x-a) + \frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=a} (x-a)^2 + \frac{1}{3!}\left.\frac{\partial^3 f}{\partial x^3}\right|_{x=a} (x-a)^3 + \dots \tag{6.1}$$

and if we want to use these to approximate derivatives, as long as we choose a point x fairly close to a, we can start obtaining approximations.

First-order derivative approximation

When approximating a derivative, it is good to use the Taylor series form

$$f(x+\Delta x) = f(x) + \left.\frac{\partial f}{\partial x}\right|_{x} \Delta x + \left.\frac{\partial^2 f}{\partial x^2}\right|_{x} \frac{(\Delta x)^2}{2} + \left.\frac{\partial^3 f}{\partial x^3}\right|_{x} \frac{(\Delta x)^3}{3!} + \dots \tag{6.2}$$

The key for approximating derviatives is that we assume that we don't know any of the derivative terms on the right hand side, but we do know $f(x)$, at

least at some given points x (which may be a discrete set of points). We can rearrange this expression to obtain an expression for the first derivative,

$$\left.\frac{\partial f}{\partial x}\right|_x = \frac{f(x+\Delta x)-f(x)}{\Delta x} + \left.\frac{\partial^2 f}{\partial x^2}\right|_x \frac{(\Delta x)}{2} + \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{(\Delta x)^2}{3!} + \dots \tag{6.3}$$

We stated above that we need the point about which we compute our approximation to be close to the point about which we take the Taylor series. In this case, this is true if Δx is small, and if Δx is small, we might reasonably say that any term multiplied by Δx, or $(\Delta x)^n$ $(n \geq 1)$ is negligible. We can write this as

$$\left.\frac{\partial f}{\partial x}\right|_x \approx \frac{f(x+\Delta x)-f(x)}{\Delta x}. \tag{6.4}$$

However, it is often useful not to completely ignore the additional terms, but instead we could write

$$\left.\frac{\partial f}{\partial x}\right|_x = \frac{f(x+\Delta x)-f(x)}{\Delta x} + \mathcal{O}\left(\Delta x\right) \tag{6.5}$$

where we are saying that the largest term we ignore is of order Δx, since, for small values, we have

$$\Delta x > (\Delta x)^2 > (\Delta x)^3 > \dots \tag{6.6}$$

As a result of this, we call this approximation **first-order accurate** since $\Delta x = (\Delta x)^1$. Note that the $\mathcal{O}(\Delta x)$ term is linear in Δx, hence if we halve the size of Δ, we halve $\mathcal{O}(\Delta x)$ too, hence we say we halve the error.

Second-order centred derivative

Having introduced first-order error, it might not be surprising that we can also do better. In equation (6.5), we used a single Taylor expansion to compute our deriviative,

$$f(x+\Delta x) = f(x) + \left.\frac{\partial f}{\partial x}\right|_x \Delta x + \left.\frac{\partial^2 f}{\partial x^2}\right|_x \frac{(\Delta x)^2}{2} + \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{(\Delta x)^3}{3!} + \dots \tag{6.7}$$

However, we could have also used a similar Taylor expansion,

$$f(x-\Delta x) = f(x) - \left.\frac{\partial f}{\partial x}\right|_x \Delta x + \left.\frac{\partial^2 f}{\partial x^2}\right|_x \frac{(\Delta x)^2}{2} - \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{(\Delta x)^3}{3!} + \dots \tag{6.8}$$

To obtain a higher-order derivative, we can subtract these expressions,

$$\begin{aligned} f(x+\Delta x) - f(x-\Delta x) =& f(x) - f(x) + \left.\frac{\partial f}{\partial x}\right|_x (\Delta x + \Delta x) \\ &+ \left.\frac{\partial^2 f}{\partial x^2}\right|_x \frac{(\Delta x)^2 - (\Delta x)^2}{2} \\ &+ \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{(\Delta x)^3 + (\Delta x)^3}{3!} + \dots \\ =& \left.\frac{\partial f}{\partial x}\right|_x 2\Delta x + \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{2\,(\Delta x)^3}{3!} + \dots \end{aligned} \tag{6.9}$$

Rearranging this, we have

$$\begin{aligned} \left.\frac{\partial f}{\partial x}\right|_x =& \frac{1}{2\Delta x}\left[f(x+\Delta x) - f(x-\Delta x) - \left.\frac{\partial^3 f}{\partial x^3}\right|_x \frac{2\,(\Delta x)^3}{3!} + \dots \right] \\ =& \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \frac{1}{3!}\left.\frac{\partial^3 f}{\partial x^3}\right|_x (\Delta x)^2 \\ =& \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} + \mathcal{O}(\Delta x^2) \end{aligned} \tag{6.10}$$

By combining these approximations, we have an approximation where the leading term in the error part is now linear in $(\Delta x)^2$. Therefore, halving Δx now results in $(\Delta x)^2 \to \frac{1}{4}(\Delta x)^2$, hence we have decreased our error by a factor 4. This is then **second-order accurate**. It would then be possible to combine additional Taylor expansions to create even more accurate approximations to the first derivative.

In figure 6.2, these first- and second-order accurate approximations are illustrated for $f(x) = x^4$ at $x = 1$. A series of different choices of Δx are taken (halving it each time) and the approximated derivative can be seen to approach the exact value in each case. However, the rate at which the second-order accurate approximation approaches the true derivative is clearly faster, as we would expect.

6.1.2 Higher derivative approximations

Note also that equations (6.7) and (6.8) can be combined to provide an approximation for the second derivative. Adding these two equations cancels the first derivative terms and gives

$$f(x+\Delta x) + f(x-\Delta x) = 2f(x) + \left.\frac{\partial^2 f}{\partial x^2}\right|_x (\Delta x)^2 + \left.\frac{\partial^4 f}{\partial x^4}\right|_x (\Delta x)^4 + \cdots \tag{6.11}$$

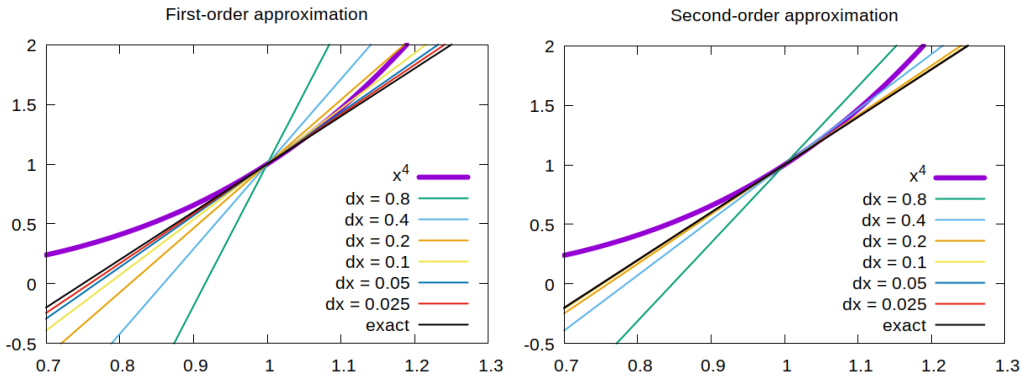


Figure 6.2: Approximations of the derivative of $f(x) = x^4$ at $x = 1$ for different choices of Δx. The left hand figure shows the first-order approximation, and the trend to the true derivative is clear. The right hand figure shows the second-order approximation. It is clear that this approaches the exact value much faster than the first-order approximation

which can be rearranged to give

$$\left.\frac{\partial^2 f}{\partial x^2}\right|_x = \frac{f(x+\Delta x) - 2f(x) + f(x-\Delta x)}{(\Delta x)^2} + \mathcal{O}(\Delta x)^2. \tag{6.12}$$

6.1.3 Quantifying order of accuracy

In figure 6.2, we demonstrate that the second-order accurate derivative approaches the exact result faster. In general, just stating that it is faster is probably not enough, we need to check that it is actually behaving as it should be. For example, from this figure, we can't actually tell by eye if the second-order accurate method truly halves each time we double Δx - it could decrease by a factor of 1.8 instead, perhaps, or 2.1. In this simple case, this might be unlikely (given how easy it is to compute the derivatives of x^4 anyway), but in general, we need to quantify error obtained from a numerical computation. If we don't do this, we can't be certain we have implemented our method correctly.

In this course, we are not going to go into detail about how you obtain the order of accuracy of a numerical method - this will be introduced with methods themselves in other courses. However, how you use this information is common across applications, hence we shall focus on this aspect of error quantification. In general, though, much like our derivative approximations, computational methods have an order of accuracy which we expect our simulation results to achieve.

If we consider our example shown in figure 6.2, for a given Δx, we have:

- The exact value of the derivative:

$$\left.\frac{\partial f}{\partial x}\right|_{x=1} = 4x^3\big|_{x=1} = 4 \tag{6.13}$$

- The first order approximation:

$$\left.\frac{\partial \hat{f}_1}{\partial x}\right|_{x=1}(\Delta x) = \left.\frac{f(x+\Delta x) - f(x)}{\Delta x}\right|_{x=1} = \frac{(1+\Delta x)^4 - 1}{\Delta x} \tag{6.14}$$

- The second order approximation

$$\left.\frac{\partial \hat{f}_2}{\partial x}\right|_{x=1}(\Delta x) = \left.\frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}\right|_{x=1} = \frac{(1+\Delta x)^4 - (1-\Delta x)^4}{2\Delta x} \tag{6.15}$$

In order to compute the accuracy of these approximations, we first need to know the error, E, which is easy to compute, we have two possible values,

$$E_1 = \left| \left.\frac{\partial f}{\partial x}\right|_{x=1} - \left.\frac{\partial \hat{f}_1}{\partial x}\right|_{x=1}(\Delta x) \right| \tag{6.16}$$

for the first-order method and

$$E_2 = \left| \left.\frac{\partial f}{\partial x}\right|_{x=1} - \left.\frac{\partial \hat{f}_2}{\partial x}\right|_{x=1}(\Delta x) \right| \tag{6.17}$$

for the second-order method. In general, the error, E, could be defined as

$$E = |g - \hat{g}| \tag{6.18}$$

where g is an analytic expression of a function, and $\hat{g}$ is an approximation.

The order of accuracy for a method, $n_{\mathcal{O}}$, satisfies

$$E(\Delta x) \leq C(\Delta x)^{n\mathcal{O}} \tag{6.19}$$

where C is a constant. For our expressions, we hope that we could find

$$E_1(\Delta x) \leq C(\Delta x)^1 \quad \text{and} \quad E_2(\Delta x) \leq C(\Delta x)^2 \tag{6.20}$$

and for simple derivations, such as Taylor series, we can say we have shown this to be the case mathematically. However, this was based on some assumptions, such as the differentiability of $f(x)$; in general, if we cannot be certain these assumptions hold, we need to compute, or at least, check, $n_{\mathcal{O}}$ by numerical experiment.

Equation (6.19) is not necessarily the most natural expression to work with; we don't necessarily know what the constant, C, actually is (and we usually don't want to try to evaluate it, even if we could). However, clearly we could consider the ratio for two different values of Δx,

$$\Delta E = \frac{E(\Delta x_1)}{E(\Delta x_2)} = \frac{C(\Delta x_1)^{n_{\mathcal{O}}}}{C(\Delta x_2)^{n_{\mathcal{O}}}} = \left(\frac{\Delta x_1}{\Delta x_2}\right)^{n_{\mathcal{O}}}. \tag{6.21}$$

Technically, the central equals sign is not correct, given that $n_{\mathcal{O}}$ was related to E through an inequality. In this case, we expect this relationship to hold in the limit $\Delta x_1 \to 0$.

Actually computing $n_{\mathcal{O}}$ is then a matter of rearranging this expression,

$$\ln(\Delta E) = \left(\frac{\Delta x_1}{\Delta x_2}\right)^{n_{\mathcal{O}}} = n_{\mathcal{O}} \ln\left(\frac{\Delta x_1}{\Delta x_2}\right) \tag{6.22}$$

or

$$n_{\mathcal{O}} = \frac{\ln(\Delta E)}{\ln\left(\frac{\Delta x_1}{\Delta x_2}\right)}. \tag{6.23}$$

When computing the order of accuracy, it is particularly common to use $\Delta x_2 = 2\Delta x_1$, or $\Delta x_1 = 2\Delta x_2$, i.e. the spacing is halved each time. For this, we then have

$$n_{\mathcal{O}} = -\frac{\ln(\Delta E)}{\ln 2} \tag{6.24}$$

for $\Delta x_2 = 2\Delta x_1$ or

$$n_{\mathcal{O}} = \frac{\ln(\Delta E)}{\ln 2} \tag{6.25}$$

for $\Delta x_1 = \Delta x_2$.

Computing order of accuracy - an example

We can use these definitions to check the accuracy of our approximations to the derivative of x^4, i.e. quantify the behaviour in figure 6.2. This is shown in tables 6.1 and 6.2. For the first order method, it is clear that initially, the error is very large, but this does converge towards the expected value. Due to the very large initial error, and the nonlinear nature of the function we

Δx	$\frac{\partial \hat{f}_1}{\partial x}\Big\vert_{x=1}(\Delta x)$	$E_1(\Delta x)$	$\Delta E_1(\Delta x)$	$n_{\mathcal{O}}$
0.8	11.872	7.872	-	
0.4	7.104	3.104	0.3943	1.34
0.2	5.368	1.368	0.4407	1.18
0.1	4.641	0.641	0.4684	1.094
0.05	4.310125	0.310125	0.4838	1.047
0.025	4.1525	0.1525	0.4918	1.024

Table 6.1: Checking the order of accuracy for a first-order approximation to the derivative of x^4 at $x = 1$. The first column shows the spacing and the second column shows the estimated value of the derivative; it is clear that this approaches the expected value of 4. The third column shows the absolute error, which is clearly decreasing as Δx decreases, and the fourth column shows the ratio between two successive values of Δx. This is converging towards $\Delta E_1 = 0.5$, and, similarly, the order of accuracy converges towards $n_{\mathcal{O}} = 1$.

Δx	$\frac{\partial \hat{f}_2}{\partial x}\Big\vert_{x=1}(\Delta x)$	$E_2(\Delta x)$	$\Delta E_2(\Delta x)$	$n_{\mathcal{O}}$
0.8	6.56	2.56	-	-
0.4	4.64	0.64	0.25	2
0.2	4.16	0.16	0.25	2
0.1	4.04	0.04	0.25	2
0.05	4.01	0.01	0.25	2
0.025	4.0025	0.0025	0.25	2

Table 6.2: Checking the order of accuracy for a second-order approximation to the derivative of x^4 at $x = 1$. The first column shows the spacing and the second column shows the estimated value of the derivative; it is clear that this rapidly approaches the expected value of 4. The third column shows the absolute error, which is clearly decreasing as Δx decreases, and the fourth column shows the ratio between two successive values of Δx. This is consistently $\Delta E_1 = 0.25$, and, similarly, the order of accuracy consistently is the expected value of $n_{\mathcal{O}} = 2$.

are approximating, we see that the order of accuracy initially appears to be slightly greater than the expected $n_{\mathcal{O}} = 1$. However, as Δx decreases, it is clear that we are tending towards the correct value. For the second order method, the behaviour is almost perfectly as expected; the error decreases consistently according to $n_{\mathcal{O}} = 2$.

Another common way to demonstrate order of accuracy is through plot-

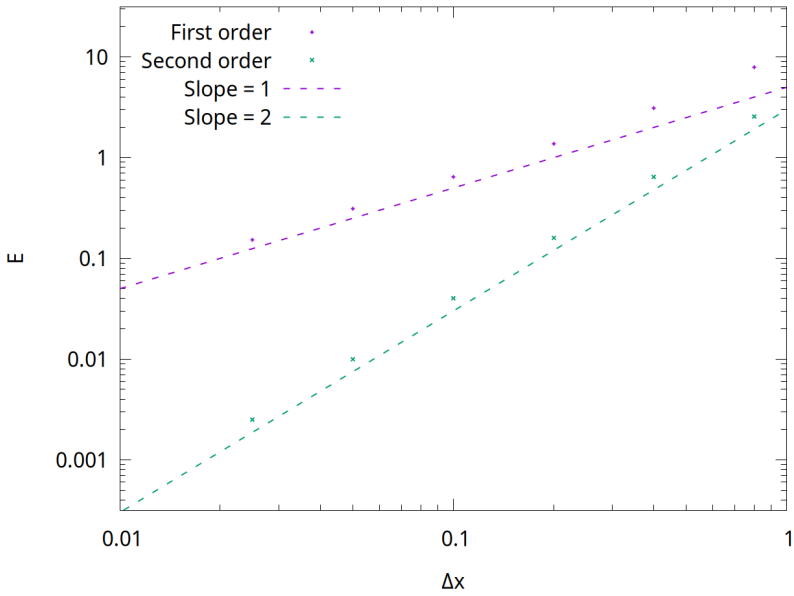


Figure 6.3: Demonstrating a graphical representation of the order of accuracy of different approximations for the derivative of x^4 at $x = 1$ by comparing error, E, for a range of Δx. The computed results are shown by the points whilst the lines show slopes of 1 and 2. A method that converges at first-order should have slope 1 on this log-log plot, and we see that the results do tend towards this line as Δx decreases. Note that decreasing Δx involves reading from right-to-left in this sort of plot. Similarly, the second order method follows the line of slope 2 as expected. A final note is that we don't need the points to actually overlap with the line to demonstrate convergence - only the slope is important, and in practice, we allow for a slight offset between points and lines to help distinguish between the two.

ting the error, E, as it changes with Δx. This is demonstrated in figure 6.3, where the error is plotted for both first- and second-order methods. Because the order of accuracy appears as an exponent in equation (6.19), these plots should be made with a log-log scale. This means we are effectively plotting according to

$$\ln(E) = \ln C + n_{\mathcal{O}} \ln(\Delta x) \tag{6.26}$$

and the slope of the points should reveal the order of accuracy (and, technically, the value of C). In practice, we only care about the order of accuracy, and we simply compare the points to lines of the appropriate slope.

p-norms and error norms

So far, we have considered error computed for a single value. Scientific computing, however, rarely involves finding a single value as accurately as possible, and we are typically working with larger systems, such as clusters of atoms or discrete points in a computational mesh. Additionally, we may not have a single source of error; for a PDE, we need to discretise time and each spatial dimension, and each of these could contribute error to the system. In these cases, being able to identify convergence to the correct solution (assuming you even know the correct solution) can be harder, and quantitative analysis is important.

In general, complex numerical methods will have a known order of accuracy (and for many, you won't want to derive this yourself), but practical implementations may not provide the expected decrease in error. There are many reasonss for this, including

1. There are other errors (e.g. from other derivatives) you have neglected in the analysis
2. The method is actually unstable - order of accuracy and stability are not the same thing
3. The tests you are working with may not be as well behaved as the theory behind the method's analysis
4. There is a bug in your code

The first of these is a matter of oversight, and is often obvious once you investigate your system. The second of these will be investigated in the context of specific numerical methods in other courses. The third of these is an related to accuracy and stability, but is worth discussing. In many cases, order of accuracy analysis (and stability analysis) take place assuming linear, or at least, differentiable behaviour. In strongly nonlinear problems, we may find that some error approximations do not provide good results, and we shall discuss what we mean by different error approximations shortly.

The final point in the list above is one of the key reasons for conducting this analysis; it is possible for small bugs to exist in code such that reasonable answers are produced, but they do not behave correctly under increased accuracy - sometimes unphysical features may grow, or errors never get smaller. Such issues are often difficult to identify without this quantitative analysis.

When we considered error for a single point, there was only one way to approach this. However, if we assume we have a set of points, $a_1, a_2, \ldots a_n$,

we can consider a vector of points, and as such, a vector of errors. In which case, if we want an overall error measurement, we need to consider the size of this vector, or, in more mathematical terminology, we need to compute a vector norm. There are some obvious options, for example, we often talk about the magnitude of a vector,

$$|\mathbf{a}| = (\mathbf{a} \cdot \mathbf{a})^{1/2} = \left(a_1^2 + a_2^2 + \ldots + a_n^2\right)^{1/2}. \tag{6.27}$$

Alternatively, we could just consider a simple sum,

$$||\mathbf{a}||_1 = |a_1| + |a_2| + \ldots + |a_n|. \tag{6.28}$$

Here, we have introduced some notation, $||\cdot||_p$, which is a general form for a p-norm, here, with $p = 1$. From these two expressions, we can start to see the general form for a p-norm;

$$||\mathbf{a}||_p = \left(|a_1|^p + |a_2|^p + \ldots + |a_n|^p\right)^{1/p}. \tag{6.29}$$

By itself, the p-norm of a vector is not a good measure of error. If we applied a method to a system of 100 atoms, and later to a system of 1000 atoms, we would expect to be able to record the same error in the method if it was working correctly. However, using p-norms alone would result in many more terms being considered in the summation, and thus a larger overall norm for the vector. As a result, we need to weight, or normalise, the norm, and this is the concept of error norms.

As an example, we will consider that the components of $\mathbf{a}$ are an approximate solution to a function $A(x)$ computed at positions $x = (x_1, x_2, \ldots, x_N)$. We know, then, that the error at a given point, x_i is

$$E_i = |B(x_i) - b_i|. \tag{6.30}$$

In this case, we can weight our error by the spacing between points. The precise method for doing this can be dependent on how the data is discretised at these points, so for this example, we shall assume that each point is uniformly distributed between x_1 and x_N. An error norm is then given by

$$||\mathbf{E}||_p = \left(\frac{x_N - x_1}{N} \sum_{i=1}^{N} |E_i|^p\right)^{1/p}. \tag{6.31}$$

In theory, any error norm could provide a useful measure of error, but it is unlikely that you will encounter anyone computing the 4-norm, or the 23-norm of a solution. There are generally three useful error norms:

- The 1-norm,

$$||\mathbf{E}||_1 = \frac{x_N - x_1}{N} \sum_{i=1}^{N} |E_i| \,. \tag{6.32}$$

This is particularly useful in non-linear problems, where discontinuities and sharp changes are encountered.

- The 2-norm,

$$||\mathbf{E}||_2 = \left(\frac{x_N - x_1}{N} \sum_{i=1}^{N} |E_i|^2 \right)^{1/2} . \tag{6.33}$$

This is more common in linear problems, and, overall, if error is stated without the appropriate norm being given, it is probably the 2-norm. The reason for this is that this norm is most commonly associated with magnitude, and thus is often considered a 'natural' choice. Additionally, outside of direct error analysis, the 2-norm has useful geometric properties.

The reason, then, that non-linear problems don't use the 2-norm needs addressing. The squaring of the error in the 2-norm puts a greater emphasis on the larger error. Overall, we would expect error to decrease everywhere, and this not to be an issue, but this is not always true around strong nonlinear features and discontinuities. In fact, some methods need to locally decrease their order of accuracy around such features to maintain stability. As a result, we may find that a few points within the solution don't obey the correct order of accuracy, but **globally**, the correct behaviur is observed. As a result, using the 2-norm could give misleading results to the global behaviour of the method, and hence the 1-norm is common in these cases.

- The ∞-norm,

$$||\mathbf{E}||_\infty = \max_i |E_i| \,. \tag{6.34}$$

This norm is obtained in the limit $p \to \infty$, where it is noted that

$$\lim_{p\to\infty} \sum_{i=1}^{N} |E_i|^p = \left(\max_i |E_i| \right)^p \tag{6.35}$$

and

$$\lim_{p\to\infty} \left(\frac{x_N - x_1}{N} \right)^{1/p} = 1 \tag{6.36}$$

hence

$$\lim_{p\to\infty}\left(\frac{x_N - x_1}{N}\sum_{i=1}^{N}|E_i|^p\right)^{1/p} = \lim_{p\to\infty}\left[\left(\max_i|E_i|\right)^p\right]^{1/p} = \max_i|E_i|\,. \tag{6.37}$$

This is the strictest measure of error - if it doesn't decrease everywhere, then the expected order of convergence will not be achieved. As a result, it is used primarily in smooth problems, where no nonlinear effects are expected.

Once we have an error norm, the order of accuracy can be computed in the same manner as if we had a single error value, i.e. equation (6.19) becomes

$$||\mathbf{E}||_i(\Delta x) = C(\Delta x)^{n_\mathcal{O}} \tag{6.38}$$

where Δx is now directly related to the scaling that went into the error norm, for example, in the case above, we have

$$\Delta x = \frac{x_N - x_1}{N}. \tag{6.39}$$

We then also have

$$||\Delta E||_p = \frac{||E(\Delta x_1)||_p}{||E(\Delta x_2)||_p} = \frac{C(\Delta x_1)^{n_\mathcal{O}}}{C(\Delta x_2)^{n_\mathcal{O}}} = \left(\frac{\Delta x_1}{\Delta x_2}\right)^{n_\mathcal{O}}. \tag{6.40}$$

The final thing to note in this section is that we have always assumed that there is an exact solution available. This is not always going to be the case, and we may still want to compute an order of accuracy. In this case, we need a reference solution, e.g. a solution with a small Δx, certainly smaller than Δx_1 and Δx_2. This can identify order of accuracy, but it may not guarantee that our method is actually approaching the true solution to the underlying equations, and more work will be needed to demonstrate this.

6.2 Fourier series

We have discussed how we might find the order of accuracy of a method. We will find, in practice, that knowing the order of accuracy is not a guarantee that the method will actually provide a solution. For that, additional analysis is needed, and this might require other approximations. Recall that Fourier series represent a periodic function, with periodicity L, through

$$g(x) = A_0 + \sum_{n=1}^{\infty}\left(A_n\cos\left(\frac{2\pi n x}{P}\right) + B_n\sin\left(\frac{2\pi n x}{P}\right)\right) \tag{6.41}$$

where

$$A_0 = \frac{1}{P}\int_{-P/2}^{P/2} g(x)\mathrm{d}x, \tag{6.42}$$

$$A_n = \frac{2}{P}\int_{-P/2}^{P/2} g(x)\cos\left(\frac{2\pi n x}{P}\right)\mathrm{d}x \tag{6.43}$$

and

$$B_n = \frac{2}{P}\int_{-P/2}^{P/2} g(x)\sin\left(\frac{2\pi n x}{P}\right)\mathrm{d}x. \tag{6.44}$$

The overall idea is that it could be possible to represent error through a Fourier series, and find out how A_n and B_n can be chosen such that the error is **bounded**, i.e. it doesn't grow. If this is true for all n, then our numerical technique can tell us that our numerical method **will converge** towards a solution. Note that we don't know whether this is the correct solution, at this stage, that requires additional work still.

These techniques will be discussed in other courses, in the context of specific methods, here, we want to note one general thing - it is not ideal to have to work with the coefficients A_n and B_n in convergence analysis. Ideally, we'd like to do a single calculation to show a method will converge, and in this form, we'd need to do two. Fortunately, we can write a Fourier series as a sum over an exponential function.

6.2.1 Exponential form of the fourier series

In order to derive a more useful form of a Fourier series for convergence analysis, we define new coefficients,

$$C_0 = A_0, \quad C_n = \frac{1}{2}(A_n - iB_n), \quad C_{-n} = \frac{1}{2}(A_n + iB_n). \tag{6.45}$$

We will see that the definition of the C_{-n} coefficient (rather than another variable, D_n, is useful. Note we can also write the original Fourier series coefficients as

$$A_0 = C_0, \quad A_n = C_n + C_{-n}, \quad B_n = i(C_n - C_{-n}). \tag{6.46}$$

If we now consider the typical Fourier series expansion of a function $f(x)$,

and substitute in these new constants, we have

$$\begin{aligned} f(x) &= A_0 + \sum_{n=1}^{\infty} \left(A_n \cos\left(\frac{2\pi nx}{P}\right) + B_n \sin\left(\frac{2\pi nx}{P}\right)\right) \\ &= C_0 + \sum_{n=1}^{\infty} \left[(C_n + C_{-n}) \cos\left(\frac{2\pi nx}{P}\right) + i(C_n + C_{-n}) \sin\left(\frac{2\pi nx}{P}\right)\right] \\ &= C_0 + \sum_{n=1}^{\infty} C_n \left(\cos\left(\frac{2\pi nx}{P}\right) + i \sin\left(\frac{2\pi nx}{P}\right)\right) \\ &\quad + \sum_{n=1}^{\infty} C_{-n} \left(\cos\left(\frac{2\pi nx}{P}\right) - i \sin\left(\frac{2\pi nx}{P}\right)\right) \end{aligned} \tag{6.47}$$

For the C_n term in this expression, we note that we can use

$$e^{ix} = \cos x + i \sin x \tag{6.48}$$

whilst for the C_{-n} term, we first make the substitution $n = -m$, giving

$$\begin{aligned} f(x) = C_0 &+ \sum_{n=1}^{\infty} C_n \exp\left(\frac{2\pi inx}{P}\right) \\ &+ \sum_{m=-\infty}^{1} C_m \left(\cos\left(\frac{-2\pi mx}{P}\right) - i \sin\left(\frac{-2\pi mx}{P}\right)\right). \end{aligned} \tag{6.49}$$

Now using the fact that cos is an even function, and sin is an odd function, we get

$$\begin{aligned} f(x) &= C_0 + \sum_{n=1}^{\infty} C_n \exp\left(\frac{2\pi inx}{P}\right) \\ &\qquad + \sum_{m=-\infty}^{1} C_m \left(\cos\left(\frac{2\pi mx}{P}\right) + i \sin\left(\frac{2\pi mx}{P}\right)\right) \\ &= C_0 + \sum_{n=1}^{\infty} C_n \exp\left(\frac{2\pi inx}{P}\right) + \sum_{m=-\infty}^{1} C_m \exp\left(\frac{2\pi inx}{P}\right) \\ &= \sum_{n=-\infty}^{\infty} C_n \exp\left(\frac{2\pi inx}{P}\right) \end{aligned} \tag{6.50}$$

where, for the final step, we renamed $m = n$, and used the fact that in this expression, when $n = 0$, $e^0 = 1$, so the C_0 term can be brought inside the summation.

In this form, convergence analysis could find that C_n is bounded, for the general case.

6.3 Interpolation

Working with discrete quantities is useful for identifying error and convergence, but sometimes we are forced to work with these quantities when we'd rather not. For example, material properties may be known only for specific data points, e.g. at specific temperatures. In a simulation, however, we are likely to need information between these points - we need to interpolate values. Common uses of interpolation include:

- Identifying the location of a material interface
- Plotting or finding contours
- Determining material properties from tabulated data
- Transferring data between two different meshes

Many methods for interpolation exist, and these can be directly related to Taylor series, as in order to interpolate information between two points, we need to approximate the derivative between these points. For example, if we have a function $y(x)$, and we have two known values, $y(x_1) = y_1$ and $y(x_2) = y_2$, then we can approximate our derivative

$$\frac{\mathrm{d}y}{\mathrm{d}x} \approx \frac{y(x_2) - y(x_1)}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \tag{6.51}$$

which we know is first-order accurate. This expression could be from expanding $y(x)$ about $x = x_1$, and then inserting the values at x_2, or vice versa, we'd get the same expression either way. Interpolation is effectively finding the equation of the line which passes through (x_1, y_1) and (x_2, y_2). This is given by

$$\hat{y}(x) \approx y_1 + (y_2 - y_1)\,\frac{x - x_1}{x_2 - x_1} \tag{6.52}$$

which can be easily obtained from the general equation of a straight line, $y = mx + c$, and it is also clear that this expression yields $\hat{y}(x_1) = y_1$ and $\hat{y}(x_2) = y_2$. For any value $x_1 \leq x \leq x_2$, $\hat{y}$ is a **linear interpolation** of the value.

Note that this equation could also be written as

$$\begin{aligned}\hat{y}(x) \approx & y_1 + (y_2 - y_1)\frac{x - x_1}{x_2 - x_1} \\ &= y_1\frac{x_2 - x_1}{x_2 - x_1} + (y_2 - y_1)\frac{x - x_1}{x_2 - x_1} \\ &= \frac{x_2y_1 - x_1y_1 + xy_2 - xy_1 - x_1y_2 + x_1y_1}{x_2 - x_1} \\ &= \frac{x_2 - x}{x_2 - x_1}y_1 + \frac{x - x_1}{x_2 - x_1}y_2 \end{aligned} \tag{6.53}$$

Higher-order approximations do exist, for example, if we know $y(x)$ and three points, we could obtain a second-order accurate approximation to the derivative. This derivative effectively takes information from three points, though, for example, if we have equally spaced $x_1 < x_2 < x_3$, and we know y_1, y_2 and y_3, then we could approximate the derivative at x_2 using equation (**??**) as

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x_2} = \frac{y(x_3) - y(x_1)}{2(x_3 - x_1)} = \frac{y_3 - y_1}{2(x_3 - x_1)}. \tag{6.54}$$

The interpolation would now come from a line through (x_1, y_1) and (x_3, y_3) which has the gradient given above at x_2; we need to fit three pieces of information, in general, a quadratic equation could do this. More generally, spline interpolation exists for higher orders of accuracy.

In practice, many applications will not worry about higher-order extrapolation, and linear will be sufficient. Reasons for this include

- Interpolation is often used as part of a computational method - it may need to be done repeatedly, and linear interpolation is the most efficient (requires the fewest calculations)
- Linear interpolation has no issues at the edge of the data range - higher order methods need more points, and these may not always be defined at the extrema
- Some applications, for example, interpolation of material interfaces, are done where numerical methods may be locally first-order accurate anyway, additional interpolation accuracy doesn't achieve anything
- When interpolating from tabulated data, it is often worth creating high-resolution data in the table, such that absolute error is low, even for linear interpolation (this is related to efficiency)

- If data contains discontinuities, higher-order functions may show unphysical overshoots or undershoots which could actually lead to more error in values than linear interpolation

This will, of course, not always be the case, for example, interpolation between meshes may need to be done at greater-than-first-order to avoid adding additional error to a simulation.

6.3.1 Interpolation in multiple dimensions

Linear interpolation extends naturally to higher dimensions, effectively by considering one dimension at a time. Consider the two-dimensional function $z = z(x, y)$, and assume that we know four values, $z_{11} = z(x_1, y_1)$, $z_{12} = z(x_1, y_2)$, $z_{21} = z(x_2, y_1)$ and $z_{22} = z(x_2, y_2)$. If we want to interpolate the value $z(x, y)$, with $x_1 \leq x \leq x_2$ and $y_1 \leq y \leq y_2$, we can do so by first treating this as a 1D problem at y_1, and y_2. For this, we can interpolate a value at (x, y_1) through equation (6.53)

$$z(x, y_1) \approx \frac{x_2 - x}{x_2 - x_1} z(x_1, y_1) + \frac{x - x_1}{x_2 - x_1} z(x_2, y_1), \tag{6.55}$$

and similarly at (x, y_2) as

$$z(x, y_2) \approx \frac{x_2 - x}{x_2 - x_1} z(x_1, y_2) + \frac{x - x_1}{x_2 - x_1} z(x_2, y_2). \tag{6.56}$$

This then gives us two values for z and (x, y_1) and (x, y_2), so we can simply perform another linear interpolation between them,

$$z(x, y) \approx \frac{y_2 - y}{y_2 - y_1} z(x, y_1) + \frac{y - y_1}{y_2 - y_1} z(x, y_2). \tag{6.57}$$

This is **bilinear interpolation**, and we note that it does not matter whether we performed the initial linear interpolation treating the y-direction as constant or the x-direction as constant, we would get the same expression either way. And in 3D, the concept extends naturally to **trilinear interpolation**.

Note that in setting up this problem, we assumed that our four points were the vertices of a rectangle. This is not a requirement for the bilinear interpolation; any convex quadrilateral mesh could be used, but in these cases, a mapping back to a rectangular grid may be needed.

It is again possible to go beyond first-order accurate in higher dimensions, typically bicubic interpolation would be used in 2D and tricubic in 3D. These do come with greater algorithm costs; the number of points needed squares in these algorithms (16 instead of 4 points in 2D, and 64 instead of 8 in 3D).

6.4 Root finding

Throughout the MPhil in Scientific Computing, you will encounter various numerical methods and algorithms, many of which are directly tied to applications, and involve complex systems of PDEs or ODEs. These methods will be lectured in dedicated courses, and are too specific to focus on here. However, there are other standard numerical techniques which are common across all areas of scientific computing, and often provide some fundamental understanding of more advanced techniques. One of the simplest such methods is root finding - solving an equation to find x such that $f(x) = 0$.

Such methods are important, since physical variables do not always follow straightforward relationships. In this course, we have regularly used an ideal gas as an example, considering relationships such as

$$pV = nRT. \tag{6.58}$$

Root finding on such an equation could be finding the value of p which satisfies

$$nRT - pV = 0 \tag{6.59}$$

and clearly, for this equation, this is trivial. However, the ideal gas is a simple expression (which is why it comes up in so many examples). Many materials have much more complicated expressions for their properties, for example

$$\varepsilon = \frac{K_0}{2\alpha_R^2}\left(I_3^{\alpha_R/2} - 1\right)^2 + c_V T_0 I_3^{\gamma_R/2}\left(e^{s/c_v} - 1\right) + \frac{b_0^2}{2} I_3^{\beta_R/2}\left(\frac{I_1^2}{3} - I_2\right) \tag{6.60}$$

which relates specific internal energy, ε, in some solids to entropy, s, and density, where I_1, I_2 and I_3 are invariants of a matrix related to density. Everything else in this expression is a known constant, which is typically computed experimentally. Now, if we wanted to find the value of I_3 which satisfies this expression, is is not straightforward, since the expression contains both $I_3^{\beta_R/2}$ and $I_3^{\gamma_R/2}$.

This is where iterative root finding techniques are valuable; if we can't solve for I_3, we can make a guess, and keep refining this guess until we are happy with the accuracy.

Before we look at any methods, we first consider what we mean by being 'happy with the accuracy'.

If our equation is simple to solve, e.g. the ideal gas law, we can identify an **analytic solution**, and this theoretically contains no error. We say theoretically, since computational round off errors could be an issue if the solution required, e.g. multiplication and division by a mix of very large

numbers and very small numbers. The computational lectures in the MPhil contain more details about this issue. Here, we shall assume that an analyitic solution is known perfectly (or, at least, to machine precision).

An equation like equation (6.60) has an **exact solution** for e.g. I_3 (if all other quantities are known), but this may not be analytic. This means that there is a single number which could satisfy the equation, but we may not be able to rearrange an equation to obtain it.

Iterative root finding methods are then methods such that this root can be found to an arbitrary degree of accuracy, and this is what we mean by being happy with the solution. We are tasked with determining when a solution is accurate enough, where, computationally, this will often be as close to machine precision as we can achieve.

A lot of root finding methods exist - we shall consider some of the most commonly encountered ones.

6.4.1 Bisection method

The bisection method is perhaps the first root finding method that is normally encountered, it is both simple and reliable, though the downside is that it is not the most efficient root finding approach.

We will assume that we are trying to solve equation (6.60) for I_3 (with all other variables known). From the root-finding perspective, we need to find

$$f(I_3) = \varepsilon - \frac{K_0}{2\alpha_R^2}\left(I_3^{\alpha_R/2} - 1\right)^2 + c_V T_0 I_3^{\gamma_R/2}\left(e^{s/c_v} - 1\right) + \frac{b_0^2}{2} I_3^{\beta_R/2}\left(\frac{I_1^2}{3} - I_2\right) = 0. \tag{6.61}$$

To start the algorithm, we need to two find values of I_3, one where $f(I_3) \geq 0$ and one where $f(I_3) \leq 0$. Note that if we manage to get equality in either of these guesses, we've found our root without any iterations needed! Generally, this is unlikely.

Once we have these two values, we know that providied our function is continuous between these ranges, at least one root exists between them. The bisection algorithm is guaranteed to then find one root to this equation.

If we assume we have found a such that $f(a) < 0$ and b such that $f(b) > 0$ (note, this doesn't mean $a < b$), the algorithm then works by:

1. Take the average value of a and b,

$$c = \frac{1}{2}(a + b) \tag{6.62}$$

2. Evaluate $f(c)$

3. Check whether $f(c)$ is sufficiently close to zero. This can be done either through
$$|f(c)| < \epsilon \tag{6.63}$$
where ϵ is the desired accuracy of your root, or
$$\left|\frac{f(c)}{c}\right| < \epsilon. \tag{6.64}$$
These two options exist because it can be good to define your accuracy relative to the size of c. If we set $\epsilon = 10^{-12}$, then the first option, the absolute difference, is significantly harder to reach for $c \approx 10^5$ that if $c \approx 0.01$.

   If you are satisfied c is accurate enough, you have found the root and can stop the bisection algorithm.

4. Otherwise, you will replace either a or b with c.
$$\begin{cases} f(c) < 0: & \text{set } a = c \\ f(c) > 0: & \text{set } b = c \end{cases} \tag{6.65}$$
   In either case, we still have one value greater than zero, and one value less than zero, so a root still exists between the values, but either of the new ranges, $[a, c]$ or $[c, b]$, will be smaller than $[a, b]$, thus we are closer to finding our root.

5. Now, we restart the algorithm from the top, until we do converge.

In figure 6.4, the bisection algorithm is illustrated. Here, each step of the method is clearly narrowing the region which contains the root. Note that it is not general behaviour that the new guesses, c, will always be closer to the root. In the final image in figure 6.4, if the root is not sufficiently accurate, the next guess will actually be further from the true root. This does not mean the method is diverging, though, but does show some of the potential inefficiency - it could be quick to reduce the range around the root from one side, but comparatively slow from the other.

A couple of things are worth mentioning about the practical use of the bisection method. Firstly, the guarantee that it will find a root may seem unhelpful in cases where there are multiple roots. However, in practice, multiple roots to a function for physical relationships are less common, but not unheard of. Many physical relationships are **monotonically increasing** or **monotonically decreasing**, and, as such, we always know we have one

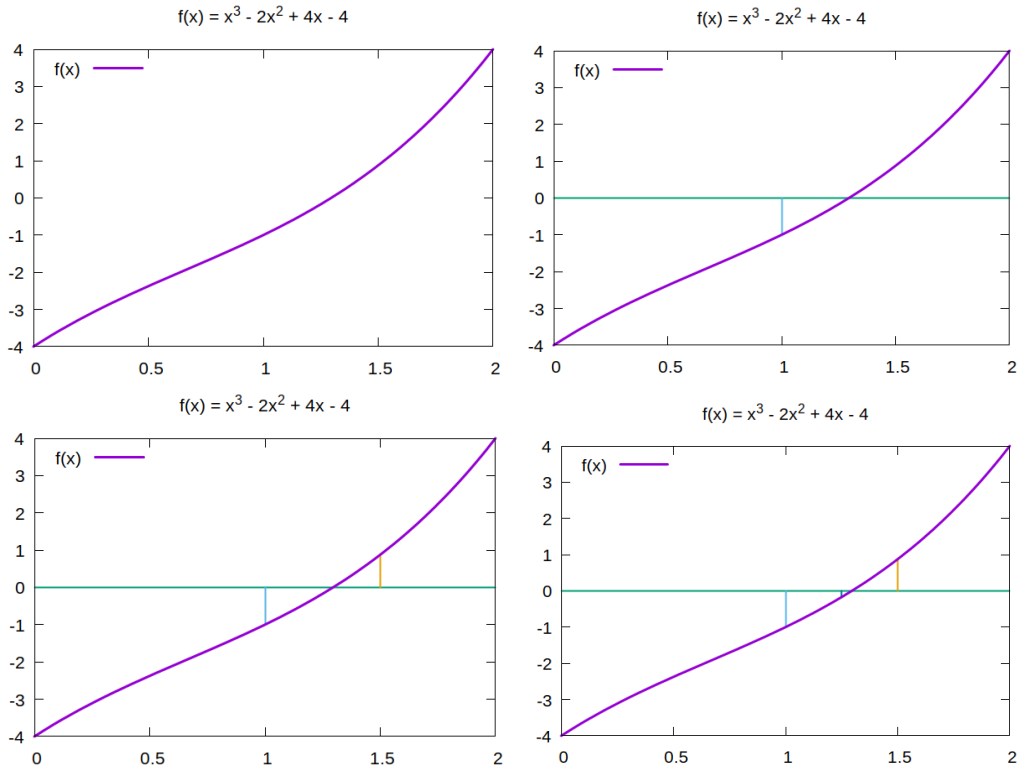


Figure 6.4: First stages of a bisection algorithm finding the roots to $x^3 - 2x^2 + 4x - 4 = 0$. Each vertical line represents the guess at the average of the two known values which contain a root.

root available. In cases where this isn't true, then additional information will always be needed to ensure your root is physical. For example, the density of water at temperatures close to freezing is non-monotonic. For a given pressure, two possible densities exist for different temperatures. Solving the roots for such an equation then requires additional information about another variable, such as temperature, to ensure you have the correct value. In such a case, if you find the incorrect root, you may be able to use this as a starting point to finding the correct root.

The second thing to mention is how do you choose your roots. The closer you are to the true root, the quicker the algorithm will be, in general. Typically, you can assume that you know a previous value of your variable, e.g. you know I_3 from earlier in your simulation. This is a good option for one of your roots. You will probably also know some information about the behaviour of the function - is it increasing or decreasing? How quickly does this happen? This can give you information about how to choose the second

root. For example, if it is monotonically increasing, you might guess a second value of $2I_3$, and test whether $f(I_3)$ does have the opposite sign. The worst case is you then have to double your guess a few more times until you find your starting values.

6.4.2 Newton's method

The relative inefficiency of the bisection method can be a serious issue. Typically, if a numerical method requires root finding, it does not require this once, but needs it repeatedly (thousands, or even millions, of times per simulation). Efficiency is a big issue in this case. Under certain measures, the bisection method can be considered the best method that is **guaranteed** to find a root. But if we have a method that is reasonably reliable, but significantly faster, this would mean more efficient simulations. Such methods do exist, and one of the most popular is Newton's method. The idea behind this method is that we can follow the slope of the function (i.e. its derivative) and this should bring us closer to the true root. Of course, this does come with the provision that the function we are trying to find the root of is differentiable.

Add NR pictures here?

Simple derivation

Newtons method requires a single initial guess, so if we are trying to solve $f(x) = 0$, we start with our guess, x_n, which has a value $f(x_n)$. If the derivative, $f'(x_n)$, is non-zero, then we have the tangent line to $f(x)$ defined by

$$t(x) = f'(x_n)x + c = f'(x_n)x + (f(x_n) - x_n f'(x_n)) \tag{6.66}$$

where we have chosen the constant to satisfy $t(x_n) = f(x_n)$. There is then a point, x_{n+1}, where $t(x_{n+1}) = 0$, or

$$0 = f'(x_n)x_{n+1} + (f(x_n) - x_n f'(x_n)) \tag{6.67}$$

or

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \tag{6.68}$$

Once we have a new x_{n+1}, we can check if this is sufficiently close to the root, and if not, we use this to compute x_{n+1}, and so on until the method converges.

Newton's method is illustrated in figure 6.5, where we see that the convergence to the correct root is indeed very rapid.

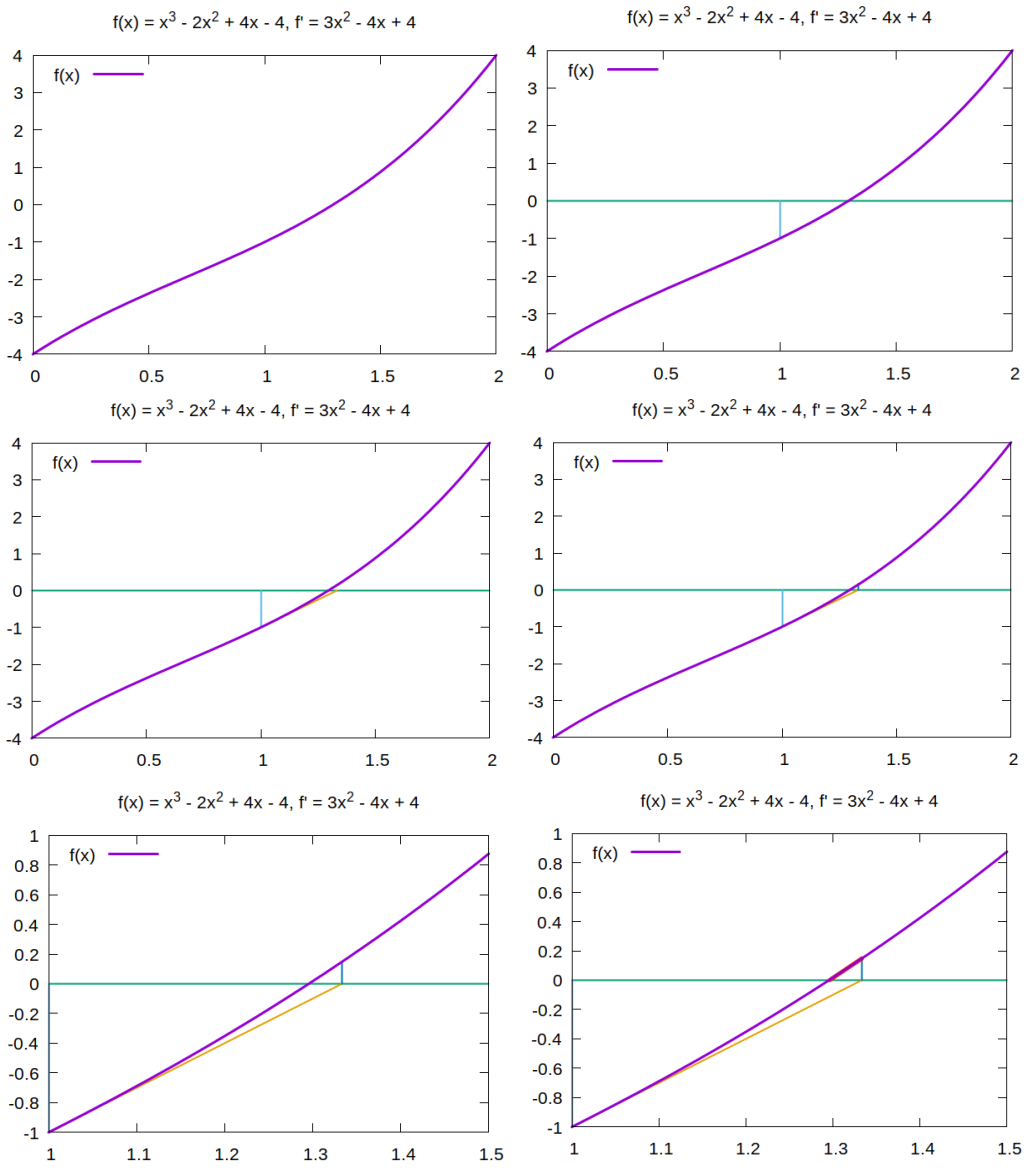


Figure 6.5: Subsequent iterations of Newton's method. The illustrations show first the identifcation of a point, and then following the tangent line back to the $y = 0$ axis. The fourth and fifth plots are the same result, but one is zoomed in to show how close to the root we are.

Derivation from Taylor series

In addition to the intuitive derivation for the Newton-Raphson method given in lectures, it can also be derived through a Taylor series expansion. If we consider our initial guess, x_0, and make a taylor expansion about this guess,

we get

$$f(x) = f(x_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0} (x - x_0) + \frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x_0} (x - x_0)^2 + \frac{1}{3!}\left.\frac{\partial^3 f}{\partial x^3}\right|_{x=x_0} (x - x_0)^3 + \dots \tag{6.69}$$

and we now consider a new point, $x = x_1$ such that $f(x_1) = 0$. Keeping only the first few terms, we can say

$$f(x_1) = 0 \approx f(x_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0} (x_1 - x_0) + \dots \tag{6.70}$$

which can then be rearranged to give an approximate value for x_1 as

$$x_1 = x_0 - \frac{f(x_0)}{\left.\frac{\partial f}{\partial x}\right|_{x=x_0}}. \tag{6.71}$$

We then repeat this process, obtaining the same formula as given in equation (6.68),

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \tag{6.72}$$

Convergence

In order to compute the order of convergence of Newton's method, we first note that the Taylor series also exists for the derivative of a function, i.e.

$$f'(x) = f'(a) + (x - a) f''(a) + \frac{1}{2}(x - a)^2 f'''(a) + \frac{1}{3!}(x - a)^3 f^{(4)}(a) + \dots \tag{6.73}$$

We now consider the root to be denoted by $\bar{x}$, i.e. $f(\bar{x}) = 0$. The error in iteration $n + 1$, ϵ_{n+1}, is defined by the distance between x_{n+1} and $\bar{x}$, i.e.

$$\epsilon_{n+1} = x_{n+1} - \bar{x}. \tag{6.74}$$

Inserting the equation for Newton's method, equation (6.68), gives

$$\epsilon_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} - \bar{x} = \epsilon_n - \frac{f(x_n)}{f'(x_n)}. \tag{6.75}$$

Now we replace $f(x)$ and its derivative with Taylor expansions about $\bar{x}$,

$$\begin{aligned} f(x_n) &= f(\bar{x}) + (x_n - \bar{x}) f'(\bar{x}) + \frac{1}{2}(x_n - \bar{x})^2 f''(\bar{x}) + \dots \\ &= \epsilon_n f'(\bar{x}) + \frac{1}{2}\epsilon_n^2 f''(\bar{x}) + \dots, \end{aligned} \tag{6.76}$$

where we use $f(\bar{x}) = 0$ and equation (6.74), and

$$f'(x_n) = f'(\bar{x}) + \epsilon_n f''(\bar{x}) + \dots \tag{6.77}$$

Inserting these expressions in equation (6.75), and factoring out a multiple of $f'(\bar{x})$ on the bottom of the fraction gives

$$\epsilon_{n+1} = \epsilon_n - \frac{\epsilon_n f'(\bar{x}) + \frac{1}{2}\epsilon_n^2 f''(\bar{x}) + \dots}{f'(\bar{x})\left[1 + \epsilon_n \frac{f''(\bar{x})}{f'(\bar{x})}\right] + \dots} \tag{6.78}$$

and we can then use the binomial expansion on the term in square brackets, which gives

$$\epsilon_{n+1} = \epsilon_n - \left(\epsilon_n + \frac{1}{2}\epsilon_n^2 \frac{f''(\bar{x})}{f'(\bar{x})} + \dots\right)\left(1 - \epsilon_n \frac{f''(\bar{x})}{f'(\bar{x})} + \dots\right) \tag{6.79}$$

$$= \epsilon_n - \left(\epsilon_n + \frac{1}{2}\epsilon_n^2 \frac{f''(\bar{x})}{f'(\bar{x})} - \epsilon_n^2 \frac{f''(\bar{x})}{f'(\bar{x})} + \mathcal{O}\left(\epsilon_n^3\right)\right) \tag{6.80}$$

$$= \frac{1}{2}\epsilon_n^2 \frac{f''(\bar{x})}{f'(\bar{x})} + \mathcal{O}\left(\epsilon_n^3\right) \tag{6.81}$$

where only terms up to ϵ_n are kept in the simplifications above. Therefore the error for iteration $n+1$ is proportional to the square of the error for iteration n. Provided these errors are small, (and the derivatives exist, and we are not at a stationary point where $f'(\bar{x}) = 0$), then we have quadratic convergence.

6.4.3 Bad choices for Newton's method

The convergence of Newton's method was dependent on ϵ_n in equation (6.81) being small. If this is not the case, we might find convergence to be very slow, or non-existent. This is particularly likely if the derivative points away from the root.

If we consider a variant of the function investigated in figure 6.5,

$$g(x) = x^3 + 2x^2 - x - 4 \tag{6.82}$$

we see that it is possible to have a case where convergence takes a very long time to occur. We enter a near cyclical behaviour which drifts a sequence of three guesses slowly further from the root. This drift continues until all three guesses in the cycle are beyond the local minimum of the function, at which point we can finally break the cycle and converge.

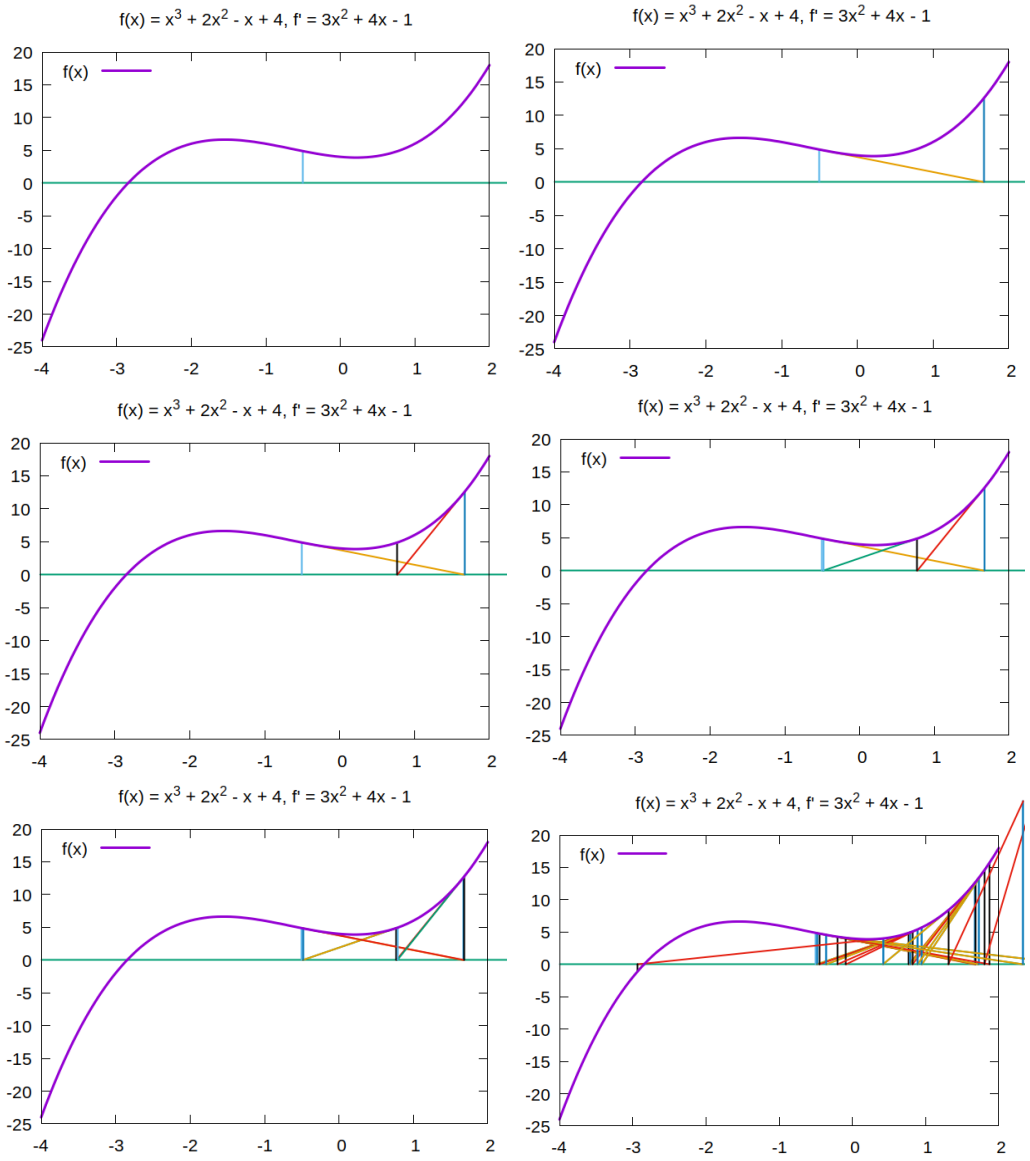


Figure 6.6: Illustration of a bad choice for the initial guess for Newton's method. The derivative at our initial guess points away from the root, and this leads to our second guess now being further away. The third guess is now slightly closer, and so is the fourth guess (only just). This leads to near cyclical behaviour around three guesses, we get further away, then closer, then closer again, but each time there is an overall drift outwards of this cycle. This continues until there the third part of this cycle actually passes the local minimum, at which point, its derivative now points in the correct direction, and we do eventually converge.

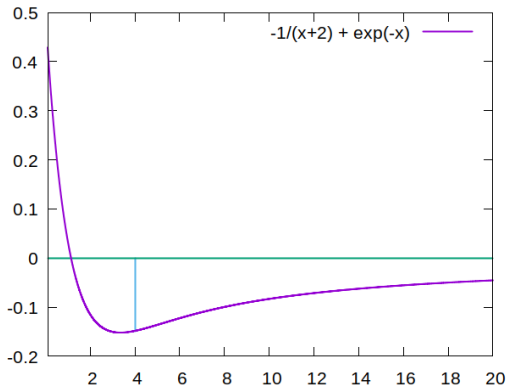


Figure 6.7: An illustration of an initial guess for a function that will never converge. Beyond the local minimum at $x \approx 3$, the derivative of the function always points away from the single root, so the algorithm will never converge.

Figure 6.7 demonstrates a worse case; an initial guess that will never converge. If a function has a region where the slope is such that the next guess will always be further from the root, it can never converge.

Although Newton's method can fail, or take a long time to converge, it is still extremely valuable in scientific computing. Firstly, a lot of functions are well-behaved so that roots are fairly easy to find. This is particularly true if you have a starting guess taken from the solution to the last time the method was used. Additionally, because Newton's method is quick to converge, when it works, it is likely that you will identify rapidly when it is failing to converge, and at this point, you can switch to a safer method like the bisection search. For example, if Newton's method hasn't converged in 10 iterations, then, in many applications, it should probably be stopped, and a different approach tried.

6.4.4 Root finding and physics

Other root finding approaches do exist, but probably not as widely used as those introduced here. One final note, though, is that the root finding algorithms we consider do not know about the physics of your problem. This means that guesses may be outside a range of physically permissible values. For example, a density guess could be negative, and it may be that the underlying function cannot be evaluated with negative density (e.g. it contains the sound speed, which contains a square root). This is generally avoided by applying limits to values. For example, if a negative density was obtained, it should be corrected within the algorithm to the minimum

admissible value that density can take.

6.5 Runge-Kutta methods

We have mentioned differential equations, especially PDEs, frequently in this course. Ordinary differential equations are also commonly encountered, and, in many cases, the key aspect of a numerical method for PDEs is to compute quantities such that an ODE can be solved. For example, if we have a PDE of the form

$$\frac{\partial u_i}{\partial t} + \nabla_j f_{ij}\left(\mathbf{u}\right) = 0, \tag{6.83}$$

then the **method of lines** can be employed which requires our numerical method provides an approximation to the spatial derivative,

$$\mathcal{F}_i(\mathbf{u}) = -\nabla_j f_{ij}\left(\mathbf{u}\right). \tag{6.84}$$

Here, the approximation is such that $\mathcal{F}_i = \mathcal{F}_i(\mathbf{u}, t)$, but there is no dependency on space any more. And other courses in the MPhil, as well as various textbooks, will spend a lot of time working out how you compute $\mathcal{F}_i$, and very little time dealing with what happens next. This is because once you have an equation of the form

$$\frac{\mathrm{d}u}{\mathrm{d}t} = F(u, t) \tag{6.85}$$

we have an ordinary differential equation, and there are a wide variety of standard algorithms to solve such cases.

In this course, we consider **Runge-Kutta** methods, one of the most standard families of methods for solving ODEs. They are an iterative method for solving an intial value problem, i.e.

$$\frac{\mathrm{d}u}{\mathrm{d}t} = F(u, t), \quad u(t_0) = u_0 \tag{6.86}$$

where u_0 is a known value for some initial time t_0.

Suppose we want to find a solution to our ODE at some future time, t_N, which we will denote u_N. Note that in general, we have $u_N \neq u(t_N)$, due to numerical error in the iterative approach. The idea is that, because we know the derivative of u, we can use this to predict the behaviour of u a short time into the future. In general, we expect the derviative to also change with time, or the value of u, so after this short time, we would need to recompute

the derivative and estimate u at another short time in the future. We would do this until our esimates reach t_N.

In general, each step of this process for a Runge-Kutta method can be written

$$u_{n+1} = u_n + h \sum_{i=1}^{s} b_i k_i \tag{6.87}$$

where u_n is the solution we know, u_{n+1}, and is an estimate of the solution $u(t_{n+1}) = u(t_n + h)$, and thus h is the size of the 'short jump' in time, often known simply as the **time step**. Within the summation are terms which are used to represent the derivative function, $F(u, t)$, allowing us to use the derivative from multiple times between t_n and t_{n+1}. This is, itself, sometimes challenging, since we may not actually know an approximation for u at these times. Within this summation, b_i is a constant for a given i, and k_i is an evaluation of $F(u, t)$, defined by

$$k_1 = F(u_n, t_n) \tag{6.88}$$

$$k_2 = F(u_n + h(a_{21}k_1), t_n + hc_2) \tag{6.89}$$

$$k_3 = F(u_n + h(a_{31}k_1 + a_{32}k_2), t_n + hc_3) \tag{6.90}$$

$$\vdots$$

$$k_s = F(u_n + h(a_{s1}k_1 + a_{s2}k_2 + \ldots + a_{s,s-1}k_{s-1}), t_n + hc_s) \tag{6.91}$$

where additional constants c_i and a_{ij} are used. These constants cannot be chosen arbitrarily, Runge-Kutta methods have particular values of these constants which are known to provide good solutions.

The number of steps is often directly related to the order of accuracy of your solution - in the same way that a higher-order approximation to a derivative requires information from more known points, a higher-order solution to an ODE requires more evaluation of the function $F(u, t)$.

6.5.1 The Euler method

In general, deriving the constants for a Runge-Kutta method is a lot of effort, and requires various analytical techniques which we will not consider in this course. However, the simpler ones (lower-order accurate) can be derived directly. This begins with a first-order method, the **Euler method**. This comes directly from replacing the derivative with a first-order accurate approximation, i.e. we use

$$\frac{\partial u}{\partial t} \approx \frac{u(t+h) - u(t)}{\Delta t} = \frac{u_{n+1} - u_n}{h}. \tag{6.92}$$

Depending on the application, some sources will use a variable like h to denote the step size, some will explicitly define it as a change in the variable that is changing, e.g. Δt.

At $t = t_n$, our ODE is

$$\left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n} = F(u_n, t_n) \tag{6.93}$$

hence substituting our first order derivative gives

$$\frac{u_{n+1} - u_n}{h} = F(u_n, t_n) \tag{6.94}$$

or

$$u_{n+1} = u_n + hF(u_n, t_n). \tag{6.95}$$

If we compare this to the general form of a Runge-Kutta method, equation (6.87), we see that we have $s = 1$ and $b_1 = 1$, and since $k_1 = F(u_n, t_n)$, no further constants need defining.

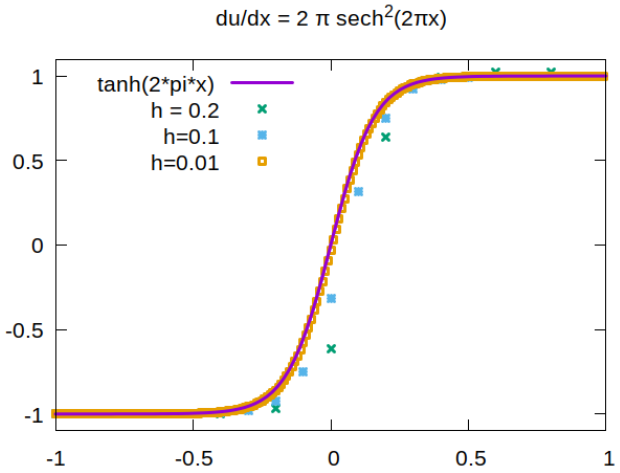


Figure 6.8: Evaluation of the Euler method for solving an ODE. Three different choices of h are shown, and it is clear that for small h, an accurate solution is obtained, but even for large h, a reasonable approximation is made.

In figure 6.8, a sample solution to the problem

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2\pi\mathrm{sech}^2(2\pi x), \quad u_0 = u(-1) = \tanh(-2\pi) \tag{6.96}$$

is shown (note we use x instead of t for this example). Three choices of h are shown, and it is clear that the smaller h is, the more accurate the solution, but the overall behaviour is predicted in each case.

Convergence of the Euler method

We know that the error in equation (6.92) is first order accurate, i.e. $\mathcal{O}(h)$. However, this does not, itself, guarantee that a solution u_{n+1} will actually converge. It could be that these errors accumulate in an unbounded manner. If we assume that we take N steps of the Euler method to obtain a solution at $t_0 + t_N$, the total error, E, is given by

$$E = \sum_{i=0}^{N} E_i \tag{6.97}$$

where E_i is the error of step i of the Euler method. We need to know that these single-step errors do not grow unboundedly, and, fortunately, this can be checked.

We consider the error for a given iteration of the Euler method, E_n,

$$E_n = u_n - u(t_n), \tag{6.98}$$

i.e. the difference between the approximate and the exact solution at time $t = t_n$. We now investigate the error at $t = t_{n+1} = t_n + h$,

$$E_{n+1} = u_{n+1} - u(t_{n+1}) = u_n + hF(u_n, t_n) - u(t_{n+1}) \tag{6.99}$$

where in the final step, we use the definition of u_{n+1} from equation (6.95). We now take a Taylor expansion of $u(t_{n+1})$ about t_n,

$$E_{n+1} = u_n + hF(u_n, t_n) - \left(u(t_n) + h \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n} + \mathcal{O}_(h^2) \right) \tag{6.100}$$

and then note that the $\mathcal{O}(h^2)$ term, although unknown, can be calculated for all t_n from t_0 to t_N. This will provide a bound on this term, i.e. we can say

$$\mathcal{O}(h^2) \leq ah^2 \tag{6.101}$$

for some positive constant a. For example, at iteration n, we could define this error term as $\mathcal{O}_n$, and we then know that

$$a = N \max_{i=N}(\mathcal{O}_i) \tag{6.102}$$

would bound this term.

In order to quantify the error, we only care about the magnitude (absolute value) of the error, not its sign. Therefore we have

$$|E_{n+1}| \leq \left| u_n + hF(u_n, t_n) - \left(u(t_n) + h \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n} + ah^2 \right) \right|. \tag{6.103}$$

We can now use the triangle inequality,

$$|a+b| \leq |a| + |b|\,, \tag{6.104}$$

as well as noting that by the definition of the equation we are solving $\frac{\mathrm{d}u}{\mathrm{d}t}\big|_{t_n} = F(u(t_n), t_n)$ to get

$$\begin{aligned} |E|_{n+1} &\leq \left|(u_n - u(t_n)) + h(F(u_n, t_n) - F(u(t_n), t_n)) + ah^2\right| \\ &\leq |u_n - u(t_n)| + \left|h(F(u_n, t_n) - F(u(t_n), t_n)) + ah^2\right| \\ &\leq |u_n - u(t_n)| + h\,|F(u_n, t_n) - F(u(t_n), t_n)| + ah^2. \end{aligned} \tag{6.105}$$

In order to further simplify this, we need to make an additional statement about the function $F(u(t_n), t_n)$; that it is **Lipschitz continuous**.

A Lipschitz continuous function $f(x)$ is one such that there exists a constant, K, such that for any two points, x_1 and x_2,

$$|f(x_1) - f(x_2)| \leq K\,|x_1 - x_2|\,. \tag{6.106}$$

In the case $x_1 = x_2$, this is trivially satisfied. Otherwise, we could state instead that

$$K \geq \frac{|f(x_1) - f(x_2)|}{|x_1 - x_2|}. \tag{6.107}$$

Roughly speaking, for a function to be Lipschitz continuous, we need it's derivative to be bounded. To see this, consider the equation above in the limit $x_2 \to x_1$, and we recover an expression for the derivative of f.
If we now return to equation (6.105), we note that the in the term $|(F(u_n, t_n) - F(u(t_n), t_n))|$, both entries have the same value of t_n, i.e. this is kept constant, they are functions of one variable. Therefore, if they are Lipschitz continuous, then we have

$$|(F(u_n, t_n) - F(u(t_n), t_n))| \leq |u_n - u(t_n)| \tag{6.108}$$

so equation (6.105) becomes

$$|E|_{n+1} \leq |u_n - u(t_n)| + hK\,|u_n - u(t_n)| + ah^2. \tag{6.109}$$

We now use equation (6.98) to replace the $u_n - u(t_n)$ terms with the error E_n,

$$|E|_{n+1} \leq |E_n| + hK\,|E_n| + ah^2 = (1 + hK)\,|E_n| + ah^2 \tag{6.110}$$

We now need to find an expression for E_n, which we find by considering the first few error values:

$$|E_0| = 0 \tag{6.111}$$
$$|E_1| = (1+hK)\,|e_0| + ah^2 = ah^2 \tag{6.112}$$
$$\begin{aligned}|E_2| &= (1+hK)\,|e_1| + ah^2 = (1+hK)\,ah^2 + ah^2 \\ &= ah^2\,[1+(1+hK)]\end{aligned} \tag{6.113}$$
$$|E_3| = (1+hK)\,ah^2\,[1+(1+hK)] + ah^2 \tag{6.114}$$
$$= ah^2\left[1+(1+hK)+(1+hK)^2\right] \tag{6.115}$$

From this, we can see that

$$|E_n| = ah^2 \sum_{i=0}^{N-1} (1+hK)^n\,. \tag{6.116}$$

Note that this is a geometric series, which, if, for simplicity, we define $r = (1+hK)$, we have

$$\sum_{i=0}^{N-1} r^n = r^0 + r^1 + \cdots + r^n \tag{6.117}$$

and, multiplying this by r gives

$$r\sum_{i=0}^{N-1} r^n = r^1 + r^2 + \cdots + r^{n+1}. \tag{6.118}$$

Subtracting this from the original sum gives

$$\sum_{i=0}^{N-1} r^n - r\sum_{i=0}^{N-1} r^n = (1-r)\sum_{i=0}^{N-1} r^n = \left(1-r^{n+1}\right) \tag{6.119}$$

or

$$\sum_{i=0}^{N-1} r^n = \frac{(1-r^{n+1})}{1-r^n} = \frac{1-(1+hK)^n}{1-(1+hK)} = -\frac{1-(1+hK)^n}{hK}. \tag{6.120}$$

This means that our error in equation (6.110) can be written

$$\begin{aligned}|E|_{n+1} &\leq ah^2 \sum_{i=0}^{N-1} (1+hK)^n + ah^2 \\ &\leq -\frac{ah^2}{hK}\left(1-(1+hK)^n\right) + ah^2 \\ &\leq \frac{ah}{K}\left((1+hK)^n - 1\right) \\ &\leq \frac{ah}{K}\,(1+hK)^n\,.\end{aligned} \tag{6.121}$$

The next thing we need is the Taylor expansion of e^x about $x = 0$,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots \tag{6.122}$$

and note that if $x \geq 0$, every term would be positive, hence in this case, we would have

$$e^x \geq 1 + x. \tag{6.123}$$

This is useful, since K is a positive constant, and h is positive, hence we know

$$(1 + hK)^n \leq \left(e^{hK}\right)^n = e^{hKn}. \tag{6.124}$$

And again, since h and K are positive constants, if we have $n \in [1, N]$, then we know that

$$e^{hKn} \leq e^{hKN} \tag{6.125}$$

and, since $t_N = t_0 + hN$, we have

$$e^{hKn} \leq e^{K(t_N - t_0)}. \tag{6.126}$$

This means our error is now bounded by

$$|E|_{n+1} \leq \frac{ah}{K} e^{K(t_N - t_0)} + ah^2. \tag{6.127}$$

Since a, K and $t_N - t_0$ are all positive and constant, we can say

$$|E|_{n+1} \leq bh + ah^2 = bh + \mathcal{O}(h^2) \tag{6.128}$$

where

$$b = \frac{a}{K} e^{K(t_N - t_0)}. \tag{6.129}$$

This means that as $h \to 0$, the error $|E|_{n+1}$ decreases at first order.

6.5.2 Midpoint method

We can derive a second-order accurate method for solving ordinary differential equations by starting with a second-order accurate approximation to our time derivative. From equation (6.10), we have

$$\left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_t \approx \frac{u(t + \Delta t) - u(t - \Delta t)}{2\Delta t}. \tag{6.130}$$

If we now set $t_{n+1} = t + \Delta t$, $t_n = t - \Delta t$ and $h = 2\Delta t$, we can write this as

$$\left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n + h/2} \approx \frac{u_{n+1} - u_n}{h} \tag{6.131}$$

where we note that from our definition,

$$t = t_n + \Delta t = t_n + \frac{h}{2}. \tag{6.132}$$

Rearranging equation (6.131), we get

$$u_{n+1} = u_n + h \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n+h/2}. \tag{6.133}$$

We now need to know what this derivative term is. From our original equation,

$$\frac{\mathrm{d}u}{\mathrm{d}t} = F(u,t), \tag{6.134}$$

analysing this at $t = t_n + h/2$ gives

$$\left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{t_n+h/2} = F(u_{n+h/2}, t_{n+h/2}) \tag{6.135}$$

and hence our numerical update is given by

$$u_{n+1} = u_n + hF(u_{n+h/2}, t_{n+h/2}). \tag{6.136}$$

However, since we are only approximating the solution at integers of h, we don't know $u_{n+h/2}$. However, here, we can approximate this using the Euler method, i.e.

$$u_{n+h/2} = u_n + \frac{h}{2}F(u_n, t_n). \tag{6.137}$$

This gives an update formula of

$$u_{n+1} = u_n + hF\left(u_n + \frac{h}{2}F(u_n, t_n), t_{n+h/2}\right). \tag{6.138}$$

Effectively, the midpoint method works by trying to guess the derivative of u not from the known value, u_n, but from the point halfway between u_n and u_{n+1}. This should better identify the behavioural trends between these two points, and formal analysis does show that it is second order accurate.

If we relate this now to the general Runge-Kutta formulation in equation (6.87), we have the case $s = 2$, i.e.

$$u_{n+1} = u_n + h\sum_{i=1}^{2} b_i k_i \tag{6.139}$$

which means we have

$$k_1 = F(u_n, t_n) \tag{6.140}$$
$$k_2 = F(u_n + h(a_{21}k_1), t_n + hc_2) \tag{6.141}$$

and we can now define the rest of the constants. The important thing to note here is that only a single term appears in equation (6.138) where we might have expected a sum, in other words, $b_1 = 0$ and $b_2 = 1$. Matching the coefficients inside k_2 is straightforward, and we have $a_{21} = \frac{1}{2}$ and, since $t_{n+1/2} = t_n + \frac{h}{2}$, $c_2 = \frac{1}{2}$.

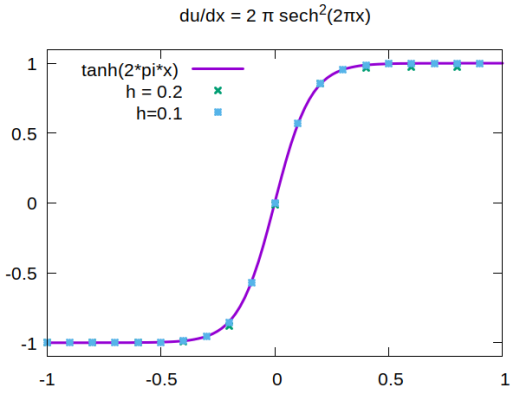


Figure 6.9: Illustration of the midpoint method for solving an ODE. This can be contrasted with figure 6.8, where for the two lowest choices of h in this previous figure, we see that the midpoint method is now consistently close to the exact solution.

An illustration of the midpoint method for the initial data given in figure (6.96) is shown in figure 6.9. Here, the greater accuracy of the midpoint method is immediately apparent, since even the larger choices of h give good results.

6.5.3 Butcher tableaux

The generic form of a Runge-Kutta method, described in equations (6.87) to (6.91), can be rather awkward to write down for each method. In practice, we are actually only interested in the quantities s, b_i, c_i and a_{ij}. Butcher tableaux are relatively simple tables designed to make the reading of, and certainly the algorithmic implementation of, Runge-Kutta methods more

straightforward. For a given method, the a Butcher tableau has the form

$$\begin{array}{c|cccccc} 0 & & & & & & \\ c_2 & a_{21} & & & & & \\ c_3 & a_{31} & a_{32} & & & & \\ \vdots & \vdots & & \ddots & & & \\ c_s & a_{s1} & a_{s2} & \dots & a_{s,s-1} & & \\ \hline & b_1 & b_2 & \dots & b_{s-1} & b_s \end{array} \tag{6.142}$$

This is a table with s columns and rows, which defines all coefficients, where we note that $c_1 = 0$ for all methods, and a_{ij} only exists for $j < i$.

Some common Runge-Kutta methods, expressed in Butcher tableaux, are:

- Euler method (first-order accurate):

$$\begin{array}{c|c} 0 & \\ \hline & 1 \end{array} \tag{6.143}$$

- Midpoint method (second-order accurate):

$$\begin{array}{c|cc} 0 & & \\ \frac{1}{2} & \frac{1}{2} & \\ \hline & 0 & 1 \end{array} \tag{6.144}$$

- Heun's method (second-order accurate):

$$\begin{array}{c|cc} 0 & & \\ 1 & 1 & \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array} \tag{6.145}$$

- **The** Runge-Kutta method (fourth-order accurate) - this is the method that ended up giving its name to the family of methods, also known as the original Runge-Kutta method

$$\begin{array}{c|cccc} 0 & & & & \\ \frac{1}{2} & \frac{1}{2} & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ 1 & 0 & 0 & 1 & \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array} \tag{6.146}$$

- 3/8-rule (fourth-order accurate):

$$\begin{array}{c|cccc} 0 & & & & \\ \frac{1}{3} & \frac{1}{3} & & & \\ \frac{2}{3} & -\frac{1}{3} & 1 & & \\ 1 & 1 & -1 & 1 & \\ \hline & \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{array} \tag{6.147}$$

Having introduced a number of Runge-Kutta methods, the obvious question might now be 'why are there so many?' In particular, there are multiple methods at a given accuracy. The answer to this is because accuracy is not the only criteron through which you would choose a method; stability and efficiency are often essential things to consider too.

First, we consider accuracy versus effiency. Two methods with a given order of accuracy are do not necessarily produce the same error in an approximate solution. At a given choice of h, there will be a given error, ϵ, which will decrease according to the order of accuracy, but there is no reason that two given methods have the same initial ϵ, which is referred to as the **absolute error**. When looking at the error of the Euler method, we bounded the error with several positive constants; other methods could have larger (or smaller) constants here.

Considering the two fourth-order accurate methods we considered, the 3/8-rule has lower absolute error thant the original Runge-Kutta method. However, the large nuber of zeroes present in the Butcher tableau for the original method offers an efficiency advantage. If we return to the expression for a general k_s, equation (6.91),

$$k_s = F(u_n + h(a_{s1}k_1 + a_{s2}k_2 + \ldots + a_{s,s-1}k_{s-1}), t_n + hc_s), \tag{6.148}$$

we see that if all a_{sj} are non-zero, we need to store all previous k_i values in order to compute k_s. However, in the case of the original Runge-Kutta method, only the $a_{s,s-1}$ coefficient, which multiplies k_{s-1}, is non-zero. All other k_i values are not used, thus do not need to be stored. This makes the method computationally efficient, and this efficiency can often outweigh the slight decrease in absolute error.

Stability of a Runge-Kutta method is another important aspect; error in a solution is ok if it is reduced, hopefully according to the order of accuracy of the method. An unstable method, though, has growing, or oscillatory error, and if this exists, the method would be an undesirable choice. We showed that, for the Euler method, we know that decreasing h will result in decreasing error, however, this does not tell us whether a given value of h will actually be stable. We will not go into details as to how stability is computed for a given h; this is often dependent on the form of $F(u,t)$, and may require analysis of this equation. We shall simply note a practical example where unstable behaviour can be seen.

If figure 6.10, the solution to

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -15u, \quad u_0 = u(0) = 1, \tag{6.149}$$

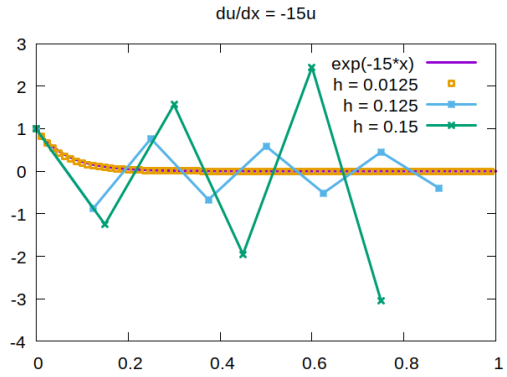


Figure 6.10: Example of a solution to an ODE with the Euler method where, for large choices of h, the method is unstable. For small h, the method recovers the exact solution reasonably well, but for moderate h, oscillations start to appear, and as h grows even larger, these then grow unboundedly.

is shown. This ODE can be solved analytically, with an exact solution

$$u = e^{-15x}. \tag{6.150}$$

The Euler method is used to solve this equation too, and we see that for large h, oscillatory behviour appears, and, as h increases, the oscillations grow unboundedly.

Although we are not going to compute stability for Runge-Kutta methods, this highlights how it might be considered in practice. Ideally, you want the limit on a stable choice of h to be as large as possible, and you'd like your solution to be solution to be unsatisfactory because the error is too high (e.g. the largest choice of h in figure (6.8)) before it becomes unsatisfactory due to stability issues. This means that you can achieve the best solution at t_N with the fewest steps taken to reach it. Stability can often be directly related to efficiency as a result - the more stable the method, the more efficient your calculations can be.